

What Do Populist Voters Want?

A Nonparametric Revealed Preference Approach

Nina Nikiforova*

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Abstract

This paper proposes a nonparametric revealed-preference approach for identification and estimation of voter preferences from aggregate election returns in a multidimensional spatial voting model. I apply it to Germany's four Bundestag elections between 2013 and 2025, modeling political competition along left-right and anti-elite dimensions to study the rise of populism. The results reveal a transformation in the basis of support for the radical-right populist party Alternative for Germany (AfD): what began primarily as anti-elite voting became increasingly aligned with the party's right-wing positions. By 2021, the left-right dimension accounts for as much as 70% of AfD support. Counterfactuals in 2025 suggest that a modest rightward shift by the center-right Christian Democratic Union (CDU) attracts about one-fifth of AfD voters if the AfD's position is fixed. When allowed to respond strategically, the AfD moves further right and reverses these gains; in equilibrium, the CDU does not move right. These results may explain why establishment parties struggle to reclaim populist voters despite apparent ideological openings.

*Bocconi University. E-mail: nina.nikiforova@unibocconi.it. I am grateful to Guido Tabellini for his constant support and guidance; and to Jérôme Adda, Cailin Slattery, Pietro Tebaldi and Francesco Trebbi for their generous advice and feedback. I thank Christian Cox, Catherine De Vries, Francesco Decarolis, Sarah Eichmeyer, Rafael Jiménez-Durán, Thomas Le Barbanchon, Nicolas Longuet-Marx, Massimo Morelli, Eduard Osipov, Carlo Schwarz, Piero Stanig, Edoardo Teso and Christopher Turansick for their helpful comments and suggestions.

1 Introduction

Western democracies are experiencing profound political transformations, with the rise of populism being one example. The populist vote share in Western democracies has roughly tripled since the 1990s and now exceeds a quarter of the electorate (Guriev and Papaioannou, 2022). This trend shows no signs of reversing: Trump returned to the White House in 2024, Marine Le Pen’s National Rally topped the first round of France’s 2024 legislative elections, and the Alternative for Germany (AfD) doubled its vote share between the 2021 and 2025 federal elections. Existing work links this phenomenon to issues such as trade (Autor et al., 2020; Colantone and Stanig, 2018b), immigration (Dustmann, Vasiljeva and Piil Damm, 2019), and identity (Norris and Inglehart, 2019; Bonomi, Gennaioli and Tabellini, 2021). Yet evidence on individual drivers does not by itself provide a unified empirical account of how voters’ preferences have evolved (Margalit, 2019).

In this paper, I ask and empirically investigate *what are the preferences of populist voters?* And *how should mainstream parties respond to the rise of populism?* To answer these questions, I develop an empirical model of multidimensional spatial voting (Downs, 1957) that can be identified and estimated using aggregate vote shares. The model accommodates instrumental variables, minimizes reliance on functional-form and distributional assumptions, and enables counterfactual analysis of spatial competition. Recent empirical work in political economy develops econometric models to estimate voters’ preferences (Berry, Cox and Haile, 2025; Danieli et al., 2026; Longuet-Marx, 2026). I connect this empirical agenda to spatial voting theory.¹

I apply this framework to study the rise of populism in Germany, where the radical-right populist party Alternative for Germany (AfD) leads the national polls and has won several state elections. I begin by recovering the preferences of AfD voters over 2013–2025 along the Left-Right and Anti-Elite dimensions. I find that the party’s earlier support was grounded in anti-elite sentiment, whereas its more recent support is rooted in its radical-right stance. Holding the AfD’s position fixed, counterfactuals for 2025 suggest that a modest rightward shift (for instance, on immigration policy) by the center-right Christian Democratic Union (CDU) could attract about a fifth of AfD voters. However, in the equilibrium of a sequential repositioning game with the CDU moving first, the AfD responds to a rightward CDU move by moving further right, largely reversing these gains. Anticipating this response, the CDU typically stays put.

In the first part of the paper, I develop an approach to identifying and estimating voters’ ideological preferences using variation across localities in aggregate electoral returns, political supply, and demographic composition. I adopt a spatial voting model in which voter i ’s valuation of candidate j depends only on the weighted Euclidean distance between the voter’s ideal point (bliss point) X_i and the candidate’s position Y_j in the policy space. Weights, represented by a diagonal weight matrix W , capture the importance voters attach to each dimension, and are assumed to be the same across i in a given region. Voters are grouped into types defined by observable characteristics, and the distribution of ideal points is assumed to be the same

¹Merlo and De Paula (2017) pioneered estimation of spatial voting models, but did so under a specific set of functional form and distributional assumptions, and without tackling endogeneity arising from the correlation between voters’ preferences and candidates’ positions.

within each type but allowed to differ between types. The approach relies solely on the geometry of choice implied by the multidimensional spatial voting model, imposes no distributional assumptions, and exploits variation in candidate positions across choice environments. In my application, this variation comes from differences in candidate positions across electoral districts within a state. The approach applies to any setting in which voters face different ballots in the same election, such as simultaneous elections for several offices (the United States, Italy), ranked-choice voting (Australia, Ireland), or district-based representation.

Two observations are key. First, the weight matrix W determines *which* joint voting patterns are feasible across different electoral districts: since a single type-specific distribution of ideal points must generate that type’s vote shares in every district, a given W can rationalize only some joint patterns and not others. Building on this variation, weights are set-identified. Second, there are generally infinitely many ideal-point distributions consistent with the data, as the data could pin down only the mass of voters on each joint voting pattern, not the exact distribution that generated it.² Point identification of a single distribution would therefore require additional structure, either on the choice model or on unobserved heterogeneity.³ Instead, I bound the target parameters, building on the methodology of [Tebaldi, Torgovitsky and Yang \(2023\)](#). Target parameters can describe both voters’ preferences and counterfactual voting behavior: examples include the share of voters whose ideal points lie in a given area of the policy space, or the share who would abandon a party if it shifted its position by a given amount.

In the second part of the paper, I apply the framework to Germany, where the electoral landscape has undergone a pronounced transformation over the past two decades, with the traditional establishment parties losing ground to the populists. The central empirical object is the preferences (W, X_i) of the electoral base of the populist party Alternative for Germany (AfD), which by October 2026 leads national polls with 10-percentage-point margin.⁴ I estimate these preferences separately for each federal election over 2013–2025 to trace their evolution, and use the recovered preferences to conduct counterfactual analyses of party repositioning that would otherwise be infeasible.

I assume that ideal-point distributions are common across electoral districts within each demographic group, state, and year, while weights are common across all individuals within each region (East/West) and year.⁵ I use municipality-level demographic data and electoral returns, together with rich within-state variation in the political supply of local candidates, to estimate these objects. On the voter side, the unit of analysis is a demographic group, defined as an age–gender–education bin within a state and year. I obtain municipality-level joint demographic data for 24 such groups.⁶ On the candidate side, I measure positions on the Left–Right and Anti-Elite dimensions, using the GLES Candidate Survey and a nonpartisan *Whom*

²Each feasible joint voting pattern corresponds to a convex cell of the partition of the policy space induced by the weights and candidate positions, restricting where voters’ bliss points can lie without determining their exact locations within that cell.

³This is the approach taken, for example, by [Merlo and De Paula \(2017\)](#); [Iaryczower, Montero and Kim \(2024\)](#).

⁴Politico, National Parliament Voting Intentions – [Link](#). Accessed: 6 October 2026.

⁵This is because estimating the weights requires substantial variation. I discuss this point in more detail in the approach section below.

⁶Since the model abstracts from turnout decisions, I use administrative turnout rates by state, year, and age–gender group to reconstruct the demographic composition of actual voters.

to *Vote For?* candidate survey.⁷ Because the same survey items recur across election years, the resulting positions are comparable over time. I combine these data with municipality-level party-list vote shares for the six main parties, allowing the model to relate observed vote shares to latent group-specific preference distributions.

Germany’s mixed electoral system combines a plausibly sincere measure of party choice with geographic variation in political supply. Voters cast one vote for a district candidate and another for a party list. I focus on party-list vote shares, treating choices among the six major parties as sincere. Under this assumption, each vote reveals which party is closest to the voter’s ideal point, allowing observed vote shares to inform the underlying preference distributions. I use district candidates’ positions as proxies for locally perceived party positions, allowing the same party to be perceived differently across districts. I provide evidence supporting this proxy by relating voters’ perceived party positions from survey data to local candidates’ positions within state and election year, and by showing that the observed candidate positions yield a better model fit than randomly reassigned positions in a permutation test.

The identifying variation comes from differences in candidate positions across electoral districts within a state and year. The identifying assumption is that the within-state variation in candidate positions is unrelated to unobserved voter heterogeneity across districts, conditional on the observed age–gender–education composition of municipalities. This is consistent with institutional details of Germany’s mixed electoral system: because the proportional allocation of Bundestag seats is determined by party-list votes, district candidacies are embedded in a broader state-level process of selecting candidates for parliamentary representation, rather than being solely tailored to the preferences of specific local electorates. I test the robustness of my results to this assumption in four ways: first, by focusing on very small municipalities, which are least likely to be targeted by candidates; second, by restricting the analysis to districts with non-contender AfD candidates; third, by allowing for preference heterogeneity between electoral districts in which AfD candidates are more or less conservative; fourth, by focusing on municipalities near district borders within states. The results remain quantitatively similar across all specifications.

I start by estimating the weight of the Anti-Elite dimension relative to the Left–Right dimension among AfD voters. The results suggest that in Western Germany the Left–Right dimension carries roughly three to five times the weight of the Anti-Elite dimension throughout the period. The same does not hold in Eastern Germany, where the Anti-Elite dimension carries substantial weight in every year and is estimated to outweigh the Left–Right dimension in 2013 and 2025.

I then recover the *motivations* behind the AfD vote, in an exercise that combines the estimated weights with the underlying preference distribution. Suppose the AfD split into two parties: one keeping its radical-

⁷These dimensions connect to how the political science literature conceptualizes populism. A body of work distinguishes the role of *thin* and *thick* ideologies in populist support, that is, anti-elite appeals and substantive ideological stances. See [Mudde and Kaltwasser \(2017\)](#); [Vries and Hobolt \(2020\)](#) for a discussion, [Gurieva and Papaioannou \(2022\)](#) for an overview of the political economy literature, and my companion paper [Nikiforova \(2026\)](#) for the formal test on the importance of different dimensions in vote choices in Germany over 2013–2022.

right position but moving towards the establishment, the other keeping its anti-establishment appeal but moderating its Left–Right position. What share of voters would each attract? I find that, over time, AfD support has become increasingly grounded in the party’s conservative Left–Right positions rather than in its anti-elite appeal, with up to 70% of voters supporting the AfD in 2021–2025 due to its radical-right ideology.

I next run two partial-equilibrium repositioning exercises to understand the intensity of competition between the establishment parties and the AfD. In the first, I explore the sensitivity of the AfD electorate to shifts in the AfD’s own position towards more pro-elite or more moderate Left–Right stances (a form of own-position *elasticity*). The results suggest that until 2025 voters were relatively insensitive to small shifts in party positions. This indicates that competition between the AfD and other parties used to be relatively weak: few of its voters lay near the boundaries with other parties’ attraction regions. The picture changes in 2025, when even a small moderation in the AfD’s Left–Right position would lead to substantial electoral losses.

In the second exercise, I ask how establishment parties could compete with populist challengers. I focus on the largest center-right establishment party, the CDU, and, holding the AfD’s position fixed, estimate how much of the AfD electorate it could capture by shifting towards more conservative and more anti-elite positions (a form of cross-position *elasticity*). The results suggest that a small repositioning on the Left–Right dimension would capture no more than 10% of the AfD electorate before 2025. In 2025, however, the same shift would allow the CDU to capture about 20% of the AfD electorate. Crucially, hardening the party’s anti-elite positions is not nearly as effective in any year.

Finally, I compute the equilibrium of a game with strategic interactions for 2025: both the CDU and the AfD can reposition, while the positions of the remaining parties are held fixed. I consider a Stackelberg game in which the CDU leads and the AfD follows. If the AfD stays put, it is optimal for the CDU to move rightward, because the gains on the right flank (from the AfD) exceed the losses on the left. The AfD’s optimal response to a rightward move by the CDU, however, is to move further right itself, recapturing the most right-wing (and relatively pro-elite) voters. Therefore, the Stackelberg equilibrium is that the CDU does not reposition. This exercise offers one answer to the question of why mainstream parties do not adjust their platforms as much as a static calculation would suggest: the anticipation of the populists’ response.

I conclude by discussing the choices underlying the framework and what they imply for the results. Among other things, I show that (i) the model can be read as a probabilistic voting model in which the error distribution is induced by the distribution of bliss-point shocks rather than assumed directly; (ii) measurement error in candidate positions shifts the bounds towards the uninformative composition; and (iii) the consequences of omitting a dimension depend on its relevance and on its correlation with the included ones. I also re-estimate the model with Economics and Culture as the two dimensions, and find a rise in the role of cultural concerns relative to economic ones.

Related Literature. This paper is related to several strands of literature. First, it contributes to a growing body of research that estimates structural models of voting in elections from observed election returns (Kawai and Watanabe, 2013; Ujhelyi, Chatterjee and Szabó, 2021; Iaryczower, Montero and Kim, 2024; Kawai and Sunada, 2025; Berry, Cox and Haile, 2025; Longuet-Marx, 2026). This paper contributes by developing a novel nonparametric approach to identifying and estimating a multidimensional spatial voting model from aggregate election results, in which individuals vote for the candidate closest to them in the weighted policy space. Of these papers, only Iaryczower, Montero and Kim (2024) estimates a spatial voting model, and only in one dimension. As noted in Merlo and De Paula (2017), estimation of a single-dimensional model is considerably easier, since the problem of separately identifying *weights* from bliss points does not arise.

The paper is closely related to Degan and Merlo (2009), who first formalise the partitioning of the policy space under the multidimensional spatial voting model using Voronoi cells. Merlo and De Paula (2017) exploit this partitioning to prove point identification of the bliss-point distribution and the weights under a large-support assumption on candidates’ positions, and jointly estimate the best-fitting preference distribution and weights by minimum distance with a sieve density. The large-support assumption requires that any profile of candidate positions—including candidates clustered arbitrarily close to any two points, or arbitrarily far apart—eventually be observed. It delivers point identification, but is untenable in most electoral settings.

This paper, in turn, relaxes the large-support assumption, at the cost of set rather than point identification, and asks what the observed candidate configurations alone identify and how to estimate the resulting objects without imposing a functional form on the distribution of bliss points. Building on Tebaldi, Torgovitsky and Yang (2023) and Lieber, Tebaldi and Torgovitsky (2024), I establish set identification of the dimension weights and bliss-point distributions and adapt a linear-programming procedure for estimating sharp bounds on parameters describing both voters’ preferences and counterfactual voting outcomes. The approach exploits a similarity between the geometry of choice induced by spatial voting in my setting and that induced by quasilinear demand models in their applications: both partition the space of latent preferences into regions that imply the same choices across choice environments. The model in this paper is a pure-characteristics model, in which the latent component is not a valuation for each alternative but the voter’s bliss point, which enters utility non-linearly through its weighted distance to candidates’ positions.⁸ The weights and bliss-point distributions are informed by distinct features of this geometry. The framework can also accommodate instrumental variables.

Second, this paper is related to a literature that employs a range of methodologies to recover voters’ preferences, including the use of survey responses on political attitudes and self-placement, which are then mapped into ideological positions (Fiorina and Abrams, 2008; Algan et al., 2017; Hangartner et al., 2019; Gethin, Martínez-Toledano and Piketty, 2022; Desmet, Ortuno-Ortín and Wacziarg, 2024; Draca and Schwarz, 2024; Danieli et al., 2026). Such methods face two common shortcomings: voters’ views do not nec-

⁸The dimension weights therefore enter the partition through the metric of the latent space rather than through an index that is linear in observed characteristics. After rescaling the policy space by the weights, the identification of the weights has a structure similar to that of the enumeration problem in Lieber, Tebaldi and Torgovitsky (2024) (Appendix B).

essarily translate into voting choices, and some views may be stigmatized or socially desirable, and thus not reported truthfully. Survey-based stated preferences can therefore differ substantially from the revealed preferences inferred from electoral outcomes, and the extent of this divergence can only be assessed by estimating preferences from actual election data.

Third, this paper contributes to the large and growing literature on the causes of the rise and success of populist parties and leaders in the Western world. An extensive body of work examines the various drivers of populist voting, including economic factors such as trade and the 2008–2009 Global Financial Crisis ([Algan et al., 2017](#); [Colantone and Stanig, 2018b,a](#); [Autor et al., 2020](#); [Guiso et al., 2021](#); [Dippel et al., 2022](#)), immigration ([Dustmann, Vasiljeva and Piil Damm, 2019](#); [Alesina and Tabellini, 2024](#)), and cultural factors ([Algan et al., 2018](#); [Margalit, 2019](#); [Gennaioli and Tabellini, 2025](#)), among others. While these drivers can be mapped onto distinct ideological dimensions (see, e.g., [Rodrik, 2021](#), for a clear formalization of this logic and a meta-analysis of mechanisms related to globalization shocks), they do not reveal the relative importance of each, or their interplay. This paper addresses this gap by estimating voters’ ideal-point distributions and the relative weights of policy dimensions in a multidimensional spatial model, and tracing how preferences evolve over time.

In this respect, the paper is closely related to [Danieli et al. \(2026\)](#), who recover the preferences of voters of radical-right populist parties in Europe. Unlike the present paper, however, that study relies on survey-based self-reported vote choices and issue preferences; moreover, while its conceptual framework can be cast as a bliss-point model, it cannot separate a strengthening association between individual characteristics and ideal points on a given dimension from a change in the weight on that dimension.

I also contribute to the literature on how established parties respond to new entrants and to niche or populist challengers ([Palfrey, 1984](#); [Meguid, 2005](#); [Abou-Chadi and Krause, 2020](#); [Spoon and Klüver, 2020](#); [Krause, Cohen and Abou-Chadi, 2023](#)), by estimating best responses and the Stackelberg equilibrium of a platform-choice game between an establishment party and a populist challenger.

Lastly, this paper contributes to the debate on the relative importance of ideological and anti-elite dimensions in populist voting. A large part of the literature emphasizes the role of right (or left) positions in shaping populist support ([Acemoglu, Egorov and Sonin, 2013](#); [Mudde, 2013](#); [Norris and Inglehart, 2019](#); [Guenther and Nunnari, 2025](#)), whereas another strand highlights the role of populist attitudes distinct from Left–Right ideology ([Mudde, 2004](#); [Akkerman, Mudde and Zaslove, 2014](#); [Van Kessel, 2015](#)). Several studies have sought to compare the importance of these two factors directly using survey and experimental data ([Silva, Fuks and Tamaki, 2022](#); [Neuner and Wratil, 2022](#); [Neuner et al., 2025](#)). This paper shows that populist voting can take different forms, with different factors prevailing across regions and over time.

The rest of the paper is structured as follows. Section 2 lays out an approach to recover voters’ preferences, and Section 3 discusses the scope of the approach. Section 4 introduces the German institutional context and the data. Section 5 presents the estimation procedure and the identification strategy. Section 6

reports the results on the estimated preferences; Section 7 runs spatial competition counterfactuals; Section 8 studies the repositioning game between the CDU and the AfD. Section 9 discusses possible extensions and adaptations of the model. Section 10 concludes.

2 Econometric Model

2.1 Voters, candidates and elections

Elections takes place in a policy space $\mathcal{R}$, a known closed and bounded subset of $\mathbb{R}^2$.⁹ Each voter i has a bliss point $X_i \in \mathcal{R}$, known to the voter but not to the researcher, and an observed type $D_i \in \mathcal{D}$, a demographic group.

There is a finite set of elections $\mathcal{E}$. Election e is contested by a set of candidates $\mathcal{J}_e$, and candidate $j \in \mathcal{J}_e$ runs on an observed platform $Y_{ej} \in \mathcal{R}$.¹⁰ Each voter votes in one election, $E_i \in \mathcal{E}$.¹¹ Elections are grouped into states: $s(e)$ denotes the state of election e , and $S_i = s(E_i)$ the state of voter i .

Voters vote sincerely for the nearest candidate (Downs, 1957; Hinich and Munger, 1996). Distance in the policy space is the weighted Euclidean distance

$$d_w(x, y) = \sqrt{(x_1 - y_1)^2 + w(x_2 - y_2)^2}, \quad w > 0,$$

where w is the weight of the second dimension relative to the first.¹² The weight is common to all voters of a state. Voter i votes for $c(X_i, E_i)$, where

$$c(x, e) = \arg \min_{j \in \mathcal{J}_e} d_w(x, Y_{ej}). \quad (1)$$

This rule is equivalent to maximizing $U(d_w(X_i, Y_{ej}))$ for any strictly decreasing U , so no cardinal loss function is imposed.¹³ The *attraction region* of candidate j in election e ,

$$\mathcal{X}_j(e) = \{x \in \mathcal{R} : c(x, e) = j\},$$

is the set of bliss points that vote for j : the Voronoi cell of Y_{ej} under d_w , intersected with $\mathcal{R}$, a convex polygon. Attraction regions depend on w ; I suppress this until Subsection 2.4.

⁹The two dimensions anticipate the application. The approach extends in principle to more dimensions.

¹⁰*Candidate* denotes any option on the ballot: a party, an individual candidate or a referendum answer, among others.

¹¹When the same voters face several ballots at once, as under simultaneous or ranked-choice voting (Section 3), each ballot is an election and E_i is a set; nothing below changes.

¹²In matrix form, $d_w(x, y)^2 = (x - y)^\top W(x - y)$ with $W = \text{diag}(1, w)$. Only the ratio of the diagonal elements affects choices, so normalizing the first to one is without loss of generality. A non-diagonal W would allow the dimensions to interact; I do not pursue this.

¹³See Kendall, Nannicini and Trebbi (2015); Cruz et al. (2024) for the estimation of the loss function. The analysis is not invariant, however, to the notion of distance: the identification argument relies on the geometry of the regions of the policy space closest to each candidate, i.e. Voronoi cells, and alternative metrics generate different regions. For example, the sum of absolute coordinate-wise differences imposes Manhattan rather than Euclidean geometry.

Note that only relative comparisons are exploited, and the turnout decision, which would require an absolute threshold for abstention, is not modeled.¹⁴

The object of interest is the distribution of bliss points of each type in each state. I assume that the distribution of X_i conditional on observables is absolutely continuous, so that it admits a density; let $f(x | s, d)$ denote the density of bliss points of type- d voters in state s , and $\mathcal{F}$ the set of all such densities on $\mathcal{R}$.¹⁵ No restriction is placed on the shape of f .

Target parameters. Quantities of interest are functionals of f . For instance, the share of type- d voters in state s whose bliss points are left-leaning on the first dimension, $\underline{\mathcal{X}} = \{x \in \mathcal{R} : x_1 \leq \underline{x}_1\}$ for a known threshold $\underline{x}_1$, is

$$\theta(s, d; f) = \int_{\underline{\mathcal{X}}} f(x | s, d) dx.$$

Other examples are the share of a candidate's voters with bliss points in a given region, or vote shares under counterfactual platforms. I refer to such quantities as *target parameters*, and write $\theta(f)$ when the arguments (s, d) are integrated over. The goal is to characterize the values of $\theta(f)$ consistent with the data and the assumptions.

2.2 Electoral returns and identifying assumption

In most electoral settings the researcher does not observe individual votes, but electoral returns: vote shares aggregated over administrative units, which I call municipalities and index by m . Voter i lives in municipality M_i ; each municipality has one election $e(m)$, and lies in one state $s(m)$, so that $E_i = e(M_i)$. For each municipality the returns give the vote share $\sigma_j(m)$ of every candidate $j \in \mathcal{J}_{e(m)}$, and the census gives the demographic composition $q_{m,d}$, the share of type- d voters, with $\sum_d q_{m,d} = 1$. Let $f(x | m, d)$ denote the density of bliss points of type- d voters in municipality m . Under the choice rule (1),

$$\sigma_j(m) = \sum_d q_{m,d} \int_{\mathcal{X}_j(e(m))} f(x | m, d) dx. \quad (2)$$

Without further restrictions (2) is uninformative, since each municipality has its own distribution of bliss points. The identifying assumption links voters across municipalities and elections within a state.

Assumption 1a. For every state s , type d and municipality m in state s , $f(x | m, d) = f(x | s, d)$ for all x .

Conditional on type, bliss points have the same distribution in every municipality and every election of a state, while candidate platforms vary across elections. This is a selection-on-observables assumption: platforms may differ across elections for reasons related to the type composition of the electorate, but not to voters' bliss points beyond their type. It rules out that voters sort into elections according to the candidates

¹⁴In the application, I account for differential turnout across demographic groups and states. Extending the model to turnout is a separate avenue of research.

¹⁵Absolute continuity implies that ties, which occur on the boundaries between attraction regions, have probability zero.

on offer, and that parties tailor candidates to local preferences that the types do not capture.

Assumption 1a can be weakened to an instrumental-variable condition.

Assumption 1b. There is an observed variable Z_i such that, for all (s, d) , all z, z' in the support of Z_i given $(S_i, D_i) = (s, d)$, and all x ,

$$f(x \mid z, s, d) = f(x \mid z', s, d),$$

where $f(x \mid z, s, d)$ averages $f(x \mid m, d)$ over the municipalities of state s with $Z_i = z$.

Assumption 1b has the two parts of a standard instrument. *Exogeneity* is the stated condition: the distribution of bliss points does not vary with Z_i , so the instrument is unrelated to the unobserved voter heterogeneity that could be correlated with the platforms on offer. *Relevance* is required for the condition to have identifying content: the elections faced, and hence the platforms, must vary with z . Assumption 1a is the special case $Z_i = M_i$, in which the municipality is the instrument: it determines the candidates faced and, within a state and type, is unrelated to bliss points. Section 3 discusses settings in which Assumptions 1a and 1b are plausible and identification strategies from the literature that take this form, and Section 5.2 argues for Assumption 1a in the application, and discusses robustness tests.

Match the data. Under Assumption 1a, a density $f \in \mathcal{F}$ predicts the vote shares

$$\sigma_j(m; f) = \sum_d q_{m,d} \int_{\mathcal{X}_j(e(m))} f(x \mid s(m), d) dx,$$

and is consistent with the data if

$$\underbrace{\sigma_j(m)}_{\text{observed}} = \underbrace{\sigma_j(m; f)}_{\text{predicted}} \quad \text{for all } j \in \mathcal{J}_{e(m)} \text{ and all } m. \quad (3)$$

The sharp identified set of bliss-point densities is $\mathcal{F}^* = \{f \in \mathcal{F} : f \text{ satisfies (3)}\}$, and the sharp identified set of the target parameter is

$$\Theta^* = \{\theta(f) : f \in \mathcal{F}^*\},$$

the set of all values of the target parameter consistent with the data and Assumption 1a, and no others. The identifying variation is visible in (3): voters of the same type in the same state face different candidates in different elections, and their vote shares must all be generated by the same distribution $f(x \mid s, d)$.

2.3 Identification of the target parameter

$\mathcal{F}$ is infinite-dimensional, so Θ^* cannot be computed by searching over densities. Tebaldi, Torgovitsky and Yang (2023) show that whether a conjectured value belongs to Θ^* is equivalent to the feasibility of a finite-dimensional system of linear equations. This subsection adapts their argument, which rests on the geometry of the attraction regions, to the election model. Throughout, w is treated as known; Subsection 2.4

relaxes this.

Minimal relevant partition. Under Assumption 1a, voters of type d in state s are spread across the elections of the state but share the distribution $f(x \mid s, d)$. The type can therefore be treated *as if* it were a single population voting in every election of the state: each election yields a marginal distribution of the type’s votes over $\mathcal{J}_e$, while the joint distribution across elections is never observed, since no voter votes in more than one. The joint events are the *voting profiles* $\mathbf{j} = (j_e)_{e \in \mathcal{E}}$, $j_e \in \mathcal{J}_e$; the set of bliss points producing profile $\mathbf{j}$ is

$$\mathcal{X}_{\mathbf{j}} = \bigcap_{e \in \mathcal{E}} \mathcal{X}_{j_e}(e).$$

The nonempty sets $\mathcal{X}_{\mathbf{j}}$ partition $\mathcal{R}$ into convex cells. Following Tebaldi, Torgovitsky and Yang (2023), I call this partition the *minimal relevant partition* (MRP) and denote it $\mathbb{X}$.¹⁶ Voters whose bliss points lie in the same cell vote identically in every election; voters whose bliss points lie in different cells differ in at least one. The partition depends only on platforms and is therefore common to all types; what differs across types is the mass placed on each cell. For election e , let $\mathbb{X}_j(e) = \{\mathcal{X} \in \mathbb{X} : \mathcal{X} \subseteq \mathcal{X}_j(e)\}$ denote the cells contained in j ’s attraction region.

Figures 1(a) and 1(b) show the attraction regions in two hypothetical elections, between candidates A and B and between C and D . Figure 1(c) overlays them into the MRP: a voter whose bliss point lies in the light-pink cell, for instance, would vote for B if facing the first election and for C if facing the second. The MRP places no further restriction on the location of a bliss point within its cell. Figure 1(d) refines the partition by the boundary of a region T defining a target parameter, which becomes the union of $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$.

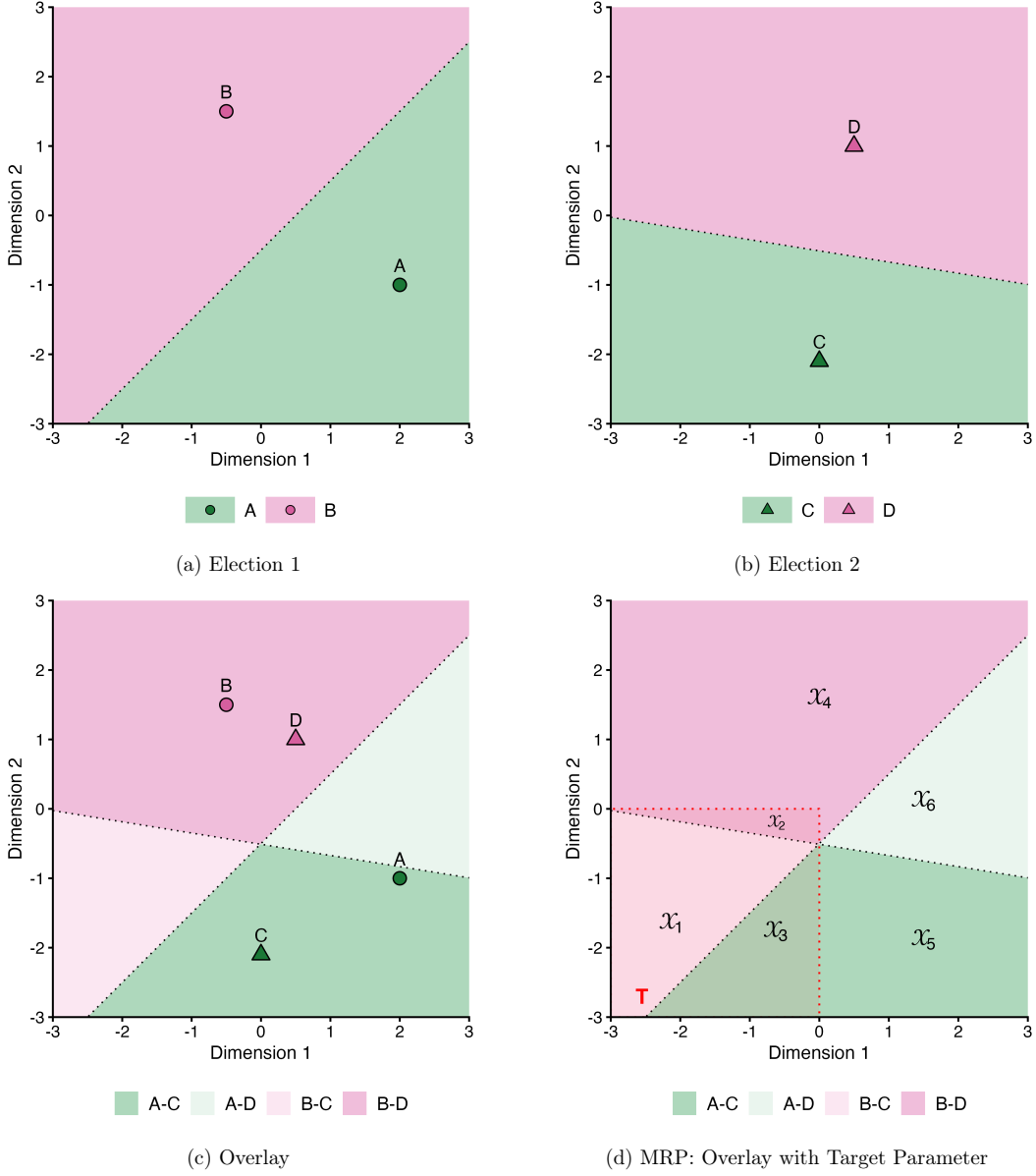
This is the structure of the application. Voters of the same type and state face different candidates in different electoral districts, and the electoral returns give how many voters, across types, voted for each candidate in each district. Demographic composition and candidate platforms provide the identifying variation: within a district, municipalities of different composition face the same candidates, so differences in their returns separate the types’ shares of each candidate; across districts, each type faces different candidates, so its shares in different districts restrict the mass it places on the cells of the overlaid partition.

From densities to cell masses. The MRP matters because the returns depend on f only through the mass it places on each cell: by (3), the share of type- d voters choosing j in election e is the mass of $f(\cdot \mid s, d)$ on the cells in $\mathbb{X}_j(e)$, and where that mass sits within a cell is immaterial. The same holds for the target parameter once the partition is refined by the regions that define it. In Figure 1(d), T is the union of $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$, so

$$\theta(s, d; f) = \int_{\mathcal{X}_1} f(x \mid s, d) dx + \int_{\mathcal{X}_2} f(x \mid s, d) dx + \int_{\mathcal{X}_3} f(x \mid s, d) dx,$$

which depends not on f per se, but only on the mass that f places on the three cells.

¹⁶When the target parameter involves hypothetical elections, such as counterfactual platforms, or regions of the policy space, such as $\underline{\mathcal{X}}$ above, $\mathcal{E}$ is augmented by those pseudo-elections and $\mathbb{X}$ is refined by the boundaries of those regions too.



Notes: The figure illustrates a hypothetical setting with two elections, each between two candidates, (A, B) and (C, D) , in a two-dimensional policy space. In panels (a) and (b), the pink region denotes the attraction area of candidates B and D , and the green region that of candidates A and C , in Elections 1 and 2, respectively. Panel (c) overlays the partitions of panels (a) and (b), yielding the MRP. Panel (d) presents a finer MRP that also includes the region corresponding to a target parameter. $\mathcal{X}_n$ denotes the unique cells that characterize observed behaviour in Elections 1 and 2 or the target parameter.

Figure 1. Example of partitions of the policy space in two elections with two parties ((a), (b)) and their overlay (c) with a target parameter (d).

Attention can therefore shift from densities to *cell masses*, $\phi(\mathcal{X} \mid s, d) = \int_{\mathcal{X}} f(x \mid s, d) dx$: for each type, a finite vector of non-negative numbers summing to one, in place of an infinite-dimensional density. Tebaldi, Torgovitsky and Yang (2023) show that this replacement preserves sharpness. A value t belongs to Θ^* if and only if some vector of cell masses reproduces the observed returns and yields t . For a scalar target parameter, Θ^* is then an interval whose endpoints are the minimum and maximum of the target parameter over the cell masses consistent with the returns; when the target parameter is linear in the masses, as in the example and in all target parameters of the application, both endpoints are solutions of linear programmes. Appendix A demonstrates these results.

Questions about the bliss-point distribution can thus be answered, with sharp bounds, without restricting that distribution to any functional form. The width of the bounds reflects exactly what the electoral returns do not identify: the location of bliss points within cells, and the set of cell-mass vectors compatible with the returns. Conclusions and counterfactuals are thereby grounded in the observed returns rather than in imposed functional forms.

2.4 Identification of the weight

The weight w enters the problem only through the attraction regions, and hence through the MRP: write $\mathbb{X}_w$ and $\mathbb{X}_{j;w}(e)$ for the partition and for the cells of candidate j under weight w . Two features of the cell-mass representation of (3) follow (see (C2) in Appendix C). First, the equations are defined over the cells available in $\mathbb{X}_w$, and these cells vary with w . Second, given the cells, the data are matched by choosing the masses $\phi(\mathcal{X} \mid s, d)$. The former is the source of identification for the weight, the latter for the bliss-point distribution: the weight is identified by *which* cells the data require, the mass function by *how much* mass those cells must carry. Appendix B formalises the identification of w , and Appendix C derives the sharp bounds on target parameters when w is set-identified. The remainder of this subsection gives the intuition.

Imagine a researcher who observes the results of two two-candidate elections (candidates A, B in the first and C, D in the second), and assume for simplicity that exactly the same voters vote in both. The goal is to learn about f and w . The researcher observes only the marginals (vote shares) $\sigma_A, \sigma_B, \sigma_C, \sigma_D$, but these are informative about the joint shares of interest, the masses of the MRP cells $\sigma_{AC}, \sigma_{AD}, \sigma_{BC}, \sigma_{BD}$.

Consider the hypothetical scenario in Table 1, in which candidates A and D win their respective elections with 70% and 90% of the vote. Intuitively, a non-negligible share of voters must vote for both A and D . The observed marginals imply Fréchet bounds on each joint cell, reported in Table 1; in particular, the A – D cell must contain at least 60% of all voters.

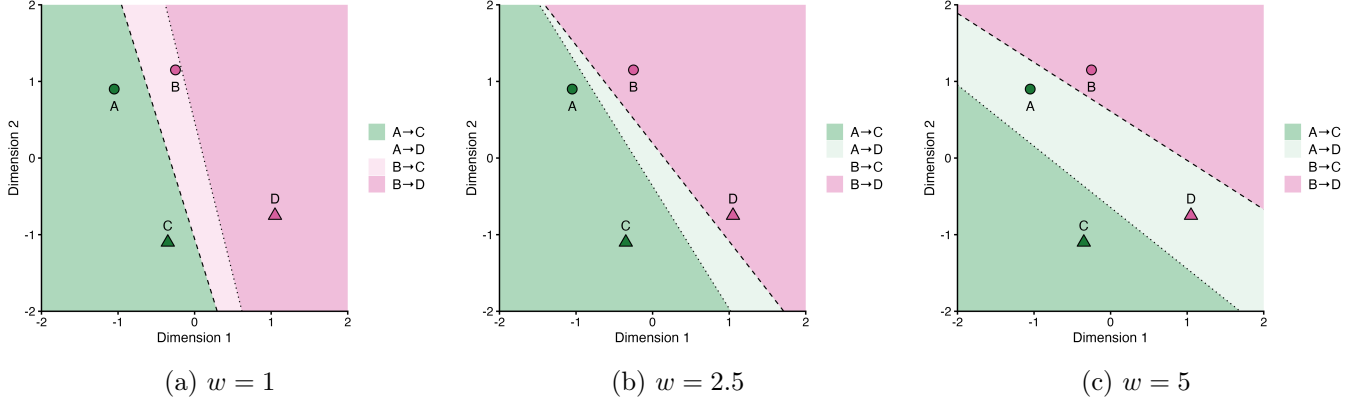
Hence the MRP of these two elections must contain a cell that rationalizes such behaviour, as otherwise the model would mispredict the votes of at least 60% of voters. The claim is that this cell (A – D) need not exist under every value of the weight, and that the weights whose partition does contain it are best suited to explain the observed behaviour.

	C	D	Total
A	$[0, 0.1]$	$[0.6, 0.7]$	0.7
B	$[0, 0.1]$	$[0.2, 0.3]$	0.3
Total	0.1	0.9	1

Notes: The Total row and column report the observed marginals (vote shares) in the two hypothetical elections. Each interior cell reports the Fréchet bounds on the joint share of voters choosing the corresponding pair of candidates, $\max\{0, \sigma_r + \sigma_c - 1\} \leq \sigma_{rc} \leq \min\{\sigma_r, \sigma_c\}$, where σ_r and σ_c denote the row and column marginals. The bold cell shows that at least 60% of voters vote for both A and D .

Table 1. Marginal (observed) shares and Fréchet bounds on the joint cell shares

Figure 2 presents the MRP induced by the two elections under $w \in \{1, 2.5, 5\}$. In this hypothetical example, the MRP with $w = 1$ does not feature an A – D cell and is therefore ill-suited to explain the data. The figure also illustrates the set-identified nature of the weight: $w = 2.5$ and $w = 5$ yield the same partition of the policy space and cannot be told apart.



Notes: The figure illustrates a hypothetical setting with two two-candidate elections, (A, B) and (C, D) . Each panel overlays the two elections in the two-dimensional policy space. The shaded regions denote the attraction areas of the joint outcomes: $A \rightarrow C$ (dark green), $A \rightarrow D$ (light green), $B \rightarrow C$ (light pink), and $B \rightarrow D$ (dark pink), where $X \rightarrow Y$ denotes voters who prefer X in Election 1 and Y in Election 2. The cells present in a given panel constitute the minimal relevant partition for that weight. Across panels, the weight w of the second dimension increases.

Figure 2. Illustration of minimal relevant partitions induced by the same candidate configurations $(A, B), (C, D)$ under different values of w .

The example shows that the weight is identified through the cells available in the MRP (and not, for example, through their size in the policy space), and that it is set- rather than point-identified. How informative the identified interval is remains an empirical question. Lieber, Tebaldi and Torgovitsky (2024) make a similar argument about the set identification of relative weights on characteristics in an additive random-utility model.¹⁷

¹⁷In practice there need not be a unique partition that best explains the data: because the number of feasible cells (over which voter mass is estimated) weakly exceeds the number of observed vote-share marginals, the problem is an under-identified linear system, and several partitions may rationalize the data equally well. However, as Lemma 1 in Appendix B shows, each cell is generated only by weights lying in an interval; identifying power then comes from Fréchet-type necessity constraints, and additional elections with distinct vote-share configurations weakly increase their number. The admissible weights must lie in the

Appendix D reports simulations comparing this approach with that of Merlo and De Paula (2017), who recover the weight jointly with the distribution of bliss points using a sieve minimum-distance estimator. The simulations suggest that the true weight belongs to the set recovered here, whereas the weight recovered jointly with the bliss-point distribution falls outside it. The intuition is that identification of the weight from the cell structure requires no assumption on the distribution of bliss points, so it is not contaminated when that distribution is misspecified; the sieve estimator must approximate the bliss-point density, and any residual misspecification is absorbed into the weight estimate. The appendix also shows that the identified set for the weight shrinks with the number of elections, and that noise in voters’ perceptions of candidate platforms leaves this conclusion unchanged: the identified region still tightens around the true weight, while the sieve estimates tend to remain outside it.

The identification argument does not rely on the weight being common across types within a state, and a similar argument goes through when w varies across types. In practice, however, type-specific weights are considerably more demanding on the data: informative type-level intervals would require either type-level choice data or far richer variation in candidate platforms than is available in the application.

3 Scope of the Approach

This section discusses settings in which the identifying assumption is plausible, highlights the approach’s broad applicability, and considers extensions, including strategic voting.

The approach offers a method to estimate voters’ preferences in a spatial voting framework from aggregate voting data. Identification rests on comparing voters who share the same preference distribution across different choice environments. This requires either that all municipalities of a state draw from the same distribution, $f(x \mid m, s, d) = f(x \mid s, d)$ (Assumption 1a), or that an instrument z shifts the candidates faced without shifting preferences, $f(x \mid z, s, d) = f(x \mid z', s, d)$ (Assumption 1b). The number of choice environments can be as small as one: a single election already partitions the policy space, and with a single environment the condition holds vacuously. Settings with one or several choice environments are pervasive, and closely related identifying assumptions are already in use in political economy. I organise the discussion around three such settings.

First, consider *simultaneous voting*, where the same voters face different choice environments because several elections take place at once. The main advantage of this setting is that Assumption 1a holds by construction: the same voters face every ballot, so the preference distribution is identical across the elections being compared (up to differences in turnout across races). Its main limitation is that the number of simultaneous elections is typically small, which yields a coarse partition of the policy space and is potentially less informative.¹⁸ Among other cases, simultaneous voting is used in the United States, where elections for

intersection of the intervals associated with every cell required to carry positive mass. Since the data were generated at the true weight, this intersection is a non-empty interval containing the truth, and it weakly shrinks as elections accumulate.

¹⁸This limitation does not apply to settings such as California, where voters face many ballot propositions on the same day. Snyder (1996) and Gerber and Lewis (2004) exploit such returns with models in the spirit of NOMINATE (Poole and Rosenthal,

multiple offices often take place on the same day; in Australia, where voters cast ballots for both chambers of parliament simultaneously; in Italy, where general elections fill both the Chamber of Deputies and the Senate on the same day; and in Switzerland, where voters may face multiple electoral ballots as well as referendum questions on the same election day.

Second, *ranked-choice voting* (RCV) provides another setting in which the same voters face varying choice environments. A ranking can be read as a sequence of choices from shrinking choice sets: the ballot contains the same candidates throughout, but a candidate who has already been ranked is no longer available at subsequent stages. Lower-ranked choices can therefore be especially informative about voters’ positions in the policy space. As with simultaneous elections, the identifying condition holds by construction. RCV is used in national elections in Australia, Ireland, Malta, and Sri Lanka, among others; in Alaska for state and federal elections; in Maine for primaries and federal general elections; and for local races in several US cities.

Third, consider *geographic representation*, in which different voters face different choice environments across districts. Such variation arises by design in many electoral systems: first-past-the-post, as in U.S. congressional elections and House of Commons elections in Canada and the United Kingdom; mixed systems combining first-past-the-post and proportional representation, as in Germany, Hungary, Japan, Mexico, and South Korea; and proportional representation with multiple districts, as in elections to Brazil’s Chamber of Deputies.

In this setting, however, the identifying condition no longer holds by construction. Political economy models tie politicians’ positions to voters’ preferences, whether through opportunistic incentives, intrinsic congruence, or voters’ responsiveness to politicians’ cues (Osborne and Slivinski, 1996; Persson and Tabellini, 2002; Callander and Carbajal, 2022). A researcher must therefore either justify the comparison across choice environments or restrict attention to a single one. I now discuss approaches in recent empirical political economy, focusing on work in which candidates’ characteristics include an ideological component.

Longuet-Marx (2026) exploits congressional district borders within states, and compares small groups of contiguous precincts that lie in different electoral districts but are otherwise very similar. In the terminology of this paper, this is Assumption 1b with the border group as the instrument: conditional on the group, observed demographics and time-invariant precinct effects, adjacent precincts share the same preference distribution while facing different candidate sets. The overlay of choice situations must then be done only within groups of precincts, which limits the available identifying variation. By considering many such groups, or by imposing additional restrictions on how preferences may differ across groups of precincts, a researcher may still recover more about the primitives of interest than from single elections alone.

Merlo and De Paula (2017) estimate voters’ ideal-point distribution from aggregate returns in the 1999 European Parliament elections, assuming party positions are independent of ideal points conditional on observed precinct characteristics. Danieli et al. (2026) combine individual survey data with manifesto-

1985), estimating proposition cutpoints jointly with voter preferences.

based party positions across European countries and over time, and estimate the weights voters place on these positions in a probabilistic voting model with party fixed effects. Both rest on the assumption that, conditional on observables, residual preference heterogeneity is unrelated to party positions.

[Iaryczower, Montero and Kim \(2024\)](#) estimate voters’ preferences from aggregate vote shares in elections to Brazil’s Chamber of Deputies. With over one hundred candidates per state-year on average and a single ideological dimension, one choice environment already partitions the policy space finely; variation across environments is used mainly to recover how preferences shift with state demographics. Their instrument addresses a different concern, correlation between positions and unobserved candidate attributes such as charisma: each federal candidate’s position is instrumented with the positions of same-party mayoral and state-level candidates in the most recent local cycle, which co-move with federal positions through the party but are plausibly unrelated to these attributes.

The model can also be extended to other features of voting behaviour. Under strategic voting, some individuals abandon their most preferred candidate if a vote for her is perceived as *wasted*, which generates different effective choice sets across voters. Those who vote for underdog candidates can be treated as sincere and as facing the full choice set; for those who support viable contenders, the choice set should be restricted to viable candidates. This coarsens the partition and may widen the bounds, but makes them robust to strategic voting. The model can also accommodate valence as an additive, candidate-specific term shifting the boundaries of the partition.

I now turn to an application to German federal elections, held under a mixed electoral system. Candidate positions vary across electoral districts within a federal state, while the preference distribution is modeled at the state level: within a state, voters of a given demographic group are assumed to draw their bliss points from the same distribution $f(x \mid s, d)$ in every district. I first argue that this condition is plausible in the German setting, and then gauge the robustness of the results.

4 Application: The Rise of Populism in Germany

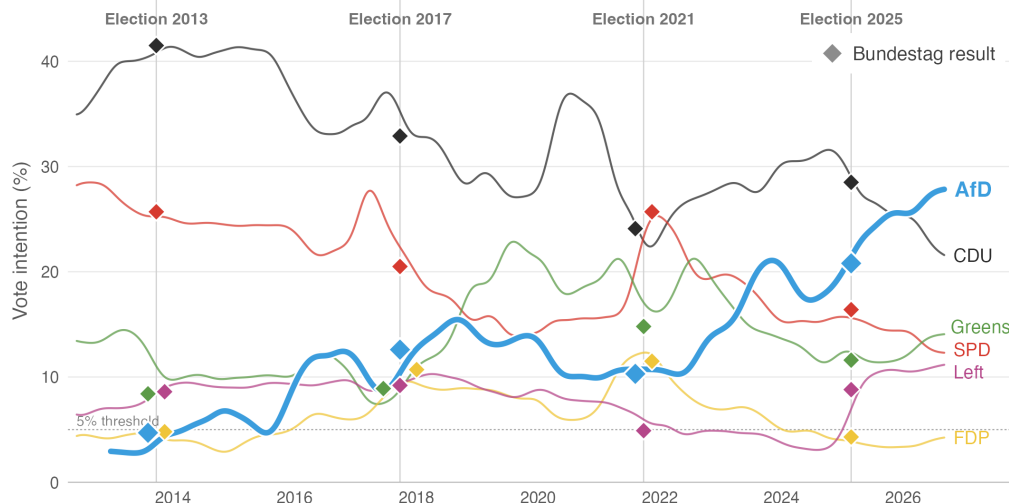
One of the main puzzles in the political economy literature is the rise of populism, and in particular radical-right populism, in Western democracies. The empirical literature has largely focused on identifying the causal mechanisms underlying the rise of populist politics, examining their effects on voters’ preferences, and analyzing the contextual and individual-level factors that shape these preference responses (see [Guriev and Papaioannou \(2022\)](#) for an overview).

While understanding specific mechanisms is essential, it is equally important to develop a broader understanding of the preferences of populist voters, so as to be able to draw policy implications and inform competition strategies for the mainstream parties. One recent attempt to recover populist voters’ preferences is [Danieli et al. \(2026\)](#), who study the rise of populist radical-right parties in Europe using survey evidence over time. To my knowledge, my paper is the first to provide revealed-preference evidence on the ideological

preferences of populist voters at large. Focusing on revealed-preference data rather than surveys is valuable because it sidesteps concerns about survey misreporting, in particular the possible underreporting of populist vote choice, and makes it possible to recover the relative importance of different dimensions of political competition, which can be difficult to identify from survey data alone.¹⁹

4.1 Context

Over recent decades, the German electoral landscape has undergone important changes, most notably the rise of the Alternative for Germany (AfD), a populist radical-right party. For the first time since the 1950s, a party to the right of the Christian Democrats (CDU/CSU) surpassed the 5% threshold and entered the Bundestag. In the most recent federal election, in 2025, the AfD finished second, ahead of several parties of the traditional political establishment, including the Social Democratic Party (SPD), one of Germany’s two major parties. Figure 3 demonstrates the rise of the AfD over 2012–2026 and plots the responses to the polling question “Which party would you vote for if the Bundestag election were held next Sunday?”, alongside the results of the four Bundestag elections used in the paper (2013, 2017, 2021 and 2025).



Notes. Lines show the share of respondents intending to vote for each party in the question “Which party would you vote for if the Bundestag election were held next Sunday?”, smoothed across 3,435 national polls by eight polling institutes (Allensbach, Forsa, Forschungsgruppe Wahlen, GMS, Infratest dimap, INSA, Kantar/Emnid, YouGov), compiled from *wahlrecht.de*. The smoothed value on a given day is a Gaussian-weighted mean of all polls with a standard deviation of 60 days. Diamonds mark the official second-vote shares in the Bundestag elections of 2013, 2017, 2021 and 2025. The dotted line marks the five per cent threshold for parliamentary representation. CDU denotes CDU/CSU; Greens and Left denote Bündnis 90/Die Grünen and Die Linke.

Figure 3. Federal voting intentions, June 2012 – September 2026.

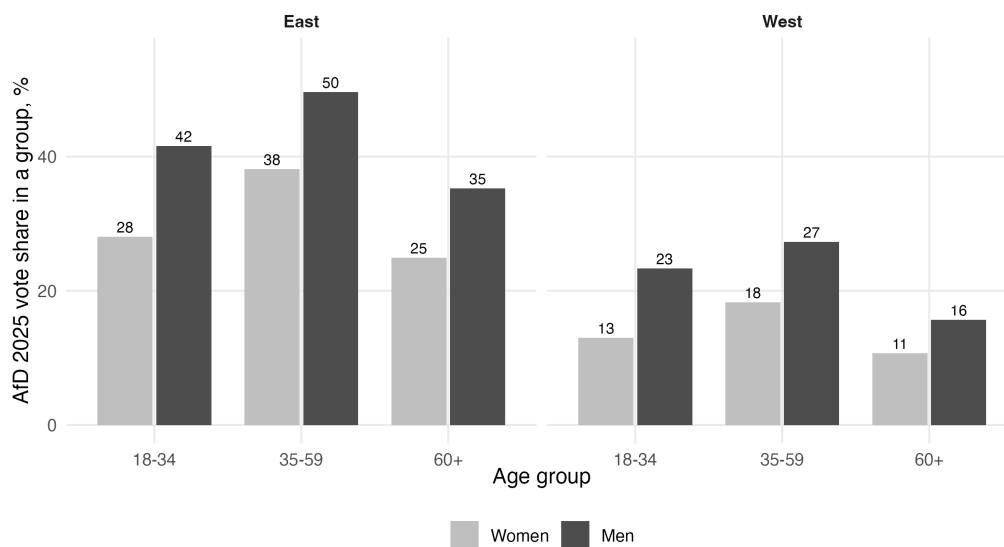
Alternative for Germany was established in 2012 to oppose the policies pursued by the German government to fight the euro crisis. From the beginning, the party was distinctive in its radical-right and anti-establishment stance. In the run-up to the 2013 Bundestag elections, its main concerns were the euro

¹⁹Voters tend to adjust their reported positions towards those of their party, particularly when the issue is not especially important to them. When relative weights are estimated using survey data, such proximity in reported positions may then be read as evidence that the dimension carries greater weight for voters than it in fact does.

crisis and the Greek bailout, as well as adherence to the rule of law and stronger participatory democracy (AfD, 2013). In 2015, the party split; many founding members resigned, and the conservative, anti-immigrant wing took over the leadership, which it has held since.

Throughout the study period (2013–2025), the AfD was consistently ranked as the most right-wing and anti-elite among Germany’s main parties (Jolly et al., 2022), and it stood out as the most conservative and anti-elite party with a realistic chance of winning seats in the Bundestag. This motivates the choice of the two dimensions of policy space in my application – Left–Right and Pro-Elite–Anti-Elite (I refer to the latter simply as *Anti-Elite* throughout).²⁰ I discuss what I mean by those dimensions below in the data section.

The support of the party is heterogeneous across population groups. Figure 4 plots administrative data on the 2025 AfD vote by region (East–West), coarse age group and gender. Support for the party is highly heterogeneous along all three dimensions: the party is much more popular in the East than in the West, among the middle-aged than among younger and older voters, and among men than among women. Figure J1 provides the same statistics for 2013, 2017 and 2021, highlighting the persistence of this heterogeneity.



Notes. Bars show the AfD share of valid second votes cast within each group. Source: *Repräsentative Wahlstatistik of the Federal Returning Officer*, an official statistic in which ballots in a random sample of polling districts are marked by the voter’s gender, birth-year group and state. The six published age groups (18–24, 25–34, 35–44, 45–59, 60–69, 70+) are pooled into three by summing votes for visualisation. East comprises Brandenburg, Mecklenburg-Vorpommern, Saxony, Saxony-Anhalt and Thuringia; West comprises the remaining states, including Berlin.

Figure 4. AfD party vote share by region, age group and gender, 2025.

My analysis draws on data from German federal parliamentary (Bundestag) elections. Germany is divided into 299 electoral districts across 16 states (e.g. Bavaria, Saxony). Under Germany’s mixed electoral system, each voter casts two votes: a first vote for a direct candidate in a single-member district, and a

²⁰This aligns with the literature in political science that highlights that populist parties build their agenda on two main dimensions: ideological and anti-system (see Mudde (2004); Guriev and Papaioannou (2022), among others), and the companion paper Nikiforova (2026), which identifies Left–Right and Anti-Elite as the two central dimensions of voting in Germany in recent years.

second vote for a state-level party list. First votes are likely to be heavily shaped by the strategic incentives of first-past-the-post competition, whereas second votes among viable parties can more plausibly be treated as sincere (more on this below). Although the two votes are distinct, the candidates appearing on district ballots and on state-level party lists largely overlap: around two-thirds of candidates on state lists also run in single-member districts, and the district candidates are on average higher-ranked on the lists (Appendix L.1 provides more detail). In what follows, I rely exclusively on second-vote party shares.

My analysis is based on the assumption that *candidates' positions in single-member districts shape perceived party positions*. In other words, I use estimated candidate positions (see below) as proxies for parties' positions in each district. This allows me to exploit cross-district variation in candidate positions while preserving the assumption of sincere voting for the party list. To support this assumption, I use survey data from the 2021 Bundestag election, which record each respondent's electoral district, and show that, within federal states and within parties, the estimated Left–Right position of a party's district candidate predicts respondents' Left–Right placement of that party (Table J1). The relationship holds for the party a respondent votes for with the second vote and is strongest among AfD voters. It is absent for parties other than the chosen one, and the candidate's position has no bearing on the respondent's own Left–Right self-placement, which speaks against the correlation reflecting local sorting of voters rather than candidate-driven perceptions. The results are robust to controlling for gender, age, and education. Additionally, survey data collected around the 2021 Bundestag election show that voters interact with and know their local candidates well (Figure J3).²¹ Both exercises use the 2021 Bundestag election, for which the richest survey data are available.

Finally, using the estimation procedure described in Section 5, I show that the observed assignment of candidates to districts fits the data better than random reassignments of the same candidates within each state. Specifically, I re-estimate the model with candidate positions randomly reassigned across districts within each state, so that every municipality keeps its own vote shares and demographics but faces another district's candidates. Across 100 such permutations for each region and election year, the observed assignment has the second-lowest fit error of the 101 values, averaged over all region-years ($p = 0.02$).²² Also note that noise in the estimated candidate positions would, if anything, bias the results towards the uninformative composition, in the spirit of attenuation bias in linear regression (see Section 9).

Two additional institutional features are important for the analysis. First, state-level party elites play a substantial role in candidate selection across electoral districts. Although German election law formally assigns nominations in single-member districts to local (district) party chapters, state party organizations

²¹In the political science literature, the influence of single-member district candidates on parties' list-tier outcomes in mixed electoral systems is known as the *contamination effect* (Ferrara, Herron and Nishikawa, 2005). It is documented across mixed systems (Herron and Nishikawa, 2001; Cox and Schoppa, 2002) and, for Germany, by Klingemann and Wessels (2001), Hainmueller and Kern (2008) and Gschwend and Zittel (2015); this literature mostly examines the effects of incumbency, candidate evaluations and campaign effort. My assumption differs in that I focus on policy positions and assume that candidates shape voters' perception of the party's position.

²²Testing each of the eight region-years separately and combining the cell-level p -values with Fisher's method gives $p = 0.04$. The higher value reflects that the observed positions are not the best fit in 2013 and in Eastern Germany in 2017 and 2025.

act as de facto gatekeepers within the state. They can influence nominations both (i) by blocking district candidates they consider unsuitable and (ii) by determining the state party list for Bundestag elections. The latter matters because candidates may enter the Bundestag either through district races or through state lists; in practice, around two-thirds of candidates on state lists also run in single-member districts. Candidate selection is moreover often uncontested, as elites tend to preselect candidates in advance—particularly when incumbents are involved (Detterbeck, 2016; Schüttemeyer et al., 2024). These considerations both motivate the choice of states as separate units for preference estimation and inform the identification argument. Appendix L.1 gives further descriptive evidence on the parties’ nomination practices, documenting persistence in nominations and the interplay between single-member districts and state lists.

Second, electoral districts are drawn primarily on the basis of population size. Unlike in the United States, redistricting in Germany is nonpartisan and district boundaries are largely stable. A permanent Constituency Commission reviews population changes each electoral cycle and recommends adjustments to the Bundestag. Constituencies should remain within 15% of the average population, must be redrawn if deviations exceed 25%, and must lie entirely within a single state. In practice, only a small share of constituencies changes in any given electoral period.²³

4.2 Data

In the application that follows, states ($N = 16$) are subdivided into electoral districts ($N=299$); and each electoral district contains multiple municipalities ($N \approx 11,000$). Voting and demographic data are observed at the municipality level. Each electoral district features a distinct set of candidates. Figure J2 illustrates this division. I now describe the data used for estimation.

The demographic group taken as the unit of analysis consists of an age–gender–education bracket in a given state, and the distribution of preferences is estimated separately for each group within each state in a given year. An example of a group is *female, aged 18–24, with higher education, in Saxony in 2021*. To proceed with the estimation I need data on age–gender–education composition and voting at the same level. I now turn to describing such data.

Municipal-level data. I collect data on turnout and second vote shares (party-list vote shares) for the six main parties at the level of municipalities^{24,25}: *AfD*, *CDU/CSU*, *FDP*, *Greens*, *Left*, *SPD*. I abstract from coalition considerations and treat those parties as *viable* in the years studied, which implies that if an individual truly prefers one of the considered parties, then there are no strategic considerations not to vote for this party.²⁶ In turn, this implies that my estimates elicit true voting preferences among the voting population.

²³See, for example, Die Bundeswahlleiterin (n.d.) for changes made prior to the 2025 Bundestag elections.

²⁴The cumulative party vote shares for the studied viable parties are 94%, 95%, 91% and 91% in 2013, 2017, 2021 and 2025 respectively. This implies that the analysis excludes only a small group of voters who supported non-viable parties.

²⁵The municipality is the smallest unit of analysis in Germany that can be geo-coded.

²⁶It also assumes that voters expected these parties to clear the 5% threshold; the exceptions across years and parties are FDP/AfD (2013), Left (2021) at 4.7–4.9%, and FDP (2025) with 4.3%.

I also collect demographic data at a municipal level for each electoral year. I construct municipality-level information on the population composition of eligible voters in 24 age–gender–education brackets, comprising 6 age groups (18–24, 25–34, 35–44, 45–59, 60–69, 70+), two genders, and a binary distinction between individuals with and without higher education.²⁷ These demographics are measured precisely for 2021, drawing on the 2022 Census. To construct demographics for 2013, 2017 and 2025, I use changes in the size of each age–gender group at the district level ($N \approx 400$) and project these changes onto municipalities within each district, using the 2021 municipal composition as the baseline.

I restrict the benchmark sample to municipalities with fewer than 10,000 inhabitants.²⁸ Smaller units retain more distinctive demographic compositions, which increases the variation available for estimating group-specific preferences. I test the sensitivity of my results to this threshold value below. Table J3 reports how voting patterns differ across municipalities below and above 10k. Small municipalities lean more strongly towards the AfD (+2.1pp) and the CDU/CSU (+4.4pp) at the expense of the other parties, but this partly reflects state and district composition effects: once those are accounted for, the gaps fall to +0.7 and +3.6pp.

Candidate positions. A key feature of the considered setting is that voters face distinct party-affiliated candidates across single-member districts. To measure candidates’ ideological positions, I combine two candidate surveys:

- 1) the GLES Candidate Study for the Bundestag elections 2013–2025, administered by the German Society for Electoral Research and the Leibniz Institute for the Social Sciences (GLES, 2023);
- 2) Wen Wählen (*Whom to Vote For?*), a nonpartisan platform that fields a standardized questionnaire to election candidates.

Both surveys elicit candidates’ own positions on a battery of policy issues, and both target candidates from parties with a high likelihood of clearing the 5% threshold. Consequently, candidates from all parties considered in my analysis were contacted for participation in each election year.

First, using the GLES Candidate Study, I construct a left–right score and an anti-elite score for each candidate from multiple survey items corresponding to each dimension (see Table L2 for the exact wording).²⁹ I treat the left–right dimension as a bundle of economic, cultural and social policies, and construct it from questions spanning economic issues (e.g., whether the government should reduce income differences, or stay out of the economy), cultural issues (e.g., whether same-sex marriage should be banned, or whether women should be free to decide about abortion), and immigration (e.g., whether immigrants should adapt to German culture, or whether immigration benefits the German economy). In line with the literature, I treat the

²⁷To do so, I use data on the age–gender composition of the population at the municipality level and complement them with data on the age–gender–education composition, which are available only for larger municipalities (with more than 10,000 inhabitants), municipal associations (*Gemeindeverbände*), or higher-level administrative units (*Kreise*).

²⁸This restriction effectively captures almost exclusively rural municipalities, making a separate urban–rural distinction unnecessary in this context. It also naturally excludes the city-states (Berlin, Bremen, and Hamburg) from the analysis.

²⁹I use the terms *anti-establishment* and *anti-elite* interchangeably.

anti-elite dimension as the axis capturing antagonism toward elites and the claim that the people (rather than professional politicians or technocrats) should hold authority, as well as general distrust in democratic institutions (Mudde, 2004; Guriev and Papaioannou, 2022).³⁰ To measure a candidate’s position on this dimension, I use items such as whether democracy offers citizens enough opportunity to participate, whether there should be nationwide referendums, and whether citizens have lost trust in politicians. See Table L2 for the list of questions used from the GLES Candidate Study. Most questions recur across election years, although some are unavailable in particular years. I standardize each item using its pooled mean and standard deviation across the years in which it appears. For each candidate, I then average the available standardized items within each dimension to obtain a single index.³¹

Participation in the candidate survey is voluntary, and the response rate among single-member district candidates remains moderate – about 35% across the years. To improve data coverage, I supplement the GLES Candidate Study with a second source, Wen Wählen (Whom to Vote For?), a nonpartisan platform that collects standardized policy questionnaires from election candidates across all major parties. Because the two surveys use distinct question sets, I bridge them using an elastic net regression (Appendix L.2 provides details on this). The 10-fold cross-validated correlation between the predicted and the GLES-based positions is about 0.87 for the Left-Right dimension and 0.73 for the Anti-Elite dimension.

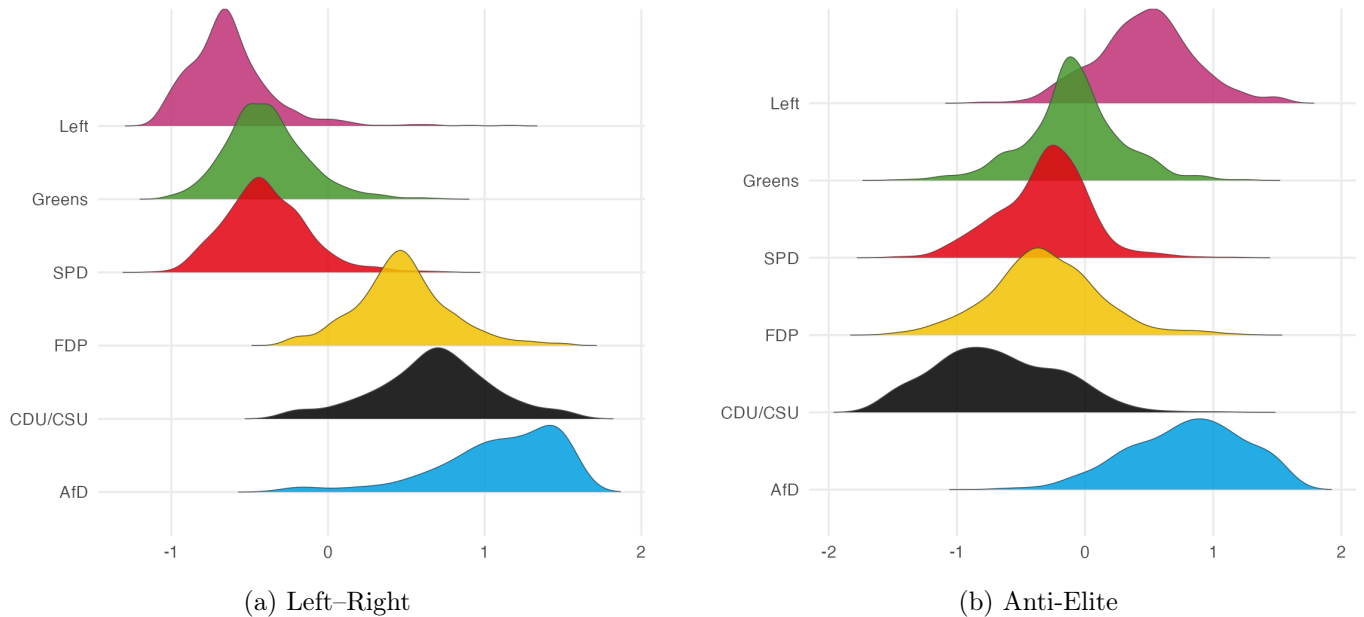
Figure 5 plots the distribution of candidate positions by party on the Left-Right and Anti-Elite dimensions, pooled across all four elections (2013–2025). Within each party candidate positions span a broad range on both dimensions, while the overall ordering of parties is preserved: the Left is furthest left and the AfD furthest right, with the Greens, SPD, FDP and CDU/CSU in between. On the Anti-Elite dimension, the Left and the AfD stand out as clearly more anti-elite, while the remaining parties sit on the pro-elite side. Figure L1 disaggregates these distributions by election year; Figure L2 plots the median values of candidate positions by party, dimension, and year.³²

Combining the two surveys yields comparable positions for about 60% of candidates across the years. Response rates differ by party: the CDU is lowest and the Greens highest (37% and 77% respectively), with about 50% of AfD candidates covered. Importantly, respondents and non-respondents obtain comparable first-vote shares and have similar winning probabilities within each party. Missing positions are imputed with the median position of the same party’s candidates in other districts of the same state; the national median

³⁰An important part of the AfD’s rhetoric is nationalism. While it could be attributed to either dimension, I treat it as belonging to the Left-Right dimension. As is clear from the imputation coefficients reported in Appendix L.2, the item most closely related to economic nationalism (“Reintroduce the Deutsche Mark!”) is a strong predictor of a candidate’s Left-Right placement in 2013.

³¹I prefer this method for its transparency. The correlation with indices based on Item Response Theory exceeds 0.85 across all years and dimensions.

³²External validation of such measures is difficult; one available benchmark is the within-party evolution of positions over time in the Chapel Hill Expert Survey (CHES) (Bakker et al., 2015). The within-party dynamics implied by my estimates are broadly consistent with CHES. In particular, the rightward and anti-elite shift of the CDU between 2021 and 2025, and the moderation of the AfD’s anti-elite positioning over the same period, are both mirrored in the change between the 2019 and 2024 CHES waves. Similarly, the hardening of AfD positions between 2017 and 2021 is consistent with the change between the 2014 and 2019 waves. Note that the CHES waves (2014, 2019, 2024) do not coincide with the German federal elections (2013, 2017, 2021, 2025), so the correspondence is approximate.



Notes. Kernel density estimates of the positions of single-member district candidates, by party, pooling the 2013, 2017, 2021 and 2025 elections and all federal states. Positions are estimated from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Each density uses a Gaussian kernel with the bandwidth selected by Scott's rule, computed separately for each party and dimension.

Figure 5. Distribution of candidate positions by party, pooled across elections

is used only when no state-level information is available, which occurs for fewer than 2% of the candidates retained. For the estimation that follows I keep only districts with at most three imputed candidates, which leaves about 70% of districts across years, in which roughly 30% of candidate positions are imputed with state-level (not national-level) fallbacks.

An important concern is that these two dimensions may be too few. While the framework in principle allows for more dimensions, the variation in candidate positions available here is too limited to yield informative bounds beyond two dimensions. Moreover, many candidate dimensions—nationalism, globalism, immigration—are highly correlated with the two dimensions considered, which makes them harder to use for preference estimation. I document the correlations between expert-estimated party positions across relevant policy dimensions in Europe (Figure L3); in my data the Left-Right and Anti-Elite dimensions are essentially uncorrelated, whereas both correlate considerably with a measure of European Union attitudes (Figure L4).³³ I discuss these issues, and how they could affect my results, in more detail in Section 9.

Other data sources. I also use administrative data on turnout by demographic group and state from the Representative Electoral Statistics, which reports turnout rates by age and gender for each state. Unlike survey-based turnout measures, these rates come from the ballots themselves: in a representative sample of polling districts the ballot papers are marked by age group and gender of the voter, so turnout by group is

³³I rely on CHES for the party positions here, because the candidate survey I use to measure candidates' positions lacks items on some important dimensions: it contains no questions on nationalism and few on the environment.

observed rather than self-reported.³⁴ The statistic does not record education, so within an age–gender group I apply the same turnout rate to voters with and without higher education. These data let me construct, for every municipality, the demographic composition of the voters who actually turned out rather than of the eligible population (see footnote 66 for details on the procedure). This adjustment matters because it allows me to abstract from the turnout decision, and the estimated preferences are those of the population that voted.

For the robustness estimation I also use survey reported data on second-best party choices aggregated by years and states as additional target moments. Figure J5 plots the evolution over time of the second-best party choices of those who indicate the AfD as their first choice, at the national level.

5 Estimation & Identification

5.1 Estimation and Model Fit

Appendix G describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how (i) the weights and (ii) the bounds on the target parameters are computed. In brief, I estimate the masses $\phi_{s,d}$ that each state s – demographic group d pair places on the cells of the policy space. I first compute the best attainable fit, $\hat{Q}$, between predicted and observed municipal AfD shares. The bounds on a target parameter θ are then its minimum and maximum over all ϕ whose total error exceeds this optimum by no more than ϵ_{abs} , as in Tebaldi, Torgovitsky and Yang (2023). Following their practice, I keep ϵ_{abs} small (10^{-5}) and verify that the bounds change little when it is raised by an order of magnitude.

When computing the target parameters, I construct a single linear program for each part of the country (East/West) in a given year, loosely coupling the preference distributions across states over coarse areas of the policy space and aggregated demographic groups. Combining information across states, even this loosely, enriches the estimation problem and allows me to explore heterogeneity both within and across them. See Appendix G for details.

Appendix G.4 discusses the fit of the model. Across municipalities, the correlation between predicted and observed AfD vote shares exceeds 0.9 for 2017–2025 and is 0.59 in 2013 (Figures G1 and G3). The model also reproduces how AfD support varies with local demographics: the correlations of predicted and observed municipal AfD shares with, for instance, the share of young or university-educated residents are close (Figure G2). This suggests that the model fits the data well and can be used for the preference estimation that follows.

I also show that accounting for demographic heterogeneity improves out-of-sample prediction: holding out 10% of municipalities within each district, the 24-group model outperforms restricted specifications that merge groups by education, age, or gender. The loss from merging education or age groups is statistically

³⁴See [Link](#) for more info.

significant ($p < 0.01$ in both cases), while the effect of merging gender groups is smaller but still significant at 5% level ($p = 0.04$).

5.2 Identification

I exploit variation in candidate positions across electoral districts within a state and year to estimate the preferences of AfD voters. Recall that x is a voter’s ideal point, m the municipality, which determines the voter’s district and hence the candidates on the ballot, s the state and d the demographic cell. The identifying assumption (Assumption 1a) is

$$f(x \mid m, d) = f(x \mid s, d) \quad \text{for every municipality } m \text{ of state } s,$$

that is, conditional on state and demographic cell, the distribution of ideal points is the same no matter which candidates the district fields. I first discuss why this assumption is plausible, and then how I assess the robustness of the results to it.

The assumption rules out two things: first, that voters sort into districts according to the available candidates; second, that parties place candidates in response to local preferences that differ across districts within a state beyond what demographics capture.

First, politically motivated relocation across districts within states is implausible in this setting. Under mixed-member proportional representation there is little scope for district-specific representation; the policy-relevant subnational tier is the state, not the district. Within-state mobility is also low and rarely political. In 2019, roughly 2% of the population moved between Kreise within the same state, mostly between cities and their surrounding areas, with comparable magnitudes throughout 1991–2024 ([Deutscher Bundestag, Wissenschaftliche Dienste, 2022](#); [Statistisches Bundesamt, 2025](#)).³⁵ Reported motives are family- and dwelling-related in over 70% of moves and job- or education-related in fewer than one in five; the SOEP records no political motive ([Hofmann and Rüger, 2024](#)).

Second, district candidacies are embedded in a state-level process of selecting candidates for the Bundestag, which limits local discretion over who runs. State party executives must co-sign district nominations and may object to a local nomination, forcing a repeat vote (§§ 20(2), 21(4) Bundeswahlgesetz), and nominations are shaped by state-list considerations, since seats are allocated through party-list votes. Candidate choice is further constrained by the relatively small pool of potential candidates (see, e.g., [Höhne, 2023](#)). District candidates are therefore selected with parliamentary representation in mind rather than tailored to the preferences of specific local electorates. Appendix L.1 documents nomination practices in more detail.

I also restrict the sample to municipalities with fewer than 10,000 inhabitants. Because the marginal electoral return to targeting a locality is low when its electorate is small, candidates have less incentive to adapt their platforms to these voters, so observed vote shares in small municipalities are less likely to reflect

³⁵Kreise are the administrative tier between municipalities and states; there are about 400 of them.

strategic positioning than those in larger, more contested urban centers.

I present two pieces of observational evidence that strengthen the assumption of little candidate response to local preferences. First, across the municipalities used in the estimation, the estimated candidate positions of a given party are largely uncorrelated with local socio-economic characteristics within a state.³⁶ Note, however, that at the district level positions do correlate with a district’s socio-economic characteristics within party and state, and with the expected signs. For example, in districts with higher unemployment or larger shares of benefit recipients, candidates are more left-leaning. Figure J6 reports the balance of socio-economic and demographic characteristics at the municipal level, and Figure J7 at the district level. Second, Table J2 demonstrates that candidates’ positions are uncorrelated with individuals’ self-reported views in surveys.

Robustness. I assess the robustness of the identification assumption in four ways, each targeting a different reason why candidate positions might be correlated with local preferences. I report the corresponding results alongside the main results below.

First, I restrict the analysis to municipalities with fewer than 5,000 inhabitants. Parties are likely to target more populous, and hence more electorally relevant, constituencies; small municipalities therefore face candidates selected with larger constituencies in mind, so their candidates’ positions are less likely to reflect local preferences.

Second, I exclude districts in which the AfD candidate was competitive. Candidates’ incentives to cater to local voters increase with their chances of winning, so positions in non-competitive districts are less likely to be tailored to the constituency. Specifically, I drop every district in which the AfD candidate trailed the winner by no more than 5 (10) percentage points in the East (West). The thresholds differ because the party is far more popular in the East. In the East, the exercise covers only 2017 and 2021: in 2013 no AfD candidate was competitive, and in 2025 almost all were, as the AfD won nearly every district. In the West, it applies only to 2025, since in earlier years no AfD candidate was competitive and the restricted sample coincides with the benchmark.

Third, I relax the assumption of preference similarity across districts within a state, following the logic of imperfect instrument in Tebaldi, Torgovitsky and Yang (2023). I classify a district as *conservative* if its AfD candidate lies above the within-state median AfD position on both the Left–Right and the Anti-Elite dimension, and as *moderate* otherwise. I then allow the mass of voters of a given demographic group in each partition cell to differ between conservative and moderate districts of the same state by at most a factor of $1 + \kappa$, with $\kappa = 2$, so that masses may differ by up to a factor of three in either direction. If candidates were placed in response to local preferences, the conservative and moderate districts would differ systematically, and this exercise accommodates such differences.

Finally, I restrict the analysis to municipalities on the boundaries between electoral districts within a

³⁶Correlations remain significant for three variables after controlling for state-by-year and party-by-year fixed effects. Two of these are demographic characteristics, which I therefore account for in the estimation.

state, and replace the assumption that preferences are identical across the whole state with a local one. I group municipalities by border segment, defined as the boundary between two adjacent districts, and require the preference distribution of a given demographic group to be identical only among municipalities on either side of the same segment (the median number of municipalities per segment is seven). Across segments within a state, preference distributions may deviate from the state mean by up to 20% in either direction. Neighbouring municipalities on either side of a border are similar but face different candidates, so district assignment acts as an instrument: it shifts the candidates a voter faces (*relevance*) but, within a segment, is assumed independent of preferences (*exogeneity*). The identifying variation therefore comes from comparisons in which preferences are most plausibly similar.

In the analysis that follows I leverage the variation in estimated preferences not only within states, but also across states. This is possible because candidate positions are constructed from survey items harmonised and standardized across elections, so the estimated bliss-point distributions lie in a common policy space and can be compared and aggregated across states, and compared across elections. Target parameters can therefore be defined relative to national benchmarks, such as the national median AfD position, and traced over 2013–2025 (Section 6 onwards). None of these comparisons requires candidate positions to be exogenous across states: bliss-point distributions remain state-specific, and states may occupy different regions of the policy space.

6 Results: Voters’ Preferences

The main goal of this section is to provide new insights into the rise of populism using the revealed-preference evidence from the approach developed above. To do so, I first estimate voters’ preferences: the weight on the Anti-Elite dimension relative to the Left–Right dimension, w_{AE} , and the distribution of ideal points in a policy space defined by Left–Right and Anti-Elite dimensions separately for demographic groups d in states s , $f(x \mid s, d)$. I next use those estimates for the counterfactual exercises.

I present results separately for eastern and western Germany. The distinction is motivated by different histories and present-day institutional environments, as well as by very different levels of AfD support: as Figure J9 shows, in 2025 the AfD led the party-list vote in every eastern state and in no western one.³⁷ Results for the two regions of the country, analysed separately, may also inform the study of populism across countries with different institutional environments, such as eastern and western Europe.

6.1 Relative importance of dimensions

I begin by describing how the relative importance of the Left–Right and Anti-Elite dimensions has evolved over time. The weights come first because they determine the shape of the partition of the policy

³⁷At the time of writing, the AfD leads national polls at 30%, with a margin of about 10 percentage points over the second-placed party. Within-country heterogeneity is striking: in the September 2026 state election in Saxony-Anhalt (east), the AfD won with 44% of the vote, whereas in the March 2026 elections in Rhineland-Palatinate and Baden-Württemberg (west) it obtained less than 20%.

space used to evaluate the target parameters below.

Only relative weights are identified, so I normalize $w_{LR} = 1$ and search for the best-fitting w_{AE} . Since precise estimation of the weights requires rich data (Appendix D), I impose a common w_{AE} on all groups within a given year and region (East or West), while allowing each demographic group d in each state s its own bliss-point distribution. I search over a grid of relative weights $w_{AE} \in \{0.2, 0.3, \dots, 0.9, 1\}$ and their inverses.³⁸ The best-fitting weight is the one that yields the smallest model-fit error.

How well the model can discriminate between weights depends on how candidate positions co-locate along the Left–Right and Anti-Elite dimensions.³⁹ To ensure that differences in fit across weights reflect the composition of the cells rather than their number, I fix the number of cells within the AfD attraction region at 500.⁴⁰ Moreover, the correlation between candidates’ Left–Right and Anti-Elite positions, and their dispersion, differ little between Eastern and Western Germany and across years, so the regional differences reported below are unlikely to reflect party positioning and more plausibly reflect demand-side differences in the importance voters attach to each dimension.⁴¹

Figure 6 presents the results for the optimal weights. Points denote the point estimates corresponding to the best-fitting weight. The weights are set-identified, since a range of weights implies the same partition of voters and hence the same predicted vote shares; because the data are noisy, however, the optimization always yields a single best-fitting estimate. To reflect set identification, I also report the intervals of weights whose objective-function error lies within 1% of the minimum (Figure K1), as well as confidence sets calibrated from subsample re-estimations, which replace the fixed tolerance with one derived from the sampling variability of the fit (Figure K7).

The results suggest that the Anti-Elite dimension carried more weight in Eastern Germany than in Western Germany throughout the period. In Western Germany, the Left–Right dimension remains consistently the more important of the two. In Eastern Germany, the Anti-Elite dimension carried a larger weight than the Left–Right dimension in 2013 (although this estimate is imprecise once a 1% difference in the objective-function error is allowed for, see Figure K1) and in 2025. In 2017 and 2021, the Left–Right dimension is relatively more important, though the weight on the Anti-Elite dimension remains sizable.⁴²

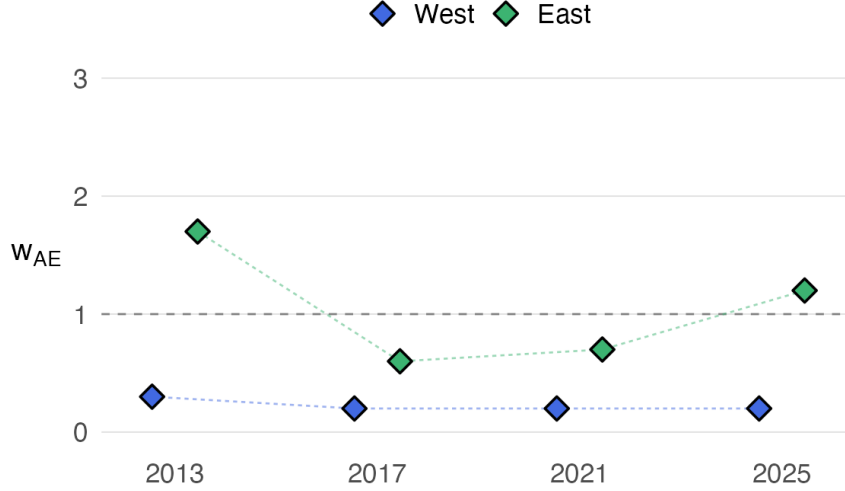
³⁸I do not extend the grid further, for two reasons. First, more extreme weights would amplify measurement error in the estimated candidate positions. Second, the weights are identified only insofar as they meaningfully change the partition of the policy space, which for each pair of candidates depends on the weight and on the ratio of their position differences, $|y_j^{LR} - y_k^{LR}|/|y_j^{AE} - y_k^{AE}|$. More than 80% of the values of this ratio across pairwise comparisons involving AfD candidates lie within $[0.2, 5]$. Whenever the best-fitting weight lies on the boundary of the grid, I extend the grid by one step: in the West over 2017–2025, adding 0.1 confirms 0.2 as the best-fitting weight in 2021 and 2025, but not in 2017.

³⁹This is analogous to a standard result in linear regression: for a fixed regressor X_1 , the improvement in R^2 from adding a second regressor X_2 is smaller the more strongly the two are correlated.

⁴⁰Specifically, I retain the 500 largest cells and merge each remaining cell into the neighbouring cell with which it shares the longest boundary. Results with 1,000 cells are very similar.

⁴¹See Figure J8 for correlations within parties (Figure J8a) and across parties (Figure J8b), and Table J5 for standard deviations.

⁴²Note that here and in what follows the evolution of preferences of AfD voters is a combination of at least two factors – (i) genuine within-person evolution, and (ii) compositional change.



Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See Figure K2 for full results over the grid. Points denote the point estimates corresponding to the best-fitting weight.

Figure 6. Weights for the Anti-Elite dimension w_{AE} relative to Left-Right $w_{LR} = 1$ across years and regions (East/West)

These results are in line with the differential role played by both dimensions across the two parts of the country. A broad literature documents the saliency with which Eastern Germans perceive themselves as neglected by federal elites, a perception grounded in measurable underrepresentation: as of 2024, Eastern Germans held only 12.1% of positional elite roles across twelve societal sectors, far below their population share of roughly 20% (Bundesministerium für Wirtschaft und Energie, 2025). The Anti-Elite dimension is much more muted in the West, where this region-wide sense of elite underrepresentation is far less pronounced.

Figure K2 plots the objective-function value for different values of the weight on the second dimension.⁴³ Figures K4 and K5 demonstrate the consistency of the estimates across subsamples by plotting the objective-function value on different subsamples of electoral districts across years and parts of the country. I interpret these results as evidence supporting the validity of the exercise.

In the following analysis I use the values for the estimated weights to define the partition of the policy space across the two regions unless stated otherwise.

6.2 Distribution of ideal points

The next preference object is the distribution of voters' ideal points $f(x|s, d)$. To provide evidence on the evolution of this object, I focus in this subsection on the location of AfD voters in the policy space.

⁴³In the West in 2017–2025 the solution 0.2 is the corner value of the grid. Figure K3 shows that when the grid is extended below 0.2 (adding 0.1), a weight of 0.2 remains the best fitting weight in 2021 and 2025, but not in 2017. For the reasons discussed above, I nonetheless retain 0.2 as the best-fitting weight for all of these years.

6.2.1 Voters' ideal points in fixed areas of policy space

Where are voters' ideal points located in the policy space? I examine this question for AfD voters in 2013–2025. I partition each policy dimension into intervals of width 0.5 and ask what share of baseline AfD voters has an ideal point in each interval.

For a given election year and region, let $\bar{Y}_j$ denote the coordinatewise median position of party j 's candidates in that year, and let w be the weight used for that region and year. The AfD's baseline attraction region is $\mathcal{X}_{\text{AfD}} := \{x \in \mathcal{R} : d_w(x, \bar{Y}_{\text{AfD}}) \leq d_w(x, \bar{Y}_j) \text{ for every } j \in \mathcal{J} \setminus \{\text{AfD}\}\}$.

Let $f(x | s, d)$ denote the density of ideal points among voters who turned out in state s and demographic group d , and let $F_{S,D}$ denote the joint distribution of states and demographic groups among those voters, constructed using the turnout-adjusted demographic shares.⁴⁴ For example, the share of baseline AfD voters whose ideal points lie in $[0.5, 1]$ on the Anti-Elite dimension is

$$\mathcal{X}_{\text{AE}, [0.5, 1]} := \mathcal{X}_{\text{AfD}} \cap \{x \in \mathcal{R} : 0.5 \leq x_{\text{AE}} \leq 1\}, \quad \theta_{\text{AE}, [0.5, 1]} = \frac{\iint_{\mathcal{X}_{\text{AE}, [0.5, 1]}} f(x | s, d) dx dF_{S,D}(s, d)}{\iint_{\mathcal{X}_{\text{AfD}}} f(x | s, d) dx dF_{S,D}(s, d)}.$$

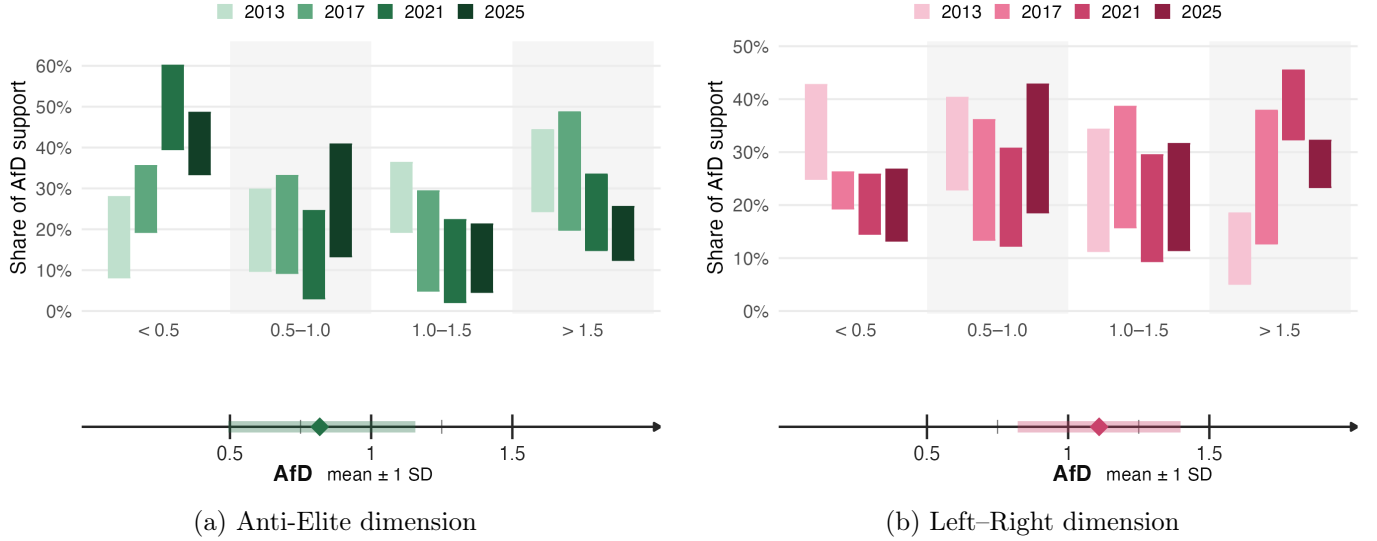
Because $f(x | s, d)$ is only partially identified from the municipal vote shares and demographic composition, I report bounds on $\theta_{\text{AE}, [0.5 - 1]}$ and other target parameters; note that the bounds reported here and below are sharp (see [Tebaldi, Torgovitsky and Yang \(2023\)](#) for details).⁴⁵ Figure 7 plots the results separately for the Anti-Elite and Left-Right dimensions for different dimension bins.

The results suggest a recomposition of AfD support between 2013 and 2025. In 2013, AfD support came predominantly from voters with strongly anti-elite views (up to 42% of AfD supporters had a bliss point above 1.5 on the anti-elite dimension) rather than from the radical right (at least one fourth of AfD supporters had a bliss point below 0.5 on the Left-Right dimension, and at most 19% lay beyond 1.5). Over the following decade this pattern reversed. The share of strongly anti-elite supporters fell to about one fifth by 2025, while the share with radical-right positions rose sharply. Notably, the party's own position moved far less than its electorate: the national AfD median remained between 0.96 and 1.30 on the Left-Right dimension and between 0.67 and 0.97 on the anti-elite dimension throughout—placing the party well below its anti-elite fringe in 2013, and to the left of much of its right support from 2021 onward. The AfD's electorate, in short, transformed around a comparatively stable platform.

A large literature in political economy relies on stated vote choices and stated issue preferences to

⁴⁴Section 4.2 provides more details on administrative data used for turnout adjustment separately for age-gender groups, states and years.

⁴⁵Also note that the fixed masks are contingent, so the lower bounds are the most informative: they give the smallest possible share of voters located in the region (since the estimated objects are the *sharp* bounds on the target parameters). The upper bounds, by contrast, may reflect both the degree to which the problem is underidentified and the mass in the cells shared by contiguous fixed masks.



Notes: Each panel plots sharp bounds on the share of AfD voters whose bliss points lie in each 0.5-wide interval of one policy dimension (the other dimension is left unrestricted): panel (a) the Anti-Elite dimension, panel (b) the Left-Right dimension. Within each interval, bars run from 2013 (lightest) to 2025 (darkest). The line below each panel places the intervals on the dimension; the diamond marks the mean position of AfD district candidates pooled over 2013–2025 and the band one standard deviation around it. Bounds are estimated without the second-choice moments, separately for East and West Germany, and aggregated; adjacent intervals share mesh cells, so within each region and year the bounds are rescaled so that the interval midpoints sum to one.

Figure 7. Distribution of AfD supporters' bliss points along each dimension, 2013–2025

estimate voters' preferences and their relationship to electoral behavior (Le Pennec and Pons, 2019; Gethin, Martínez-Toledano and Piketty, 2022), including in the context of support for populist politicians (Guiso et al., 2017; Algan et al., 2017; Danieli et al., 2026; Bartels, 2023). A natural question, then, is how the preferences inferred from survey data differ from those recovered using the revealed-preference approach developed in this paper. Appendix E.1 presents this analysis. The patterns are broadly similar: AfD voters are more often to the left on the Left-Right dimension in earlier years, and more radical-right in later years. The revealed-preference estimates, however, place more voters at extreme positions, consistent with underreporting of extreme views in surveys. The survey's Anti-Elite measure is less informative, but shows a similar trend towards less anti-elite attitudes over 2017–2025. Exact shares and trends are difficult to compare, given differences in measurement and potential misreporting of both vote choices and stated preferences.

Figure K8 reports the sampling uncertainty behind the bins: I resample municipalities within districts (50 replications), re-estimate the bounds and construct 95% confidence intervals around the identified sets.

These patterns constitute a first piece of evidence on the distribution of AfD voters' bliss points and its evolution, pointing to a changing composition of the party's electorate. While instructive and intuitive, this evidence does not speak directly to voters' motivations for choosing the AfD, nor to how establishment parties might win them back. I now turn to the exercises that resolve this.

6.2.2 Voters' ideal points and their motivations for AfD support

The AfD combines a radical-right stance with an anti-elite appeal, so a ballot cast for the party does not reveal which of the two motivated it. Recovering that motivation matters because it speaks to the policy demands the vote expresses and could help policymakers craft more appropriate responses. I address this ambiguity with a counterfactual: suppose the AfD split into two parties, one retaining its radical-right position on the Left–Right dimension but moving toward the elite, and the other retaining its anti-elite appeal but moderating its right-wing position. What share of baseline AfD voters would be closer to each?

Using the nationwide median AfD position defined above, the two shifts of size δ partition the baseline AfD attraction region $\mathcal{X}_{AfD}$ into an anti-elite region $\mathcal{X}_{A-E}$ (green) and a Right region $\mathcal{X}_{Right}$ (pink). Voters in each region are closer to the hypothetical successor that retains, respectively, the AfD's anti-elite or Left–Right position. Figure 8a illustrates the split.⁴⁶

The target parameter is the share of baseline AfD voters in the anti-elite region:

$$\theta_{AE} = \frac{\iint_{\mathcal{X}_{A-E}} f(x | s, d) dx dF_{S,D}(s, d)}{\iint_{\mathcal{X}_{AfD}} f(x | s, d) dx dF_{S,D}(s, d)}.$$

I report bounds on θ_{AE} and θ_{Right} , where $\theta_{Right} = 1 - \theta_{AE}$ by construction.

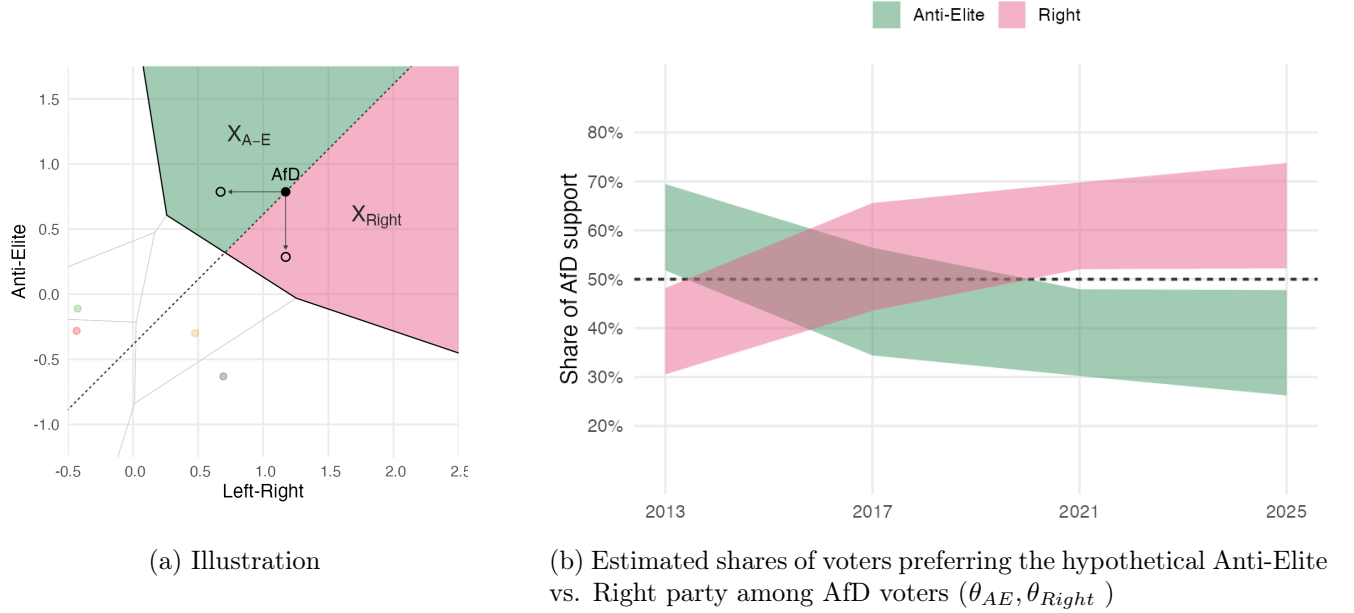
Figure 8b provides the bounds for the target parameters over years. The results indicate that AfD support has changed over time. In earlier years, AfD support was driven more by anti-elite motivations; this pattern flipped over time, and in 2021 and 2025 at least 52% of AfD support comes from individuals with stronger ideological Right motivations.

These aggregate results mask substantial regional and subgroup heterogeneity. Figure K10 reports the estimates separately for Eastern and Western Germany. The pattern for Western Germany resembles the national-level pattern presented above; in turn, voters in Eastern Germany are more strongly driven by anti-elite attitudes than voters in Western Germany.⁴⁷ The patterns are clearer and more precisely identified for Western Germany, reflecting the fact that it is larger in population and offers more electoral districts across which to exploit variation.

I highlight two subgroup-heterogeneity results among Western voters over time. Figure K11 shows the shares of AfD voters with and without higher education belonging to the Anti-Elite and Right areas of Figure 8a; Figure K12 provides the same analysis for young (under 35) and older (35 and over) individuals.

⁴⁶The benchmark uses $\delta = 1$. Figure K9 reports results for other shift sizes, which are quantitatively similar.

⁴⁷Except in 2013, when anti-elite sentiment dominated in the West. The party's anti-elite appeal then was arguably of a different nature—rooted in opposition to the Eurozone bailouts rather than in the hostility towards the establishment and democratic institutions that came to define the party after 2015. See discussion on the possibly confounding other dimensions in the Section 9.



Notes: Panel (a) illustrates party positions at their national median, and hollow circles illustrate the counterfactual AfD *shifts*. The areas of AfD voters motivated more by anti-elite (green, $\mathcal{X}_{A-E}$) or right (pink, $\mathcal{X}_{Right}$) ideological stance of AfD are plotted. The illustrative weight used is $w_{AE} = 1$; the weights used in the estimation are those estimated in Section 6.1. Panel (b) plots bounds on the shares of voters whose bliss points lie in the green and pink regions across years. The results are aggregated across the two parts of the country. Panel (b) reports the estimates without the second-choice moments; Figure K13 reports the estimates with them.

Figure 8. Illustration of a target parameter (a) and results (b)

These figures show that voters in different demographic groups support the same party for different reasons, and that the model captures how these reasons evolve over time. For voters without higher education, the party's radical-right stance came close to its anti-elite appeal only in later years, which could suggest that their preferences are more malleable, or more responsive to party messaging. Young AfD voters, in turn, are increasingly concentrated in the Right group. The growing support for radical, and in particular radical-right, movements among young people is documented in the political science literature, but its motivations remain contested (van der Brug, Hobolt and Popa, 2026); my results suggest that young voters may increasingly support these parties for their radical-right ideology rather than their anti-elite stance.

Appendix E.2 presents the same estimates using survey data. The patterns are somehow similar to the revealed-preference ones, although the survey attributes a larger role to Anti-Elite motivations in 2017–2021, again consistent with underreporting of right-wing views. The comparison concerns bliss points only: the weights are taken from the revealed-preference estimates, since they cannot be recovered from survey data.

Robustness. I now discuss the four robustness exercises anticipated in Section 5.2, which address the selection-on-observables assumption on candidate positions. The first restricts the sample to municipalities with fewer than 5,000 inhabitants (Figure K14). The second restricts it to non-contender districts, separately for Eastern and Western Germany (Figure K15). The third allows preferences to differ between districts with more moderate and more conservative AfD candidates (Figure K16). The fourth restricts it to municipalities

on district borders and requires preferences to be identical only on either side of the same border, rather than across the whole state (Figure K17). The main qualitative results hold across all four specifications, although the patterns are noticeably less precise in the border specification, which uses a much smaller sample.

The bounds reported above reflect partial identification only and do not account for any statistical uncertainty. Appendix G.5 quantifies sampling uncertainty by re-estimating the bounds on bootstrapped samples of municipalities.

The analysis so far has described the evolution of AfD voters’ preferences. The next sections use these preferences as primitives for counterfactual exercises, examining how changes in the positions of (i) the AfD and (ii) its center-right, pro-elite rival, the Christian Democratic Union (CDU), would affect the AfD vote share. These exercises speak both to how sensitive the AfD’s support is to its own position and to the intensity of competition between the two parties.

7 Partial Equilibrium: Repositioning Counterfactuals

7.1 How Many of Its Voters Would the AfD Lose by Moderating?

The party does not exist in isolation, but competes with other parties for voters. Answering the question of which dimension matters more for voters requires assessing the sensitivity of voters’ choices to counterfactual party positions.

To assess to what extent the AfD’s success is driven by right-wing ideology or by anti-elite sentiment, I combine the estimated bliss-point distributions of AfD voters, the estimated relative weights, and the median positions of all the other parties, and ask: *How sensitive are AfD voters to the party shifting its positions along the Left–Right and Anti-Elite dimensions?* Put differently, the question is how many of the AfD’s baseline supporters the party would lose if it adopted a more pro-elite or a more left-wing position.

I hold the estimated preferences fixed and shift the AfD’s position relative to its national-level median in each year, $\bar{Y}$, by 0.2/0.5 units along each dimension.⁴⁸ Figure K18 illustrates the shifts along both dimensions under median party positions across years and isotropic weights for clarity.

To interpret the steps, I compare them with the spread of each party’s candidate positions across districts in a year: the standard deviation is about 0.3–0.5 on both dimensions for both the AfD and the CDU/CSU (2013–2025). A shift of 0.2 is thus roughly half a standard deviation of the party’s own candidates, and a shift of 0.5 about one; I treat 0.2 shift as an ambitious, but realistic one.

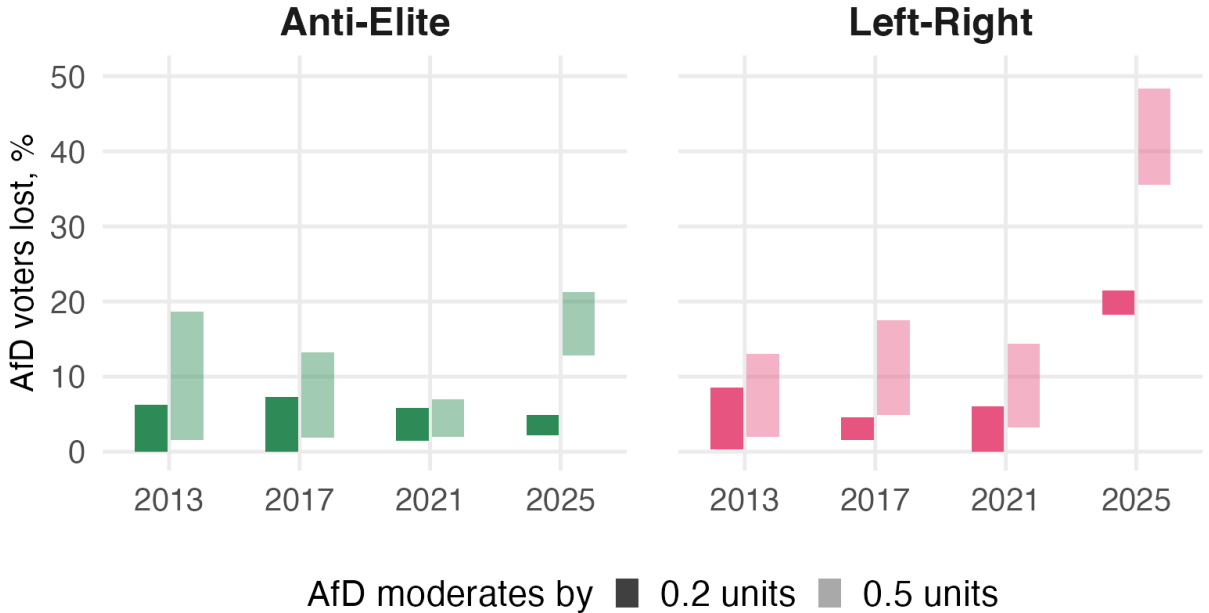
I now provide a substantive interpretation of what a shift of 0.2 or 0.5 units means in terms of policy positions. To do so, I trace back the scoring pipeline to determine what such a shift would require in terms of changes in candidates’ stated opinions. Moving a single question by one scale point shifts the dimension score by an amount specific to each question. On the anti-elite dimension, the most influential single question—

⁴⁸Positions are standardized by construction, so a given shift is comparable across dimensions and years.

Satisfaction with the way democracy works (a 1–4 item)—moves the score by about 0.2 per scale point across years, and a typical five-point item such as *Legislation reflects citizens’ interests* by 0.17. A 0.2 step therefore corresponds to revising one such question by roughly one point, and a 0.5 step to about three scale points spread across questions.

The Left–Right dimension behaves similarly: potent five-point items such as *Immigration is good for the economy* or *Marriage only for heterosexual couples* move the score by about 0.10 per point. In short, a 0.2 step is a modest adjustment of one or two questions, while a 0.5 step is a substantial repositioning that must touch several questions at once. Table J6 provides score sensitivities across questions for 2017.

The target parameter is the share of baseline AfD voters who would abandon the party had it shifted to **AfD’**.⁴⁹ Figure 9 plots the results separately for different step sizes and directions of the shift.



Notes. Each bar is the identified interval for the share of *baseline* AfD voters who would abandon the party if it moderated by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the AfD moves toward the elite along the anti-elite dimension; in the right panel (**Left-Right**) the AfD moderates along the left–right dimension. Within each year, shading denotes the size of the moderation step (0.2/0.5 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted (anisotropic) spatial metric with the estimated relative weight w_{AE} .

Figure 9. Estimated shares of AfD voters lost under counterfactual AfD shifts

The results suggest that under small step of 0.2, the share of voters abandoning the AfD is no more than 10% (20%) across years and dimensions before 2025. Shifts of a step of 0.5 yield larger losses, especially

⁴⁹Precisely, $\theta_{\text{Loss, A-E}} = \frac{\iint_{\mathcal{X}_{\text{Loss, A-E}}} f(x | s, d) dx dF_{S,D}(s, d)}{\iint_{\mathcal{X}_{\text{AfD}}} f(x | s, d) dx dF_{S,D}(s, d)}$, $\mathcal{X}_{\text{Loss, A-E}} = \mathcal{X}_{\text{AfD}} \setminus \mathcal{X}_{\text{AfD}'}$

along the Right dimension, reaching up to 48%.

The results also suggest that competition between the AfD and other parties became much tighter by 2025, in the sense that even small AfD shifts (particularly along the Left–Right dimension) would produce large defections of voters to other parties. This was not the case before 2025, indicating that the party had previously existed in relative isolation.

If competition with the establishment parties has indeed intensified, those parties should be able to capture a large mass of AfD voters by shifting their positions only slightly. I now turn to such an analysis.

7.2 How Many AfD Voters Could the CDU Win?

German establishment parties, which once commanded a large majority of the electorate, are now losing ground to the AfD, which leads the polls nationwide as of mid-2026.⁵⁰ This development is by no means unique to Germany: across Western democracies, populist and radical parties have risen sharply over the past two decades, in many cases entering national legislatures and governments, as in Italy, France, the Netherlands, Spain, and Sweden. Beyond Europe, Donald Trump’s rise is widely regarded as part of the same populist wave.

How should establishment parties compete with populist challengers? This section examines how effective it would be for them to adopt positions closer to the AfD’s in the two-dimensional space defined by the Left–Right and Anti-Elite dimensions. The natural candidate for such analysis is the Christian Democratic Union (CDU), the main establishment party on the right of the Left–Right spectrum, on which I focus throughout this section.

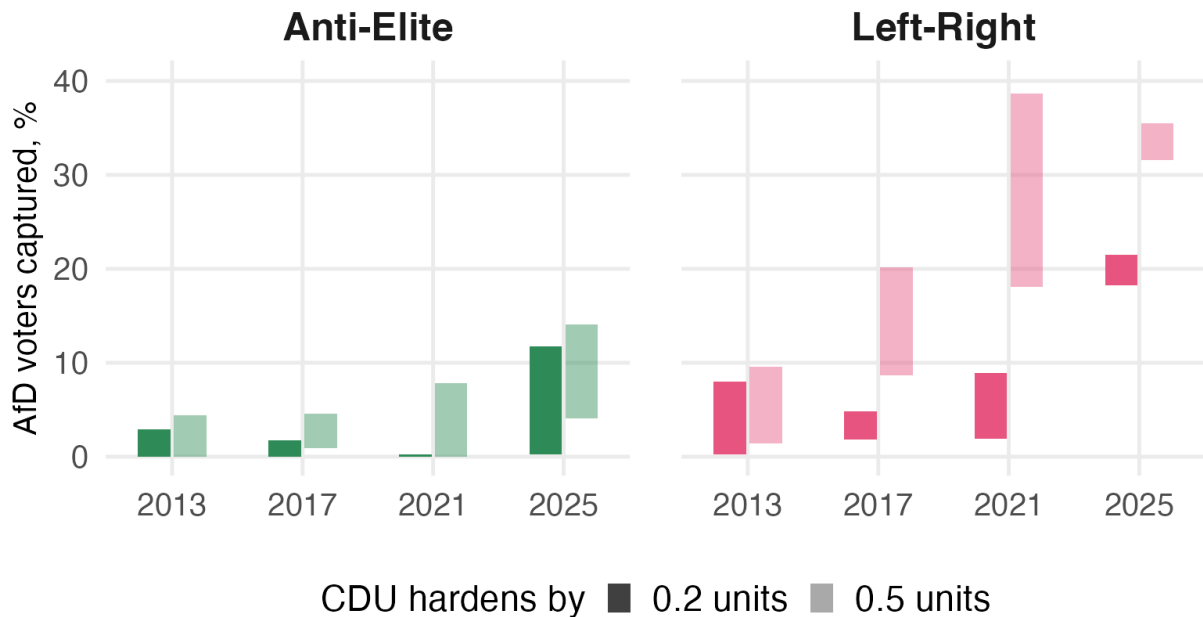
In particular, I conduct an analysis similar to that in the previous subsection, but ask the reverse question: *if the CDU moves a given step towards the AfD along the Left–Right or Anti-Elite dimension, how many AfD voters would it capture?* Figure K19 illustrates these counterfactual scenarios.

Figure 10 plots the results separately for different directions of the shift. The results suggest that a small shift (0.2) to the right along the Left–Right dimension could allow the CDU to capture up to 9% of AfD support across 2013–2021; while the shift towards more anti-elite positions along the Anti-Elite dimension would allow to capture less than 3% of AfD support.

In 2025, a 0.2 shift along the Left–Right dimension would allow the CDU to capture about 20% of the AfD electorate. Although this exercise does not solve the CDU’s vote-share maximization problem, 20% of AfD voters is a large mass by 2025, and the CDU is unlikely to lose as many voters on its left flank; the next section examines this trade-off in more detail.

Put into perspective, this result implies that had the CDU slightly hardened its position on immigration

⁵⁰As of July 2026, national polling averages place the AfD at roughly 28% and the CDU/CSU at roughly 22%. See Politico’s Poll of Polls, <https://www.politico.eu/europe-poll-of-polls/germany/> (accessed July 2026).



Notes. Each bar is the sharp identified interval for the share of *baseline* AfD voters who would abandon the party if the CDU moved by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the CDU moves towards more anti-elite positions; in the right panel (**Left-Right**) the CDU moves towards more right-wing positions. Within each year, shading denotes the size of the step (0.2 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted spatial metric with the estimated relative weight w_{AE} .

Figure 10. Estimated shares of AfD voters captured under counterfactual CDU shifts

in the 2025 election, it could have captured about a fifth of the AfD electorate. Concretely, this would mean moving from “Agree” to “Rather agree” on “*Immigrants are good for the German economy*”, from “Agree” to “Strongly agree” on “*Immigrants should adapt to German culture*”, and from 8 to 9 on “*Own position on immigration*”. The same 0.2-unit shift could alternatively be achieved through economic policy items: moving from “Neither agree nor disagree” to “Disagree” on “*Providing social security is a core government task*” and “*The government should reduce income differences*”, and from 9 to 10 on “*Own position on taxes vs. social spending*”.⁵¹

These results are aligned with the experimental findings of [Guenther and Nunnari \(2025\)](#), who find that perceiving the CDU as one unit more liberal on immigration (on a 0–10 scale) raises the intention to vote for the AfD by an average of 5 percentage points – about 25% of the AfD vote. Putting the results on the same scale, my results suggest that a 2-unit CDU hardening on immigration on the 0–10 scale would capture about 20% of the AfD electorate. The results are also consistent with the documented defection of voters between parties in 2025: about 1.2 million CDU supporters in 2021 voted for the AfD in 2025, making this the largest such flow between any two parties in the country ([Vehrkamp and Posthofen, 2026](#)). The counterfactual is thus likely driven by the fact that the CDU could win these voters back with a slightly hardened Left–Right

⁵¹All illustrative answers are given relative to the median answers of CDU candidates in the 2025 survey. See Table [J7](#) for the contribution of a one-point shift in each survey item.

position.

This section was intended to provide some intuition for what the best strategy might be for establishment parties competing with populists. Notice that the party’s hardening of its anti-elite positions for realistic step values (0.2, 0.5) does not appear to be as effective in any year—in particular, not even in earlier years, when the AfD electorate was less pronouncedly right-wing. The party selected for this exercise was a large center-right establishment party, which could be affecting the results. Nonetheless, the finding that competition with populists could be effective along the Left–Right dimension at least for this party is a reassuring one, since competing on anti-elite rhetoric could, among other things, contribute to the erosion of democratic institutions.

Overall, at least two factors determine the outcome of the repositioning exercises: the relative positions of parties, and the distribution of voters’ preferences in the two-dimensional space. As the analysis of voters’ ideal points above shows (Section 6.2.1), the AfD’s electoral base was relatively more anti-elite and less radical-right in 2013–2017 than in 2021–2025, so capturing these voters through a rightward move was unlikely in the earlier years; and the anti-elite shift that could have helped to capture them was implausible. Between 2017 and 2021, the gap between the CDU and the AfD, if anything, widened (Figure L2); the CDU’s reach nonetheless grew substantially, as the shift in the AfD’s electorate brought many of its voters within range, though only with a large step. By 2025, the CDU had moved to the right under a more conservative leadership—its current leader, Friedrich Merz, comes from the party’s conservative wing, in contrast to the long-time leader Angela Merkel—narrowing the gap and bringing a large share of AfD voters within reach of even a small step. As both parties can as of 2025 compete for the same voters, I turn to the strategic interaction between them.

8 Equilibrium: spatial competition between the establishment and the populists

One of the puzzles of the rise of populism in Europe is why establishment parties do not adjust their platforms so as to hinder the entry and growth of populist rivals. Several answers are plausible: the establishment may anticipate that the populists would move in turn, it may fear losses on its other flank, or repositioning may be costly. The framework of this paper can separate the first two, since both are governed by the distribution of voters’ preferences.

This section takes a step towards resolving the puzzle by studying a spatial competition game between the CDU and the AfD, using the estimated distribution of voters’ ideal points and weights as its primitive.⁵² Since a plausible rightward step by the CDU would have captured only a small share of the AfD’s electorate before 2025 (Figure 10), I focus in this section only on the 2025 Bundestag election. This section shows that a rightward shift could indeed be net positive for the CDU if the AfD stayed put; once the AfD can respond,

⁵²I keep AfD-specific weights, since the focus is on the competition with AfD, and keeping different weights for different parties is not consistent within the same model.

however, it most often hardens its position, and the CDU's gains disappear.

The game. All parties other than the CDU and the AfD are held at their observed 2025 positions; the set of players is $N = \{\text{CDU}, \text{AfD}\}$. Each player $n \in N$ may shift its position Y_n by at most 0.2 standardized units on each dimension, in steps of 0.1. On Left-Right both parties may move in either direction (and I separately consider a game of Left-Right moves only below). On Anti-Elite I only admit moves that are plausible for each party: the CDU may become less anti-elite but not more, the AfD more anti-elite but not less. The action sets are therefore

$$A_{\text{CDU}} = \{-0.2, \dots, 0.2\} \times \{-0.2, -0.1, 0\}, \quad A_{\text{AfD}} = \{-0.2, \dots, 0.2\} \times \{0, 0.1, 0.2\},$$

15 moves each, including the status quo $(0, 0)$.

The payoff of a profile $a = (a_{\text{CDU}}, a_{\text{AfD}}) \in A_{\text{CDU}} \times A_{\text{AfD}}$ is a *target parameter* – the change in n 's share of the electorate relative to the status quo,

$$u_n(a) = \iint_{\mathcal{X}_n(a)} f(x \mid s, d) dx dF_{S,D}(s, d) - \iint_{\mathcal{X}_n(0)} f(x \mid s, d) dx dF_{S,D}(s, d),$$

where $\mathcal{X}_n(a)$ is n 's area of attraction under the counterfactual positions $(Y_{\text{CDU}} + a_{\text{CDU}}, Y_{\text{AfD}} + a_{\text{AfD}})$, and $\mathcal{X}_n(0)$ is the same area under the status quo. The payoff is computed over the 2025 electorate with participation held fixed, and does not model the turnout margin, although a move could itself affect turnout.

Parties are assumed to know the distribution of voters' ideal points, which the researcher only knows up to the identified set. I therefore draw $K = 100$ distributions from the identified set and solve the game for each.^{53,54} For draw k the payoffs $u_n^{(k)}(a)$ define a normal-form game $G^{(k)}$, and I report how outcomes are distributed across draws.

Sequential moves. I study the sequential version of the game in which the CDU (the leader) moves first and the AfD (the follower) observes a_{CDU} before choosing a_{AfD} . The follower's best-response correspondence is

$$B_{\text{AfD}}^{(k)}(a_{\text{CDU}}) = \arg \max_{a_{\text{AfD}} \in A_{\text{AfD}}} u_{\text{AfD}}^{(k)}(a_{\text{CDU}}, a_{\text{AfD}}),$$

⁵³Each draw is a random vertex of the identified set: I minimize a linear objective with i.i.d. standard-normal coefficients over the set of voter distributions consistent with the data, which returns an extreme point of the set. Any interior point is a convex combination of vertices and its payoffs are the same combination of vertex payoffs, so the vertex draws span the full range of outcomes.

⁵⁴Solving the game draw by draw is a simplification, but still keeps the approach free of any functional-form assumption on voters' preferences — each draw is a distribution the data admit, not one I impose. Best replies and the Stackelberg outcome are similar across draws (I report the distributions below), so the equilibrium conclusions do not depend on which admissible distribution is the true one.

and the leader solves

$$a_{\text{CDU}}^{*(k)} \in \arg \max_{a_{\text{CDU}} \in A_{\text{CDU}}} \min_{a_{\text{AfD}} \in B_{\text{AfD}}^{(k)}(a_{\text{CDU}})} u_{\text{CDU}}^{(k)}(a_{\text{CDU}}, a_{\text{AfD}}),$$

the inner minimum resolving ties in the follower’s response pessimistically for the leader. A profile $a^{*(k)} = (a_{\text{CDU}}^{*(k)}, a_{\text{AfD}}^{*(k)})$ with $a_{\text{CDU}}^{*(k)}$ solving the leader’s problem and $a_{\text{AfD}}^{*(k)} \in B_{\text{AfD}}^{(k)}(a_{\text{CDU}}^{*(k)})$ is a **Stackelberg equilibrium** of $G^{(k)}$. The sequential formulation matches the question being asked, *whether the establishment party should move towards its populist rival*, and it sidesteps a well-known difficulty of simultaneous-move spatial competition: in more than one dimension a pure-strategy Nash equilibrium exists only under restrictive conditions on preferences (Osborne, 2025), and in the present game in practice it does not exist.

When the AfD stands still. I first ask whether moving right pays for the CDU when the AfD does not respond. Panel (a) of Figure 11 reports the CDU’s median net change in vote share (gray bars) for each Left–Right move, holding its Anti-Elite position and the AfD’s position fixed. The answer is yes, and increasingly so: a 0.1 step gains about 0.3 percentage points of the electorate and a 0.2 step 3.1 points, while steps to the left lose votes. The AfD’s changes (light blue bars) mirror the CDU’s: the AfD loses when the CDU moves closer and gains when it moves away. A rightward step thus gains the CDU more from the AfD than it loses to other parties on its left flank.

Panel (a) of Figure 12 shows what lies behind the 0.2 step: the CDU captures the wedge of the AfD’s area closest to it on the Left–Right dimension (dark gray) and loses a thin strip on its left flank to the FDP (hatched). The wedge is large because, as Figure 7 showed, a substantial mass of AfD voters holds relatively pro-elite, radical-right positions, and the CDU’s move brings it closer to them than the AfD is. If the AfD cannot respond, accommodation pays, and pays well.

When the AfD responds on Left–Right. The conclusion above rests on the AfD standing still while the CDU takes its voters. I first study the game in which both parties can move only on the Left–Right dimension.⁵⁵ Panel (b) of Figure 11 reports the median net changes of the CDU and the AfD after the AfD’s best reply to each Left–Right move of the CDU.

The AfD’s modal reply to the CDU’s +0.2 is a further hardening of its own, +0.1 to the right. Panel (b) of Figure 12 shows this reply and illustrates why: the AfD wins back the far-right wedge the CDU had captured (purple). Section 6.2.1 explains why this is profitable: in later years large shares of AfD voters have bliss points that are far to the right.

The game is not zero-sum, since the other parties also gain and lose voters along the way, but the AfD’s reply is aimed at the CDU. This is the possible populist party’s answer to accommodation: when the establishment party moves towards it, it does not concede the contested voters but radicalizes further,

⁵⁵This restriction reflects the view that Left–Right positions are easier to change credibly than anti-elite ones.

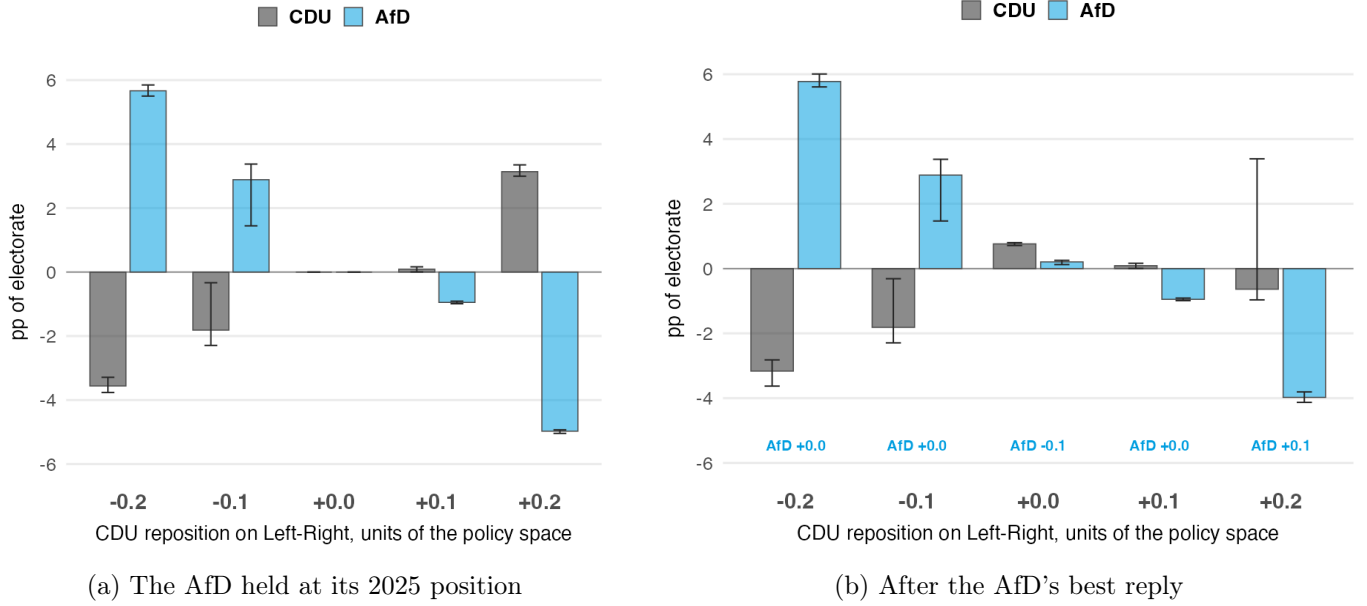


Figure 11. Estimated net CDU and AfD vote share gain relative to the status quo by CDU Left-Right moves, 2025

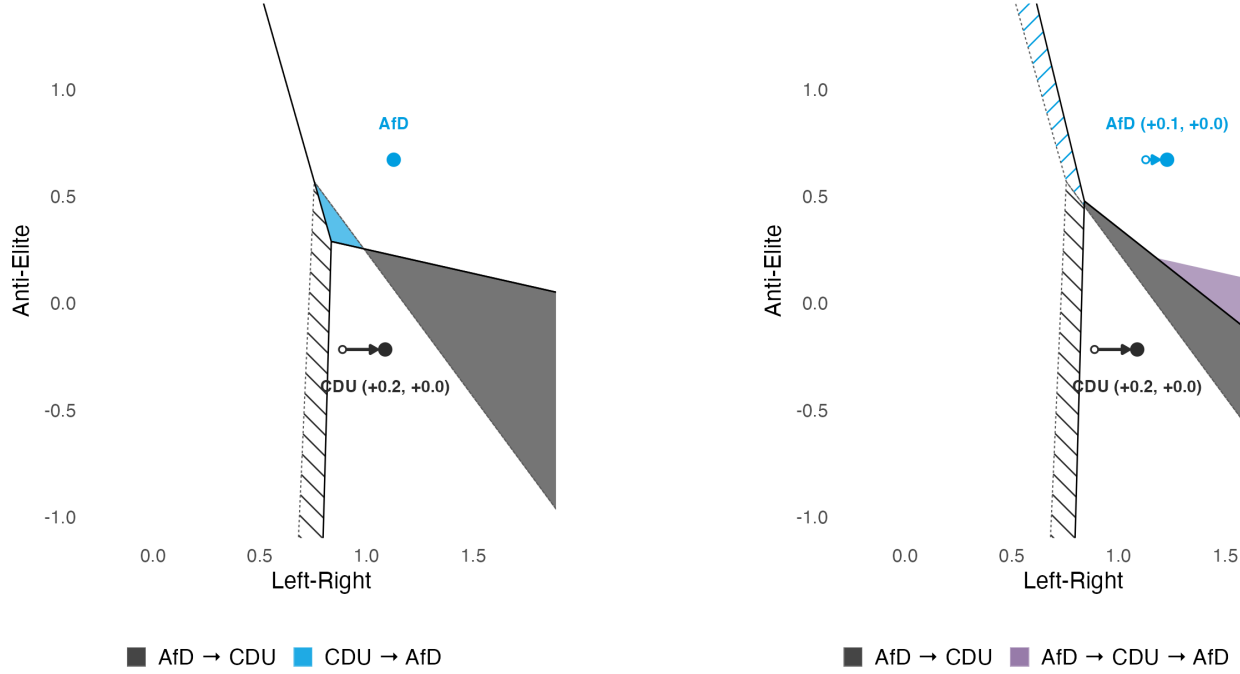
moving away from the challenger along the same dimension. The AfD's net change remains negative: the reply recovers part of its losses, not all of them. For the CDU, the +0.2 step is no longer profitable once the reply is taken into account (-0.6 points at the median).

The Stackelberg equilibrium across preference draws features the CDU staying put in 70% of draws, and moving +0.2 or -0.1 in the other 20% and 10% respectively.

Equilibrium in both dimensions. Figure K27 reports the CDU's and the AfD's net changes after the AfD's best reply when both parties can move on both dimensions. The CDU's 0.2 step to the right leaves both parties worse off: the CDU is left with -0.5 points and the AfD with -3.0 median across draws. The AfD's best reply to the CDU's $(+0.2, 0)$ is $(+0.1, +0.2)$ or $(+0.1, +0.1)$ in every draw: it hardens its Left-Right position and sharpens its anti-elite profile at the same time. The Left-Right step wins back the far-right wedge, as before; hardening on anti-elite allows the AfD to capture anti-elite voters with relatively center-right positions, at the cost of an anti-elite step that is cheap when the Anti-Elite dimension carries little weight ($w_{AE} = 0.2$ in the West).

Once the AfD's response is anticipated, moving right is no longer the CDU's best move. The Stackelberg equilibrium has the CDU staying put in 64% of draws, moving -0.1 in 34%, and moving +0.2 in only 2%.

The exercises in this subsection confirm the conventional wisdom that establishment parties can capture



(a) The CDU moves $(+0.2, 0)$, the AfD stands still

(b) The AfD replies $(+0.1, 0)$

Notes. Each panel shows the (Left-Right, Anti-Elite) policy space with the CDU (dark gray) and the AfD (blue) at their 2025 national medians and the other parties fixed; open circles and arrows mark the moves. Dotted gray lines are the boundaries of the two parties' areas of attraction at the status quo, solid black lines the boundaries after the moves. Shaded areas are voters who change parties between the two players: dark gray, from the AfD to the CDU; blue, from the CDU to the AfD; purple, captured by the CDU in panel (a) and won back by the AfD in panel (b). Hatching marks areas lost to the other parties. Partitions are drawn with the estimated West weight $w_{AE} = 0.2$.

Figure 12. Areas of attraction under the counterfactual moves, 2025

some populist voters by moving towards the populists' positions. But this logic overlooks the populists' strategic response. Once that response is taken into account, (i) in equilibrium the establishment party is better off staying put, and (ii) a rightward move by the establishment party could leave the party system as a whole more radicalized than before.

Lessons and caveats of the exercise. First, the multidimensionality of the setup is essential. In one dimension the exercise would be trivial: the populist party's reply to an approaching rival is to move towards it, not away from it – its outer flank is already its own, and hardening further would only cede the contested middle. Hardening becomes a best reply only once the rival overtakes it. In a weighted two-dimensional space this no longer holds: hardening can be a best reply even without overtaking.

Second, what matters for the results is the distribution of voters' ideal points in the policy space: parties' best replies are determined by where the mass of voters sits relative to the boundaries they can move.

Third, an important shortcoming of this equilibrium analysis is that the other parties stay put. An equilibrium featuring e.g. the AfD's move to the left is therefore partly an artifact of holding the other parties

fixed: competition on each player’s outer flank is one-sided, whereas competition between the two players is a genuine game. What is less of an artifact is the sign of the answer to the question this section asks: with the AfD able to respond, moving right is not the CDU’s best move. Considering the populists’ reaction is thus essential when assessing the establishment’s options – a channel long emphasized in the theoretical literature on party competition and entry (Palfrey, 1984) but rarely quantified.

9 Discussion

The framework developed in this paper rests on a number of choices about the model, the ideological dimensions, and the identifying assumptions. I now discuss possible extensions of the framework, as well as how it compares to alternative modelling choices.

Probabilistic Voting Model. The model considered in the main text is Downsian: voters care only about the policy platforms of the candidates. Probabilistic voting models augment this framework with another component, interpreted as an individual-specific taste for a candidate (Persson and Tabellini, 2002). I now show that the estimated model can be viewed as a probabilistic voting model with particular restrictions on the distribution of the error term.

The estimated model is a model with *random parameters*: the object of estimation is the distribution of a parameter — here, the bliss point. Let X_i be the average bliss point of type i , ϵ_i an additive shock defining the deviation from it, and Y_j the position of candidate j :

$$U_{ij}^{RPM} = -\text{dist}(X_i + \epsilon_i, Y_j).$$

The probabilistic voting model is instead a *random utility* model,

$$U_{ij}^{RUM} = -\text{dist}(X_i, Y_j) + \bar{\epsilon}_{ij},$$

with $\bar{\epsilon}_{ij}$ usually assumed to be i.i.d. across candidates and independent of X_i . For ease of exposition here and below in this section I assume the quadratic distance function and omit subscripts. Expanding the utility,

$$U_j^{RPM} = -(X + \epsilon - Y_j)^2 = -(X - Y_j)^2 + \underbrace{2\epsilon(Y_j - X) - \epsilon^2}_{\nu_j}.$$

The random-parameter model is therefore itself a random utility model, and a probabilistic voting model, $U_j^{RPM} = -(X - Y_j)^2 + \nu_j$, in which the distribution of the error is induced by the distribution of the bliss-point shock rather than assumed directly. The two models differ only in the restrictions placed on the joint distribution of the errors.

I assume the bliss-point shock is independent of both voter type and candidate positions, $\epsilon_i \perp (X_i, Y)$.

Independence from X_i requires that the distribution of deviations does not vary systematically with the type's position; independence from Y rules out candidates positioning in response to the realisation of the shock. The former just states that given the type, the deviations are pure noise. The latter is the stronger identification restriction, discussed above: it rules out that, within a type, deviations are shifted towards the candidates' positions.⁵⁶

Note that this treatment of the framework also clarifies the consequences of potential misspecifications, such as dropping the probabilistic voting term, omitting important dimensions, or measurement error in the candidate positions. I briefly discuss the effect of the first below, as it belongs to the probabilistic voting model, and turn to the others in the following paragraphs.

If there is a non-spatial *taste component* Z_{ij} orthogonal to both X_i and Y_j , dropping it does not bias the recovered bliss points, as orthogonality rules out any systematic tilt towards particular positions. It could nonetheless inflate the dispersion of the estimated distributions for the bliss points, as choices driven by *taste* are rationalised *spatially*. It could also widen the identified sets for weights, as part of the choice variation becomes uninformative about weights, flattening the criterion around its minimum; the relative weights remain unbiased.

More than two dimensions. The framework restricts attention to two dimensions, and one could argue that others matter—nationalism, anti-globalism, or attitudes towards the European Union. In principle the framework accommodates more dimensions; in practice this is infeasible with the data at hand. Moreover, the candidate surveys contain no items suitable for estimating positions on nationalism or anti-globalism, and only two items for the EU dimension.

How would my results be affected if there is another dimension which my estimates miss? Consider the cleanest case: one studied dimension with weight w and positions Y_j , and one omitted dimension with weight w^o , positions $Y_j^o = \gamma Y_j + e_j$, $\gamma > 0$ (dimensions bundled at the candidate level), and voter component X^o . Set $e_j = 0$ first.⁵⁷ True utility is

$$U_j = -w(X - Y_j)^2 - w^o(X^o - Y_j^o)^2.$$

Collecting the terms that depend on j and dropping alternative-invariant ones,

$$U_j \sim -(w + w^o\gamma^2)Y_j^2 + 2(wX + w^o\gamma X^o)Y_j = -(w + w^o\gamma^2)(Y_j - \tilde{X})^2 + \text{const},$$

⁵⁶Two properties of ν_j follow from ϵ_i being a single voter-level shock rather than a voter-candidate-level shock. First, the term ϵ^2 is common to all candidates and cancels from every utility comparison, so effectively $\nu_j = 2\epsilon(Y_j - X)$ up to a candidate-invariant constant. Second, because all ν_j are driven by the same draw of ϵ , the errors are correlated across candidates, the more strongly the closer the candidates' positions in the policy space. Consequently, whenever the number of candidates exceeds the dimension of the policy space plus one, no distribution of the bliss-point shock reproduces an i.i.d. error vector: the induced errors are confined to a lower-dimensional subspace. Exact equivalence obtains, e.g., in the case of two candidates.

⁵⁷With $e_j \neq 0$, further terms arise: a voter-level factor error $2w^oX^oe_j$, absorbed into the bliss-point shock when X^o is independent of (X, Y) within type; and candidate-specific intercepts $-2w^o\gamma e_j Y_j - w^oe_j^2$, which a pure characteristics model rationalizes spatially. All are proportional to w^o and vanish in the same limit.

where

$$\tilde{X} = \frac{w X + w^o \gamma X^o}{w + w^o \gamma^2}, \quad \tilde{w} = w + w^o \gamma^2.$$

The misspecified one-dimensional model thus retains the correct form but has distorted primitives: the recovered bliss point is $\tilde{X}$, a weighted average of the true position on the studied dimension and the omitted-dimension component, and the recovered weight is $\tilde{w} = w + w^o \gamma^2$, inflated by the omitted dimension’s importance and its correlation with the studied one. Both distortions vanish as (1) $\gamma \rightarrow 0$: dimensions along which candidates’ positions are unrelated to the studied ones can be omitted without bias; or (2) $w^o \rightarrow 0$: unimportant dimensions can be omitted without bias, regardless of bundling.

Hence, my approach is not robust to omitting relevant dimensions correlated with the included ones. The correlation patterns from the party-level CHES database (Figure L3) suggest that the Left–Right dimension, being highly correlated with plausibly relevant dimensions such as Nationalism and Immigration, may capture those dimensions. Immigration enters the Left–Right dimension by construction, as questions concerning immigration are used to build it. Nationalism, by contrast, could not be accounted for, as no survey questions capture it. Under this reading, the Left–Right results could be interpreted as those for a weighted bundle of economic, cultural, nationalist, and immigration policies. Similarly, the Anti-Elite dimension is highly correlated with the Anti-EU dimension, so its results can likewise be read as referring to a bundle.

The Anti-Elite and Left–Right dimensions are essentially uncorrelated, both across parties in the expert-survey data and across candidates, parties, and years in my sample.

Left–Right as a combination of Economics and Culture. So far the paper has treated the Left–Right dimension as a bundle of economic and cultural policies without unpacking the two. Estimating the model in three dimensions is infeasible given the data, so I instead re-estimate it in two: Economics and Culture. Omitting the Anti-Elite dimension here is relatively innocuous: it is only weakly correlated with both (Figure L3), so, following the discussion above, its omission is unlikely to affect the recovered preferences.

To do so, I first split the Left–Right items into those concerning culture and those concerning economics; see Table L2 for the classification.⁵⁸ I then follow exactly the estimation procedure of Appendix 6, applied to these two dimensions.

Figure K24 reports the estimated weights by year and part of the country, Figure K26 reports the evolution of preferences across the dimension bins, and Figure K25 reports the counterfactual motivation exercise, which splits the AfD into two hypothetical parties: one identical to the AfD on culture but economically liberal, the other identical on economics but culturally liberal. The decomposition shows a gradual realignment of voting motives, from economic ones, which clearly dominate in 2013, to cultural ones, which clearly dominate in 2025. In the West, voters attach a much larger weight to the cultural than to the economic dimension from 2017 onward; in the East the two weights are closer, although imprecisely estimated. AfD

⁵⁸Two coding decisions are worth noting: immigration items are classified as cultural (Alesina and Tabellini, 2024), and the Left–Right self-placement item is dropped.

voters' ideal points are relatively stable over time on the cultural dimension, but shift slightly to the left on the economic one.⁵⁹ These results are partly in line with [Danieli et al. \(2026\)](#), who attribute the rise of radical-right parties largely to an increase in the weight voters attach to cultural relative to economic issues.

Candidate positions estimated with an error. Assume the candidate positions are measured with error: the estimation uses $\bar{Y}_j = Y_j + \xi_j$ in place of the true Y_j . The effective utility is then

$$U_j^{RPM} = -(X + \epsilon - Y_j - \xi_j)^2,$$

so the error enters as a perturbation of the candidate's location rather than as an additional disturbance: unlike ϵ_i , which varies at the voter level, ξ_j is common across voters, and the recovered bliss points are located relative to the perturbed positions. The relevant question is therefore whether the objects of interest are stable under small perturbations of the positions. Figure [K22](#) assesses this for the West 2025 results: I add an i.i.d. $N(0, 0.01)$ shock to every estimated district-level party position (and to the national anchor of the counterfactual parties) and re-estimate the target parameter of Appendix [6.2.2](#) for each of 10 draws.⁶⁰ The shocks move the bounds without widening them, and on average pull them towards the uninformative composition,⁶¹ but preserving the overall results. Moderate error in the measured positions therefore weakens but does not systematically overturn the conclusions.

The Role of the Supply Side. To what extent the results presented above are driven by the supply side repositioning versus the demand side changes (genuine evolution of preferences, or compositional changes)? To answer this question I decompose the evolution of the values of the target parameter from the Section [6.2.2](#) into the preference evolution and supply-side repositioning, using Shapley values approach (Section [H](#) provides more details on the exercise).⁶²

The results of this exercise suggest that the evolution over time in the share of Right-motivated voters documented in this paper is not driven by parties changing their positions, but by genuine evolution of preferences of AfD electorate.

In this setup it is hard to pin down demand-side changes and attribute them to either the composition or the preference channel (see Table [J8](#) for the wide bounds of Shapley values on both). I now provide two pieces of evidence that preference evolution within demographic groups is consistent with increasing Right motivation. First, within-person evolution towards Right motivations is consistent with survey-panel evidence

⁵⁹These results also speak to the AfD's transformation around 2015. In 2013, the party, then dubbed the "professors' party" after its economist founders and centred on opposition to the euro, drew a different electorate: up to half of its voters were economically conservative, and they attached little weight to the cultural dimension.

⁶⁰A standard deviation of 0.01 is not small relative to the cell partition: about 35% of the cells are thinner than 0.02 (two s.d.).

⁶¹This resembles OLS attenuation bias: the shocks break the cross-election structure of the cell labels that the "correct" distributions exploit, and the bounds are biased towards the raw share of AfD cells in the "Right" versus "Anti-Elite" areas (39% Right; see Figure [8a](#)).

⁶²See [Shapley \(1953\)](#) for the original reference, or [Guriev et al. \(Forthcoming\)](#); [Longuet-Marx \(2026\)](#) for recent applications in political economy.

showing that repeat AfD voters (i) become both more conservative and more anti-elite over time, and (ii) move faster along the Left–Right dimension.⁶³ Second, as shown earlier, there are consistent within-group patterns of preference evolution—for instance, voters without higher education and young voters becoming increasingly Right-motivated.

Firewall against the far-right. Appendix I applies the estimated 2025 preferences to the firewall against cooperation with the AfD, comparing the two coalitions that were numerically feasible after the 2025 election, CDU–SPD and CDU–AfD, with each coalition assumed to implement the midpoint of its members’ positions. I evaluate them under two criteria: utilitarian welfare, measured by the average weighted distance from voters’ ideal points to each coalition policy, and majoritarian choice, measured by the share of voters closer to one policy than the other. The two criteria disagree. The national welfare bounds straddle zero ($[-0.18, 0.03]$), so neither coalition dominates, although the bounds are strictly negative in the East, favouring CDU–AfD. Yet for every distribution in the identified set a clear majority is closer to the CDU–AfD policy (61.9–72.8% nationally; at least 69.9% in the East and 58.5% in the West). The gap between the two criteria reflects intense opposition to a CDU–AfD coalition among voters on the left, consistent with survey evidence on coalition desirability. Whether the firewall leaves voters underrepresented therefore depends on the aggregation rule, and the tension is sharpest in Eastern Germany.

Approach of Merlo and De Paula (2017). I compare the results obtained with the method developed in this paper, which imposes no distributional assumptions on voters’ bliss points, with those from the approach of Merlo and De Paula (2017), who obtain point estimates of the same spatial model by restricting the bliss-point distribution to a semi-nonparametric class and estimating it by sieve minimum distance. Appendix F re-estimates the results of Section 6.2.2 with their estimator, applied to the same data, demographic groups, and counterfactual masks. Their point estimates lie inside or close to the sharp bounds in 2017–2025 but not in 2013, and display no systematic trend: the dominant motive switches between Anti-Elite and Right from one election to the next. Since the bounds are sharp, a point estimate outside them implies that the fitted distribution is not consistent with the data under the maintained assumptions, suggesting that the admissible bliss-point distributions do not belong to the imposed functional class. This comparison highlights the role that functional-form assumptions can play in this setting.

I also estimate the repositioning game of Section 8 under the point estimates (Figure F2). The AfD’s best reply under this specification does not depend on the move of the CDU and is the same in every column – a step to the right; in the Stackelberg equilibrium both parties move right, and the AfD’s reply undoes the CDU’s unilateral gain, leaving both parties worse off. This is what the functional form delivers: the semi-nonparametric density assigns a sizeable share of West voters to positions far to the right of every party; for the AfD, a step toward that mass is worth more than anything it can gain or lose near the CDU, so its reply is determined by the tail of the fitted density, not by the CDU’s move.

⁶³The results are based on a 2016–2021 panel of all voters, taking their self-reported issue positions as their ideal points; see Figure K23 and Table K2.

10 Conclusion

Is populist support rooted in Left–Right ideology or in protest against the elites? How could parties of the establishment respond to the rise of populism? This paper develops a novel approach to recovering voters’ preferences from aggregate electoral data in a multidimensional spatial voting model, and uses it to address exactly these questions.

The paper offers a framework for unpacking preferences over the bundle of characteristics embodied in any political option by drawing from recent developments in nonparametric demand choice literature (Tebaldi, Torgovitsky and Yang, 2023). Any voting option, whether a candidate, a party, or a referendum answer, can be seen as a product with multiple attributes, and recovering preferences over these attributes is not straightforward. Voting departs from most settings in empirical industrial organization: in a spatial model, individuals hold unobserved ideal points on each dimension, so one cannot impose the linear characteristic-utility relationship standard in that literature. The question “*Do you prefer sweeter or less sweet cereal?*” does not translate into “*Do you prefer a more or less right-wing candidate?*”: voters differ not in how much of an attribute they want, but in where their ideal point lies.

The method developed in this paper is broadly applicable and requires that voters face different sets of options. Such variation arises naturally across districts, in simultaneous elections for several offices, or under ranked-choice voting. The method can further be extended to incorporate turnout, candidate valence, or additional dimensions of conflict.

Applying the method to the rise of the radical-right populist Alternative for Germany (AfD), I find that the party’s support became increasingly grounded in its right-wing ideology rather than its anti-elite stance over 2013–2025. Counterfactuals suggest that in 2025 the main center-right establishment party, the CDU, could capture a substantial share of AfD voters by modestly hardening its Left–Right position, for instance on immigration. I then model the strategic interaction between the AfD and the CDU in 2025, and find that if the CDU hardens its position, the AfD’s best response is to move further right, winning back the voters the CDU would have gained; in equilibrium, the CDU stays put.

The insights of this paper are therefore uncomfortable: accommodation of populist policies may backfire and give rise to an even more radicalized politics. What, then, could establishment parties do? The analysis suggests at least two avenues that future research could explore. First, establishment parties could act on the salience of the anti-elite dimension. In recent years a non-negligible share of AfD voters holds relatively pro-elite attitudes yet still chooses the AfD, because it values the bundle of right-wing and anti-elite rhetoric more than the bundle offered by the CDU. If parties could increase the weight voters attach to the anti-elite dimension, for instance by stressing democratic norms and the role of national and supranational institutions, voting patterns could change. Second, parties, especially while in government, could make governing more transparent and participatory, through measures such as deliberative democracy or inclusive policy-making; both could shift voters’ ideal points towards the pro-elite end. Although derived from entirely different

premises and frameworks, these avenues mirror those in [Guriev and Papaioannou \(2022\)](#) and [Guriev \(2024\)](#).

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A Finite-dimensional problem: sharp bounds on a target parameter

This appendix adapts Propositions 1 and 2 of [Tebaldi, Torgovitsky and Yang \(2023\)](#), and their proofs, to the election model of Section 2. The weight w is fixed, and $\mathbb{X}$ is the MRP of Subsection 2.3, augmented by any hypothetical elections and refined by any regions of the policy space that enter the target parameter; its cells are the sets $\mathcal{X}_j$ of positive Lebesgue measure.

Cell masses. Let Φ denote the set of conditional mass functions on $\mathbb{X}$,

$$\Phi = \left\{ \phi \in \mathbb{R}_+^{d_\phi} : \sum_{\mathcal{X} \in \mathbb{X}} \phi(\mathcal{X} \mid s, d) = 1 \text{ for all } (s, d) \in \text{supp}(S_i, D_i) \right\},$$

where d_ϕ is the cardinality of $\mathbb{X} \times \text{supp}(S_i, D_i)$. For $f \in \mathcal{F}$, define $\phi_f \in \Phi$ by

$$\phi_f(\mathcal{X} \mid s, d) = \int_{\mathcal{X}} f(x \mid s, d) dx \quad \text{for all } \mathcal{X} \in \mathbb{X}, (s, d) \in \text{supp}(S_i, D_i). \quad (\text{A1})$$

Recall that $\mathbb{X}_j(e)$ denotes the cells contained in $\mathcal{X}_j(e)$. Since $\mathcal{X}_j(e)$ is the union of these cells up to a set of measure zero, the predicted shares of Subsection 2.2 depend on f only through ϕ_f , and (3) is equivalent to

$$\sigma_j(m) = \sum_d q_{m,d} \sum_{\mathcal{X} \in \mathbb{X}_j(e(m))} \phi_f(\mathcal{X} \mid s(m), d) \quad \text{for all } j \in \mathcal{J}_{e(m)} \text{ and all municipalities } m. \quad (\text{A2})$$

Representability of the target parameter. Assume there exists a known function $\bar{\theta} : \Phi \rightarrow \mathbb{R}^{d_\theta}$ such that

$$\theta(f) = \bar{\theta}(\phi_f) \quad \text{for all } f \in \mathcal{F}. \quad (\text{A3})$$

Since Φ depends on the MRP, and the MRP on the elections and regions used to build it, (A3) can always be satisfied by including in $\mathcal{E}$ every hypothetical election relevant for the target parameter and refining $\mathbb{X}$ by every region that enters it. In the leading example, $\bar{\theta}(\phi) = \phi(\mathcal{X}_1 \mid s, d) + \phi(\mathcal{X}_2 \mid s, d) + \phi(\mathcal{X}_3 \mid s, d)$.

Let $\Phi^* = \{\phi \in \Phi : \phi \text{ satisfies (A2)}\}$ denote the cell-mass functions consistent with the returns.

Proposition 1. Suppose (A3) holds. For $t \in \mathbb{R}^{d_\theta}$, let $\Phi^*(t) = \Phi^* \cap \{\phi \in \Phi : \bar{\theta}(\phi) = t\}$. Then $t \in \Theta^*$ if and only if $\Phi^*(t) \neq \emptyset$.

Proof. Suppose $t \in \Theta^*$, so that $\theta(f) = t$ for some $f \in \mathcal{F}^*$. Then $\phi_f \in \Phi$, since $\mathbb{X}$ is a partition of $\mathcal{R}$ up to a set of measure zero and $f(\cdot \mid s, d)$ is a density on $\mathcal{R}$; ϕ_f satisfies (A2), since f satisfies (3); and $\bar{\theta}(\phi_f) = \theta(f) = t$ by (A3). Hence $\phi_f \in \Phi^*(t)$.

Conversely, let $\phi \in \Phi^*(t)$ and define

$$\tilde{f}(x \mid s, d) = \sum_{\mathcal{X} \in \mathbb{X}} \frac{\mathbf{1}\{x \in \mathcal{X}\}}{\lambda(\mathcal{X})} \phi(\mathcal{X} \mid s, d),$$

where λ is Lebesgue measure on $\mathbb{R}^2$; every cell has positive measure by construction. The function $\tilde{f}(\cdot \mid s, d)$ distributes the mass $\phi(\mathcal{X} \mid s, d)$ uniformly over each cell. Because the cells are disjoint up to measure zero, $\int_{\mathcal{X}} \tilde{f}(x \mid s, d) dx = \phi(\mathcal{X} \mid s, d)$ for every $\mathcal{X} \in \mathbb{X}$, so $\phi_{\tilde{f}} = \phi$ and $\int_{\mathcal{R}} \tilde{f}(x \mid s, d) dx = \sum_{\mathcal{X} \in \mathbb{X}} \phi(\mathcal{X} \mid s, d) = 1$; together with non-negativity, $\tilde{f} \in \mathcal{F}$. Since $\mathcal{X}_j(e(m))$ is the union of the cells in $\mathbb{X}_j(e(m))$ up to a set of

measure zero,

$$\sigma_j(m; \tilde{f}) = \sum_d q_{m,d} \sum_{\mathcal{X} \in \mathbb{X}_j(e(m))} \phi(\mathcal{X} \mid s(m), d) = \sigma_j(m)$$

by (A2), so $\tilde{f}$ satisfies (3) and $\tilde{f} \in \mathcal{F}^*$. Finally, $\theta(\tilde{f}) = \bar{\theta}(\phi_{\tilde{f}}) = \bar{\theta}(\phi) = t$ by (A3), so $t \in \Theta^*$. $\square$

Proposition 2. *Suppose the model is correctly specified, so that $\Phi^* \neq \emptyset$. If $\bar{\theta}$ is continuous on Φ , then Θ^* is compact and connected. In particular, if $d_\theta = 1$, then $\Theta^* = [t_{lb}^*, t_{ub}^*]$, where*

$$t_{lb}^* = \min_{\phi \in \Phi^*} \bar{\theta}(\phi), \quad t_{ub}^* = \max_{\phi \in \Phi^*} \bar{\theta}(\phi).$$

Proof. Φ is a product of simplices, hence compact and convex. The equations (A2) are linear in ϕ , so Φ^* is a nonempty compact convex polytope, hence connected. By Proposition 1, $\Theta^* = \bar{\theta}(\Phi^*)$, the image of a compact connected set under a continuous map, hence compact and connected. If $d_\theta = 1$, a compact connected subset of $\mathbb{R}$ is a closed interval, whose endpoints are the minimum and maximum of $\bar{\theta}$ over Φ^* . $\square$

When $\bar{\theta}$ is linear in ϕ , as for every target parameter in the application, both bounds are solutions of linear programs.

B Identification of dimension weights

This section establishes that the identified set for the relative weight in the two-dimensional weighted spatial voting model of Section 2 is contained in a closed interval of $\mathbb{R}_+$ that contains the true weight.

The intuition behind the result is as follows. The weight w determines which cells of the MRP are non-empty, and weights are therefore identified only up to the differences in the cells they make available. The argument has two steps. First, a single cell is non-empty only for weights in an interval. Second, any weight consistent with the data must make every cell the data require non-empty; the set of such weights is therefore an intersection of availability intervals, which is again an interval.

B.1 Setting and notation

I use the model of Section 2, in which the single parameter is the weight $w > 0$ of the second dimension in $d_w(x, y)^2 = (x_1 - y_1)^2 + w(x_2 - y_2)^2$. For this appendix I take the policy space to be the box $\mathcal{R} = [a_1, b_1] \times [a_2, b_2]$. The attraction region of candidate j in election e at weight w is $\mathcal{X}_{j;w}(e) = \{x \in \mathcal{R} : d_w(x, Y_{ej}) \leq d_w(x, Y_{ek}) \ \forall k \neq j\}$.⁶⁴

Superimposing the attraction regions of every $e \in \mathcal{E}$ at weight w yields the minimal relevant partition $\mathbb{X}_w$ of Subsection 2.3. A cell of $\mathbb{X}_w$ is indexed by a *voting profile*

$$\mathbf{j} = (j_e)_{e \in \mathcal{E}'}, \quad j_e \in \mathcal{J}_e, \quad \mathcal{E}' \subseteq \mathcal{E},$$

recording the candidate chosen in each election in $\mathcal{E}'$, and equals the intersection

$$\mathcal{X}_w(\mathbf{j}) = \bigcap_{e \in \mathcal{E}'} \mathcal{X}_{j_e;w}(e). \tag{B1}$$

⁶⁴Availability below is defined by non-emptiness with weak inequalities, whereas Section 2 defines attraction regions through the strict argmin. If the cell has positive Lebesgue measure for some w , the set of such weights is the same system with strict inequalities, an open interval whose closure is $\mathcal{W}(\mathbf{j})$ below; the two differ only at the endpoints, which is immaterial for the containment in Proposition 3.

Taking $\mathcal{E}' = \mathcal{E}$ gives the cells of $\mathbb{X}_w$; a proper subset $\mathcal{E}'$ gives a *partial* profile, which indexes a union of such cells. The Fréchet constraints of Subsection 2.4 concern pairs of elections, $|\mathcal{E}'| = 2$; everything below applies to full and partial profiles alike. The profile $\mathbf{j}$ is *available* at weight w when $\mathcal{X}_w(\mathbf{j}) \neq \emptyset$; only available profiles enter the prediction of observed vote shares (cf. (A2)). A partition $\mathbb{X}_w$ is thus summarised by its set of available profiles,

$$\mathcal{C}(\mathbb{X}_w) = \{\mathbf{j} : \mathcal{X}_w(\mathbf{j}) \neq \emptyset\},$$

and two weights are observationally equivalent whenever they induce the same $\mathcal{C}(\mathbb{X}_w)$.

The cell as a system of linear constraints. By (B1), $\mathcal{X}_w(\mathbf{j})$ is an intersection of half-planes with $\mathcal{R}$: one half-plane per election $e \in \mathcal{E}'$ and rival $k \neq j_e$, for a total of $R = \sum_{e \in \mathcal{E}'} (|\mathcal{J}_e| - 1)$ constraints. The set of bliss points at least as close to an own position p as to a rival p' is the half-plane $\{x : d_w(x, p)^2 \leq d_w(x, p')^2\}$, which after expansion reads $2(p'_1 - p_1)x_1 + 2w(p'_2 - p_2)x_2 \leq (p'_1{}^2 - p_1^2) + w(p'_2{}^2 - p_2^2)$. Indexing the constraints by $r = 1, \dots, R$, each takes the form

$$\alpha_r x_1 + w \beta_r x_2 \leq \gamma_r + w \delta_r, \quad r = 1, \dots, R, \quad (\text{B2})$$

with fixed constants $\alpha_r = 2(p'_1 - p_1)$, $\beta_r = 2(p'_2 - p_2)$, $\gamma_r = p_1'^2 - p_1^2$, $\delta_r = p_2'^2 - p_2^2$, where for constraint r arising from election e and rival k the own position is $p = Y_{eje}$ and the rival is $p' = Y_{ek}$. The support adds $a_1 \leq x_1 \leq b_1$ and $a_2 \leq x_2 \leq b_2$. The key structural feature is that w enters (B2) only linearly, both in the coefficient of x_2 and in the threshold, and the same scalar w is shared across all R constraints.

The *availability set* of a profile $\mathbf{j}$ is the set of weights at which its cell is non-empty:

$$\mathcal{W}(\mathbf{j}) = \{w > 0 : \mathcal{X}_w(\mathbf{j}) \neq \emptyset\} = \{w > 0 : \exists x \in \mathcal{R} \text{ satisfying (B2) } \forall r\}. \quad (\text{B3})$$

B.2 Set identification of weights

Lemma 1. *For every voting profile $\mathbf{j}$, the availability set $\mathcal{W}(\mathbf{j})$ is an interval, closed in $(0, \infty)$.*

Proof. Fix $w > 0$ and apply the change of variables $(z_1, z_2) = (x_1, wx_2)$, with inverse $x_2 = z_2/w$. For each fixed $w > 0$ this is a bijection on $\mathbb{R}^2$, so $\mathbf{j}$ is available at w if and only if the transformed system

$$\alpha_r z_1 + \beta_r z_2 \leq \gamma_r + w \delta_r, \quad r = 1, \dots, R, \quad a_1 \leq z_1 \leq b_1, \quad wa_2 \leq z_2 \leq wb_2, \quad (\text{B4})$$

admits a solution $z \in \mathbb{R}^2$. The substitution linearises the bilinear term $w\beta_r x_2$ into $\beta_r z_2$, so w now appears only in the thresholds, including those of the support. Define the augmented set in (z_1, z_2, w) -space

$$\Omega = \left\{ (z_1, z_2, w) \in \mathbb{R}^3 : \alpha_r z_1 + \beta_r z_2 - w \delta_r \leq \gamma_r, \quad r = 1, \dots, R; \quad a_1 \leq z_1 \leq b_1; \quad wa_2 \leq z_2 \leq wb_2 \right\}. \quad (\text{B5})$$

Each defining inequality is linear in the joint variable (z_1, z_2, w) , so Ω is a polyhedron in $\mathbb{R}^3$.

Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the projection $\pi(z_1, z_2, w) = w$. By construction,

$$\mathcal{W}(\mathbf{j}) = \pi(\Omega) \cap (0, \infty).$$

The projection of a polyhedron is a polyhedron (Fourier–Motzkin elimination of z_1, z_2), so $\pi(\Omega)$ is a closed interval of $\mathbb{R}$, and $\mathcal{W}(\mathbf{j})$ is an interval, closed in $(0, \infty)$. The elimination writes $\mathcal{W}(\mathbf{j}) = \{w > 0 : \underline{w}(\mathbf{j}) \leq w \leq \bar{w}(\mathbf{j})\}$ explicitly, with endpoints that are ratios of nonnegative linear combinations of the constants in (B2) and of a_ℓ, b_ℓ . $\square$

Proposition 3. *Let $\mathcal{C}^+$ be any finite collection of voting profiles whose cells are required to carry positive mass. Then the set of weights at which every profile in $\mathcal{C}^+$ is available,*

$$\mathcal{W}(\mathcal{C}^+) \equiv \bigcap_{\mathbf{j} \in \mathcal{C}^+} \mathcal{W}(\mathbf{j}),$$

is an interval, closed in $(0, \infty)$. Moreover, if the data are generated at the true weight w_0 , then $w_0 \in \mathcal{W}(\mathcal{C}^+)$, and the identified set of weights satisfies $\mathcal{W}^ \subseteq \mathcal{W}(\mathcal{C}^+)$.*

Proof. By Lemma 1, each availability set $\mathcal{W}(\mathbf{j})$ is a closed interval, and a finite intersection of closed intervals is again a closed interval; since $\mathcal{E}$ is finite, $\mathcal{C}^+$ is finite. If the data are generated at w_0 , they are generated by a distribution supported on the cells of $\mathbb{X}_{w_0}$, so any cell that the data require to carry positive mass is available at w_0 ; hence $w_0 \in \mathcal{W}(\mathbf{j})$ for every $\mathbf{j} \in \mathcal{C}^+$, and $w_0 \in \mathcal{W}(\mathcal{C}^+)$. Finally, any weight $w \in \mathcal{W}^*$ must rationalise the data, which requires every cell in $\mathcal{C}^+$ to be non-empty at w ; therefore $\mathcal{W}^* \subseteq \mathcal{W}(\mathcal{C}^+)$. $\square$

Outer versus sharp set. $\mathcal{W}(\mathcal{C}^+)$ is an outer set: availability of the required cells does not guarantee that the mass the data leave unassigned can be placed on the cells available at w . Let $\mathcal{C}^*$ be the cells of $\mathbb{X}_{w_0}$ carrying positive mass under f and $I^* = \bigcap_{\mathbf{j} \in \mathcal{C}^*} \mathcal{W}(\mathbf{j})$. Because the predicted shares in (C2) depend only on the masses of profiles and not on the geometry of the cells at w , the true mass function reproduces the data at every w in the interior of I^* , so $I^* \subseteq \mathcal{W}^* \subseteq \mathcal{W}(\mathcal{C}^+)$. The outer set is therefore sharp whenever $\mathcal{W}(\mathcal{C}^+) \subseteq I^*$, that is, when the cells that determine the endpoints of I^* are among those the data force. This becomes more plausible as the number of elections grows: each additional election adds constraints of the form $\sigma_i(e) + \sigma_j(e') > 1$ and forces new cells, so $\mathcal{C}^+$ expands and $\mathcal{W}(\mathcal{C}^+)$ shrinks towards $\mathcal{W}^*$ (see the simulations in Appendix D). Since $\mathcal{C}^*$ is unobserved the condition cannot be verified; the grid search approach in the paper approximates $\mathcal{W}^*$ directly.

C Target Parameter Bounds With Weights

This section demonstrates that one can separately identify the weight w and the mass function of bliss points ϕ and derive the sharp bounds for the target parameters using both.⁶⁵

The two observational-equivalence conditions (3) and (A2), for f and ϕ respectively, can now be recast with the weight made explicit. The predicted vote share of candidate j in municipality m is

$$\sigma_j(m; f, w) = \sum_d q_{m,d} \int_{\mathcal{X}_{j;w}(e(m))} f(x | s(m), d) dx,$$

where $\mathcal{X}_{j;w}(e) = \{x \in \mathcal{R} : d_w(x, Y_{ej}) < d_w(x, Y_{ek}) \ \forall k \neq j\}$. It matches the data if

$$\sigma_j(m; f, w) = \sigma_j(m) \quad \text{for all } j \in \mathcal{J}_{e(m)} \text{ and all municipalities } m. \quad (\text{C1})$$

Likewise, for the equivalent finite-dimensional problem, denote by $\mathbb{X}_w$ the MRP generated by $\mathcal{E}$ under weight

⁶⁵The closest argument in the literature is Merlo and De Paula (2017), who prove nonparametric point identification of weights and distributions of bliss points in a spatial voting model. Their analysis relies on a large support assumption for candidate positions. In particular, to establish joint identification of bliss points and weights, they rely on the results of Ichimura and Thompson (1998), who prove nonparametric identification in a binary discrete choice model using the Cramér–Wold device: there exists a two-candidate configuration such that the affine hyperplane separating the two candidates delivers different election outcomes under two distinct preference distributions. Building on these results, they extend them to 1) include multiple candidate positions, and 2) introduce weights. Both 1) and 2) are based on the large support assumption, i.e. that one could sample arbitrary candidate configurations from $\mathbb{R}^2$. This assumption is infeasible in political-economy environments, where the set of candidate positions is typically finite and confined to a bounded subset of $\mathbb{R}^2$. Hence, the setup of this paper accommodates the fact that there are just finitely many observed configurations in a bounded policy space.

w , by $\mathbb{X}_{j;w}(e)$ its cells contained in $\mathcal{X}_{j;w}(e)$, and by Φ_w the set of conditional mass functions on $\mathbb{X}_w$; then

$$\sum_d q_{m,d} \sum_{\mathcal{X} \in \mathbb{X}_{j;w}(e(m))} \phi(\mathcal{X} \mid s(m), d) = \sigma_j(m) \quad \text{for all } j \in \mathcal{J}_{e(m)} \text{ and all municipalities } m. \quad (\text{C2})$$

The algorithm for obtaining bounds on the target parameter in the presence of an unknown weight is then the following. First, obtain the identified set of weights over a candidate space $\mathcal{W} \subseteq \mathbb{R}_+$,

$$\mathcal{W}^* = \{w \in \mathcal{W} : \exists \phi \in \Phi_w \text{ satisfying } (\text{C2})\}.$$

Second, for each $w \in \mathcal{W}^*$ obtain $\Theta^*(w)$ by applying Propositions 1–2 to the partition $\mathbb{X}_w$. The identified set for the target parameter is then $\Theta^* = \bigcup_{w \in \mathcal{W}^*} \Theta^*(w)$. Since each $\Theta^*(w)$ is an interval, Θ^* is contained in

$$\left[\min_{w \in \mathcal{W}^*} t_{lb}^*(w) ; \max_{w \in \mathcal{W}^*} t_{ub}^*(w) \right],$$

and these are the bounds I report. Appendix G describes this estimation procedure in the application.

D Weights estimation: simulation

This section provides a simulation comparing the weight estimator based on 1) the method used in this paper, and 2) an approach that allows for a simultaneous recovery of bliss points and weights from Merlo and De Paula (2017).

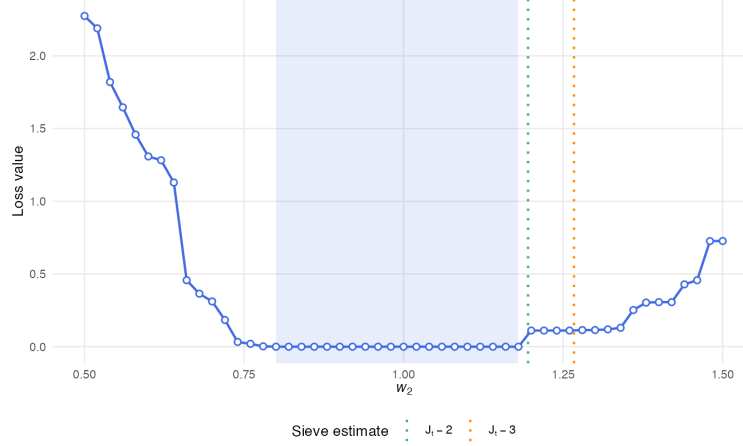
I simulate a population of 10,000 voters voting in twenty-four elections with six candidates drawn at random. Bliss points are drawn from a mixture of two dominant clusters and a diffuse background. Voters choose the candidate closest to their bliss point under the weighted Euclidean metric with true weight $w_2 = 1$; the researcher observes only the resulting candidate vote shares in each election.

For each candidate value of w_2 on a grid, the implied superimposed Voronoi tessellation determines the set of feasible cells (joint vote patterns). The question then is whether voter mass can be distributed across these cells so as to reproduce the observed shares exactly, subject to masses being non-negative and summing to one. The set of weights for which an exact fit exists is the identified set: since every cell is feasible only on an interval of weights, and concentrated vote shares force particular cells to carry mass, the weights consistent with the data are contained in an interval around the truth; in this simulation, the exact-fit set is itself an interval.

Figure D1 plots the residual fit against w_2 . The loss is exactly zero on the interval $[0.80, 1.18]$, which contains the true value $w_2 = 1$, and rises outside it. The cell geometry alone thus set-identifies the weight without any assumption on the distribution of bliss points.

The dotted lines mark the estimates obtained from the sieve estimator at polynomial orders $J_t = 2, 3$, which impose a flexible parametric form on the bliss-point density. Because a low-order polynomial correction of a Gaussian base cannot reproduce the distribution of voters' bliss points, the resulting weight estimates are far from the truth as well, at 1.20 and 1.27 for $J_t = 2, 3$ respectively.

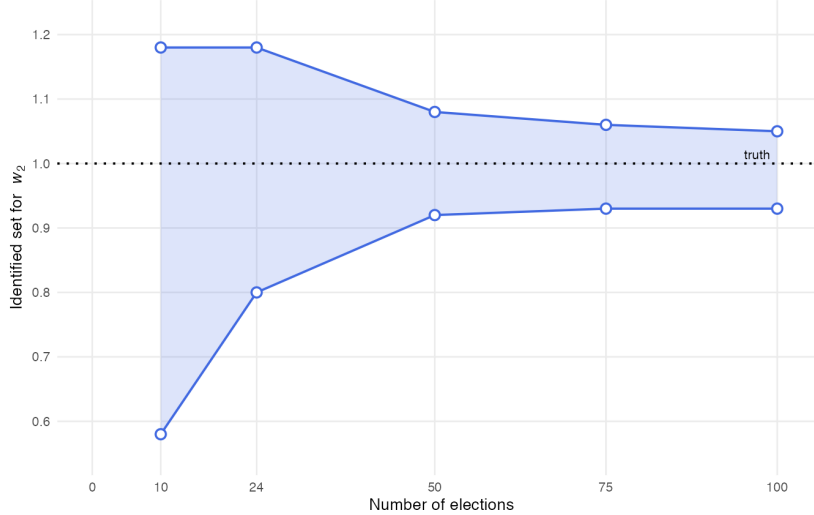
Number of Elections. The identified set shrinks as the number of elections grows. Because each additional election contributes a further set of cells that the observed shares may or may not be able to populate, more elections impose more restrictions on which weights admit an exact fit. Figure D2 traces the identified set for w_2 against the number of elections, holding the data-generating process fixed: the set narrows monotonically around the true value, from $[0.58, 1.18]$ with ten elections to $[0.93, 1.05]$ with one hundred. Note that while



Notes: Monte Carlo with 10,000 voters, true $w_2 = 1$, twenty-four six-candidate elections; bliss points follow a two-cluster mixture with a diffuse background, and only vote shares are observed. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight. The shaded region is the identified set $[0.80, 1.18]$, where the fit is exact; it contains the truth. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.20, 1.27$), both of which lie outside the identified set.

Figure D1. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017)

this simulation illustrates that the size of the identified set shrinks with the number of elections, its size at any given number of elections depends on the positions of the candidates and the distribution of preferences.

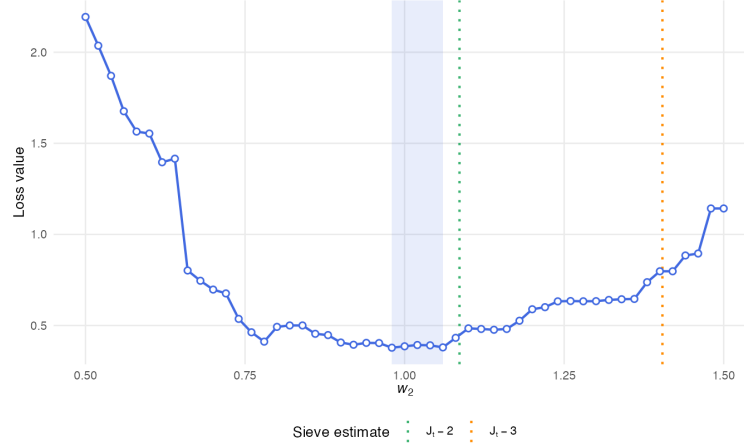


Notes: Each point pair gives the lower and upper bounds of the identified set for w_2 (the weights admitting an exact fit), computed as in Figure D1 but varying the number of elections. The dotted line marks the true value $w_2 = 1$.

Figure D2. Identified set for the weight at different numbers of elections

Noisy perceptions of candidate positions. Realistically, however, there is always noise in the data. I next simulate the behaviour of both estimators under the assumption that each voter observes every candidate's position with an idiosyncratic mislocation error, distributed $\mathcal{N}(0, 0.1^2 I)$, while the estimation procedures remain unchanged. Under noise an exact fit is no longer attainable at any weight. The question, however, is where the minimum is located and how large the basin delivering a near-identical fit is. The region of near-minimal losses becomes *smaller*: the loss now develops a strict interior minimum, and the near-feasible

set (weights whose fit is within one percentage point of share mass of the best attainable) shrinks from the noise-free interval $[0.80, 1.18]$ to $[0.98, 1.06]$, still containing the truth (Figure D3). Intuitively, the smoothed shares are best approximated by the cell structure of the true weight, so mild noise converts the flat exact-fit region into a well-defined basin centered near $w_2 = 1$. The sieve estimator yields $\hat{w}_2 = 1.09$ and 1.40 at $J_t = 2, 3$, both falling outside the near-feasible set. Note that in this simulation the results obtained with $J_t = 2$ are closer to the truth; this suggests that additional flexibility in the estimator does not necessarily translate into more precise estimates.



Notes: Same setup as Figure D1, except that each voter perceives every candidate’s position with an idiosyncratic $\mathcal{N}(0, 0.1^2 I)$ mislocation error. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight; with noise the loss is strictly positive everywhere. The shaded region is the near-feasible set $[0.98, 1.06]$ (fit within 0.01 of the minimum), which contains the true value $w_2 = 1$. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.09, 1.40$), both outside the near-feasible set.

Figure D3. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017) with voter perception noise

E Stated preferences

E.1 Preference bins

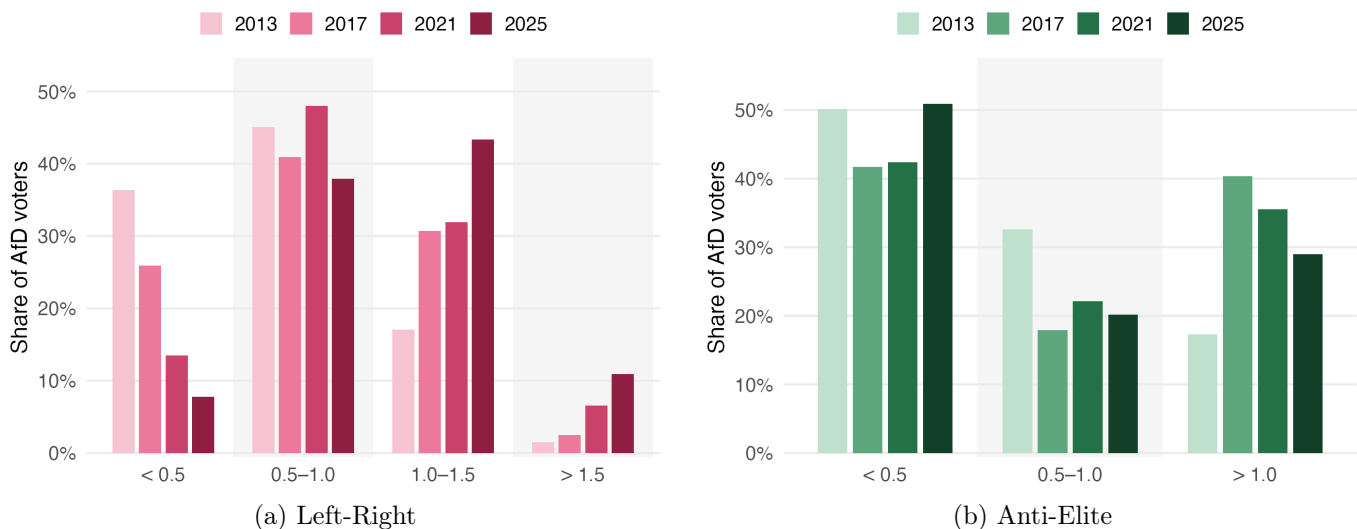
To see whether the evolution of the AfD electorate implied by the model is visible in stated attitudes, I locate the respondents of the GLES election surveys on the two dimensions. AfD voters are respondents who report an AfD party vote/party-vote intention. All shares use the GLES socio-demographic and regional weights. Voters are placed on the Left-Right and Anti-Elite dimensions with a procedure and variables similar to those used for the candidates; voter positions are hence comparable to the candidate positions conceptually, but remain different quantitatively because of the questions and samples used.

Left-Right. The questions on the Left-Right dimension in the voter surveys are many and similar to the ones in the candidate surveys. The common items are the left-right self-placement, the immigration scale, the taxes-versus-social-services scale (2021 and 2025), and the issue statements on immigrants adapting to German culture, the state staying out of the economy and reducing income differences, complemented in single years by same-sex marriage, immigration being good for the economy, defense spending and support in the euro crisis. Voter positions are therefore well reproducible from the common items among the candidate and voters surveys.

Anti-Elite. Most anti-elite items of the candidate survey have no voter counterpart, so the Anti-Elite position of voters is built from the voter survey alone. It is the mean of three standardized items: (i) “the

differences between the elite and the people are larger than the differences among the people” (2017–2025); (ii) “citizens should be able to call referendums at the federal level” (2013–2021; in 2025, when the item was not asked, it is replaced by “the people, not the politicians, should make the most important political decisions”); and (iii) “parties are only interested in votes, not in the opinions of voters” (all years).

Figure E1 reports the share of AfD voters in fixed bins of each dimension. On the Left-Right dimension, the electorate shifts steadily to the right: the share below 0.5 falls from 36% in 2013 to 26%, 13% and 8% in the following elections, while the share above 1.0 rises from 19% to 33%, 38% and 54%. On the Anti-Elite dimension, 2013 stands apart (and it is also measured with a different set of items). Among 2017–2025, the most recent election is clearly the least anti-elite for the AfD electorate: 51% of AfD voters lie below 0.5 in 2025, against 42% in 2017 and 2021, and the share above 1.0 falls from 40% in 2017 to 29% in 2025.



Notes. Bars show the weighted share of AfD voters in each bin; shares sum to 100% within a year. Data: GLES Voters Surveys, 2013–2025. AfD voters report an AfD party vote after the election or, an AfD party-vote intention before the election. Panel (a): the distribution of the Left-Right positions among AfD voters. Panel (b): the distribution of the Anti-Elite positions among AfD voters.

Figure E1. Distribution of AfD voters across fixed bins of the Left-Right and Anti-Elite dimensions, GLES voter survey 2013–2025.

E.2 Motivations Exercise

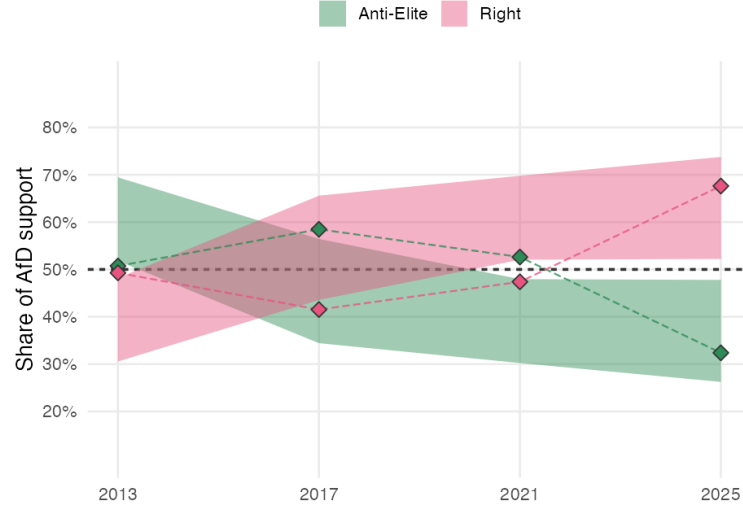
This section runs an exercise similar to the one in Section 6.2.2 using the stated-preference positions of Section E.1.

Before turning to the exercise and its results, I want to stress that recovering the weights from these data remains infeasible, so I take the estimated weights from the analysis in this paper. This is so for at least two reasons. First, the survey samples of AfD voters are small (fewer than 500 in every election before 2025, and 88 in 2013), and the positions are stated rather than revealed through a choice among the local candidates. Second, the problem of voters adjusting their preferences so as to be closer to the party is especially acute for weight estimation, since such closeness would be read as evidence that a dimension is important—which may be precisely the opposite of the truth if individuals are adjusting their stated positions on the issues that matter least to them.

Each AfD voter’s bliss point is given by the Left-Right and Anti-Elite positions of Section E.1. The AfD position is given by the national AfD candidate medians used in the main exercise. To place voters on the candidate Anti-Elite scale, the voter index is rescaled to match the mean and standard deviation of a

candidate-scale score built from the two anti-elite items common to both surveys (referendums and satisfaction with democracy); since voters are placed on the candidate Left-Right scale, the Left-Right coordinate applies directly.

Figure E2 plots the stated-preference shares over the results from Figure 8b. The survey thus reproduces the shift of the AfD electorate towards the Right region by 2025, but not the earlier change between 2013 and 2017; the latter might reflect the small 2013 survey sample and the different anti-elite items available that year, as well as voters' preference misreporting (in particular, not reporting radical-right views). The main takeaway of this exercise is that the identified patterns are broadly consistent with the survey results.



Notes: Survey benchmark constructed from the GLES surveys of Figure E1 (2013, 2017, 2021 and 2025). Each AfD voter's bliss point is given by the Left-Right position (candidate scale) and the Anti-Elite index of Section E.1. The AfD position is the national AfD candidate median; the two counterfactual positions lie one unit away from it, as in Section 6.2.2. AfD voters are assigned to the closer counterfactual position under the region-year weights w_{AE} estimated in the main analysis. The resulting shares of AfD voters are point estimates, not partial-identification bounds. The estimates are weighted throughout using the GLES socio-demographic and regional weights. The bands are the main estimates without second-choice moments.

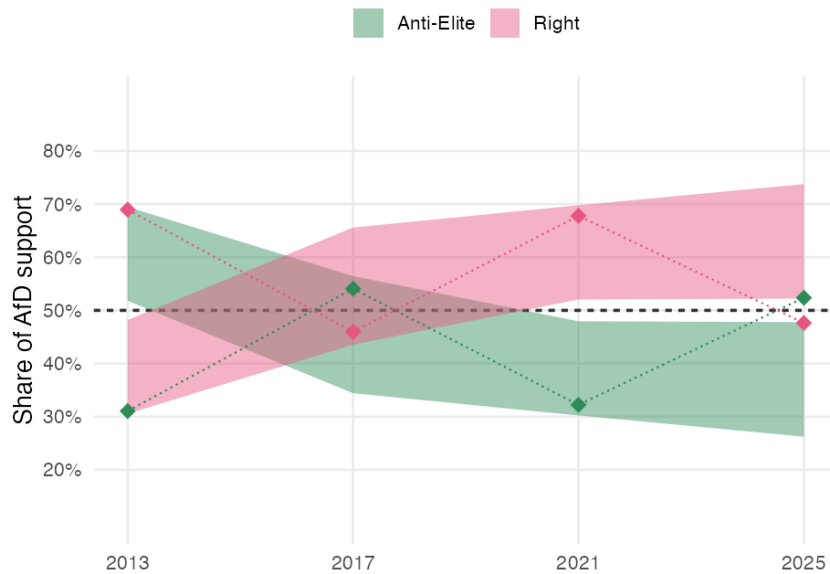
Figure E2. Comparison to the survey results

F Comparison with the Merlo–de Paula point estimates

The bounds of Section 6.2.2 are derived under no assumptions on the functional form of the distributions of bliss points. Merlo and De Paula (2017) instead obtain point estimates of the same spatial model by restricting the distribution to the semi-nonparametric (SNP) class of Gallant and Nychka (1987) and estimating it by sieve minimum distance. This section applies their estimator to the same data and target parameters. The bliss-point density of each of the 24 demographic groups is an SNP density, $f_g(t) \propto P_g(z)^2 \phi(z)$ with $z = R^{-1}(t - b_g)$, where P_g is a polynomial of degree at most two, b_g is the group mean and R a common scale matrix; the municipal vote shares are the mixture of the group densities integrated over the parties' areas of attraction, with the turnout-adjusted demographic composition as mixture weights, exactly as in the linear program. The weight matrix W is estimated jointly. Following Merlo and De Paula (2017), group means are anchored at the party centroids with a bounded deviation. The AfD composition shares are then computed from the estimated densities under the same counterfactual masks as in the main text.

Figure F1 overlays the resulting point estimates on the sharp bounds. Their point estimates lie inside or close to the sharp bounds in 2017-2025, but not in 2013, and display no systematic trend: the dominant motive switches between Anti-Elite and Right from one election to the next. They should be read as an illustration of what the parametric restriction adds rather than as an alternative estimate: the SNP class

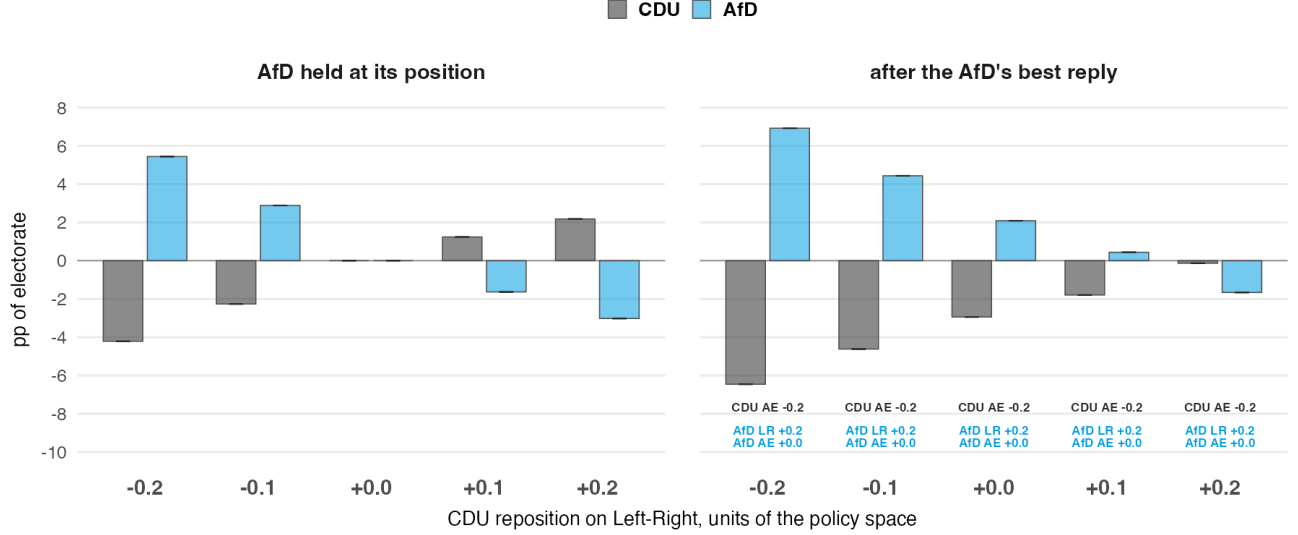
fixes the shape of each group's density, so the point estimates depend on the anchoring and on the polynomial order, whereas the bounds do not.



Notes. Bands are the sharp bounds of Figure 8b. Diamonds are the sieve minimum-distance point estimates of the same shares under the SNP restriction of Merlo and De Paula (2017) (anchored group means, polynomial order two), aggregated over East and West with AfD-vote weights. Estimation uses the same municipality sample, demographic groups and counterfactual masks as the bounds; the weight matrix is estimated jointly and agrees with the grid-searched w_{AE} of the main text.

Figure F1. Merlo-de Paula point estimates vs. sharp bounds

Figure F2 shows the net gains of the repositioning game for both CDU and AfD using fitted point densities.



Notes. Counterpart of Figure 11 with the bliss-point distribution replaced by the sieve minimum-distance point estimates of Merlo and De Paula (2017). Left: the CDU moves on the Left–Right axis, the AfD stands still. Right: the CDU pairs each Left–Right move with its best anti-elite move (restricted to steps away from anti-elite rhetoric), and the AfD replies on both dimensions (anti-elite steps upward only); the profile under each pair of bars is the AfD’s reply.

Figure F2. The repositioning game under the Merlo–de Paula point estimates

G Model Estimation

This section translates the empirical restrictions of the spatial voting model into finite-dimensional linear programmes. Section G.1 introduces the objects common to all programmes, Section G.2 covers the estimation of the relative dimension weight, and Section G.3 covers the estimation of the bounds on the target parameters given that weight.

G.1 Setup and notation

Estimation is carried out separately for each region (East, West) and year, within which the weight is fixed across all individuals, while preference distributions differ across demographic groups and states. As discussed in Appendix D, identification of the weight requires substantial variation, which pooling electoral districts within a region delivers.

States, elections and municipalities. As in Section 2, the market is the *state* $s \in \mathcal{S}$ and the submarket the *municipality* $m \in \mathcal{M}$; $\mathcal{M}(s) \subset \mathcal{M}$ collects the municipalities of state s and $s(m)$ denotes the state of m . An *election* $e \in \mathcal{E}$ is the candidate configuration of an electoral district; each municipality belongs to one election $e(m)$, and the number of elections per state is capped at 20. The choice set is the six parties. Demographic groups (types) $d \in \mathcal{D}$ are the 24 age×gender×education bins. For each municipality the data give the number of valid votes cast N_m , the turnout-adjusted share $q_{m,d}$ of group d among the voters who cast them ($\sum_d q_{m,d} = 1$), and the AfD vote share σ_m .⁶⁶ Throughout, the population is the voting population:

⁶⁶ Turnout adjustment. For municipality m the census gives the number of eligible voters N_m^0 and the shares $q_{m,d}^0$ of the demographic groups $d \in \mathcal{D}$ among them, $\sum_d q_{m,d}^0 = 1$. The Representative Electoral Statistics report, for each state s and election year, turnout rates by gender and ten age brackets; I collapse the brackets to the six age groups of the model by averaging the bracket rates and apply the same rate to the education and non-education halves of a group, since the statistic has no education split, which gives $r_{s,d}$. The turnout-adjusted share of group d in municipality m is $q_{m,d} = q_{m,d}^0 r_{s(m),d} / \sum_{d'} q_{m,d'}^0 r_{s(m),d'}$, i.e. the composition of those eligible voters in m who actually vote, with $\sum_d q_{m,d} = 1$; $N_m q_{m,d}$ is then the number of type- d voters in m .

since the model of Section 2 does not include a turnout decision, $f(x \mid s, d)$ and the mass functions below describe the bliss points of type- d voters in state s who cast a vote, and all counts and shares refer to these voters.

Metric and mesh. The weight on the Left–Right dimension is normalised to one, so the weight of Section 2 is $w = w_{AE} \in \mathbb{R}_+$, the only free parameter; w_{AE} reshapes the attraction regions $\mathcal{X}_{j;w}(e)$. Overlaying the attraction regions of all elections in $\mathcal{E}$ yields the minimal relevant partition (MRP). I restrict attention to the AfD attraction region, retain the $n = 500$ largest cells, and merge each smaller fragment into the neighbouring cell with which it shares the longest boundary. The resulting mesh is $\mathbb{X}_w = \{\mathcal{X}_1, \dots, \mathcal{X}_n\}$, indexed by cells ℓ , and is rebuilt at every w_{AE} . Fixing n across weights matters because a weight’s greater explanatory power should reflect the *composition* of the cells, not their *number*.

For municipality m and cell ℓ , let $a_{m,\ell}(w) \in \{0, 1\}$ indicate that the AfD is the nearest party on $\mathcal{X}_\ell$ in election $e(m)$, and $\mathbf{a}_m(w) \in \{0, 1\}^n$ the corresponding vector.

Unknowns and fit. The unknowns are the state-and-group mass functions over the mesh,

$$\boldsymbol{\phi}_{s,d} = (\phi_{s,d,1}, \dots, \phi_{s,d,n})^\top \in \mathbb{R}_+^n, \quad \mathbf{1}^\top \boldsymbol{\phi}_{s,d} = 1, \quad s \in \mathcal{S}, \quad d \in \mathcal{D},$$

so each state has its own bliss-point distribution over the shared mesh; $\boldsymbol{\phi}_{s,d}$ is the distribution of bliss points among the type- d voters of state s who cast a vote. $\boldsymbol{\phi}$ denotes the collection $\{\boldsymbol{\phi}_{s,d}\}$ and Φ the set of such collections. Given $\boldsymbol{\phi}$ and w_{AE} , the predicted AfD share in municipality m is the finite-dimensional analogue of (C2),

$$\sigma_m(\boldsymbol{\phi}; w) = \sum_{d \in \mathcal{D}} q_{m,d} \mathbf{a}_m(w)^\top \boldsymbol{\phi}_{s(m),d}. \quad (\text{G1})$$

In finite samples the exact restriction (C2) generically has no solution. I therefore introduce slacks $u_m \geq 0$ and impose

$$-u_m \leq \sigma_m(\boldsymbol{\phi}; w) - \sigma_m \leq u_m \quad \forall m \in \mathcal{M}. \quad (\text{G2})$$

Let $\mathcal{K}_0(w)$ be the set of pairs $(\boldsymbol{\phi}, \mathbf{u})$ with $\boldsymbol{\phi} \in \Phi$ and $\mathbf{u} \geq 0$ satisfying (G2). All programmes below are linear in $(\boldsymbol{\phi}, \mathbf{u})$ and measure fit by the total absolute error $\mathbf{1}^\top \mathbf{u} = \sum_m u_m$.

G.2 Weights

For each candidate weight, the best attainable fit is the mean absolute municipal AfD-share error,

$$\bar{Q}(w_{AE}) = \frac{1}{|\mathcal{M}|} \min_{(\boldsymbol{\phi}, \mathbf{u}) \in \mathcal{K}_0(w)} \mathbf{1}^\top \mathbf{u}.$$

This linear programme is solved for every weight on the grid

$$\mathcal{W} = \{0.2, 0.3, \dots, 0.9, 1.0\} \cup \{1/0.9, \dots, 1/0.2\},$$

symmetric around 1 in log scale, and the estimate is the fit-minimising weight

$$\hat{w}_{AE} = \arg \min_{w_{AE} \in \mathcal{W}} \bar{Q}(w_{AE}).$$

Values $\hat{w}_{AE} < 1$ mean voters tolerate larger distances on the anti-elite dimension (the left–right dimension weighs more); $\hat{w}_{AE} > 1$ the reverse. Because the data are noisy, the minimiser is generically a single point.

G.3 Bounds on target parameters

This section describes the bounds used for the motivations exercise of Section 6.2.2; the other exercises follow the same logic. The weight is fixed at $\hat{w}_{AE}$ from Section G.2; the mesh and the indicators $\mathbf{a}_m$ are those of Section G.1 evaluated at $\hat{w}_{AE}$. One feature is specific to this exercise: the state-specific masses $\phi_{s,d}$ are pooled onto the single region-wide mesh, and are tied together across states by a loose cross-state coupling restriction. The reference run samples at most 100 municipalities per election with population below 10 thousand inhabitants.

Target parameter. Let $\bar{Y} = (\bar{Y}_{LR}, \bar{Y}_{AE})$ be the AfD position (national median value in the year) on the left–right and anti-elite dimensions, as in Section 6.2.2, and define the reference points

$$C^{AE} = (\bar{Y}_{LR} - 1, \bar{Y}_{AE}), \quad C^{Right} = (\bar{Y}_{LR}, \bar{Y}_{AE} - 1).$$

The AfD attraction region is split into two target areas by which reference point is nearer under $w = \hat{w}_{AE}$: the *anti-elite* area, nearer C^{AE} , and the *right* area, nearer C^{Right} . Let $\mathcal{C}^{AE} \subset \{1, \dots, n\}$ collect the cells that intersect the anti-elite area and $\mathcal{C}^{Right}$ those that intersect the right area; a cell cut by the boundary between the two areas belongs to both sets. The mass of AfD voters in the cells of $\mathcal{C}$ is

$$W(\phi; \mathcal{C}) = \sum_{s \in \mathcal{S}} \sum_{d \in \mathcal{D}} (\mathbf{b}_{s,d}(\mathcal{C}))^\top \phi_{s,d}, \quad \mathbf{b}_{s,d,\ell}(\mathcal{C}) = \mathbf{1}\{\ell \in \mathcal{C}\} \sum_{m \in \mathcal{M}(s)} N_m q_{m,d} a_{m,\ell}, \quad (\text{G3})$$

so that $(\mathbf{b}_{s,d}(\mathcal{C}))^\top \phi_{s,d}$ counts the state- s , group- d voters who vote AfD from an ideal point in a cell of $\mathcal{C}$. The target parameter is the Right share of AfD support,

$$\theta_{Right}(\phi) = \frac{W(\phi; \mathcal{C}^{Right})}{W(\phi; \mathcal{C}^{Right}) + W(\phi; \mathcal{C}^{AE})}, \quad \theta_{AE}(\phi) = 1 - \theta_{Right}(\phi),$$

a ratio of two linear functions of ϕ .

Pooling the states of a region onto one mesh is informative only if their distributions are related, and the coupling restriction imposes this loosely. The idea is that young voters in one state of a region should not hold ideological positions wildly different from young voters in another state of the same region, and likewise for the other broad demographic groups. States are therefore compared only along six coarse marginals, $g \in \{\text{with higher education, without higher education, young, old, male, female}\}$, each a union of demographic groups $d \in g$, and only on a coarse partition of the policy space into blocks B on a 0.5 grid. Let $T_{s,d} = \sum_{m \in \mathcal{M}(s)} N_m q_{m,d}$ be the number of type- d voters in state s , and let $\rho_s(g, B)$ be the share of marginal g in state s with ideal points in block B , i.e. the mass $\sum_{d \in g} T_{s,d} \sum_{\ell \in B} \phi_{s,d,\ell}$ divided by $T_{s,g} = \sum_{d \in g} T_{s,d}$. Let $\bar{\rho}(g, B)$ be the population-weighted regional mean of these shares, an auxiliary variable of the programme. Every state's share must lie within a factor $1 + \tau_{\text{state}}$ of the regional mean,

$$\frac{\bar{\rho}(g, B)}{1 + \tau_{\text{state}}} \leq \rho_s(g, B) \leq (1 + \tau_{\text{state}}) \bar{\rho}(g, B) \quad \forall s \in \mathcal{S}, g, B, \quad (\text{G4})$$

which is linear in $(\phi, \bar{\rho})$. The benchmark uses $\tau_{\text{state}} = 2$: each state's block share may be up to three times the regional mean or as low as a third of it, so the coarse moments between two states of the same region may differ by up to a factor of nine. The coupling is loose: it operates only at the level of coarse marginals and large blocks, and allows large differences across states within each marginal–block cell.

Bounds. Let $\mathcal{K}$ be the set of $(\phi, \mathbf{u}) \in \mathcal{K}_0(\hat{w})$ that satisfy (G4) for some $\bar{\rho}$. I first solve the best-fit programme $\hat{Q} = \min_{(\phi, \mathbf{u}) \in \mathcal{K}} \mathbf{1}^\top \mathbf{u}$, the smallest total absolute error attainable under the coupling restriction. The approximately feasible set then allows the mean absolute error to exceed its minimum by at most

$\epsilon_{\text{abs}} = 10^{-5}$, i.e. $\mathbf{1}^\top \mathbf{u} \leq \hat{Q} + |\mathcal{M}| \epsilon_{\text{abs}}$, as in [Tebaldi, Torgovitsky and Yang \(2023\)](#), and the bounds are

$$\underline{\theta}_{\text{Right}} = \min \theta_{\text{Right}}(\phi), \quad \bar{\theta}_{\text{Right}} = \max \theta_{\text{Right}}(\phi), \quad \text{s.t. } (\phi, \mathbf{u}) \in \mathcal{K}, \quad \mathbf{1}^\top \mathbf{u} \leq \hat{Q} + |\mathcal{M}| \epsilon_{\text{abs}}.$$

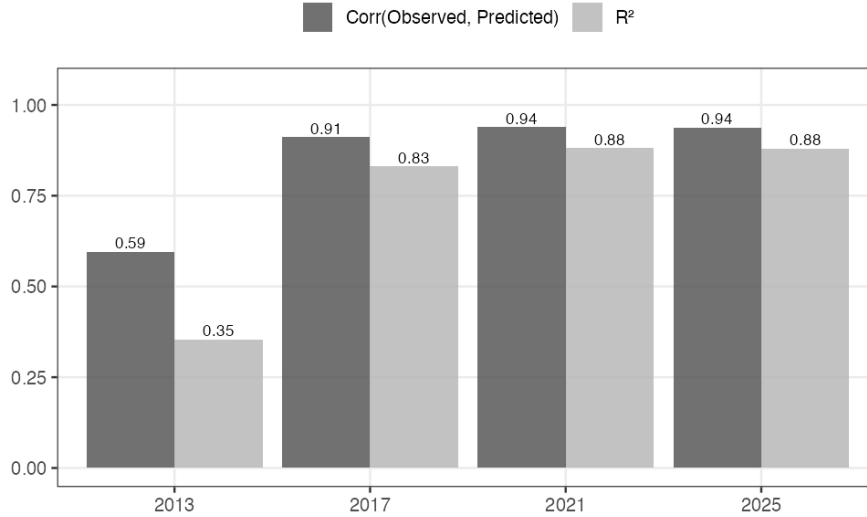
These are linear-fractional programmes, which I solve by bisection on the value of the target parameter: for a candidate value t , some feasible ϕ attains $\theta_{\text{Right}}(\phi) \leq t$ if and only if the linear programme with the additional constraint $W(\phi; \mathcal{C}^{\text{Right}}) \leq t [W(\phi; \mathcal{C}^{\text{Right}}) + W(\phi; \mathcal{C}^{\text{AE}})]$ is feasible, so $\underline{\theta}_{\text{Right}}$ is the smallest such t , found by bisection to the desired precision; $\bar{\theta}_{\text{Right}}$ is obtained symmetrically with the inequality reversed.

G.4 Model Fit

I discuss model fit. Table [J4](#) presents summary statistics on model fit at the municipality level. In every election year, the median residual is exactly 0 by construction. The standard deviation ranges from 0.016 in 2013 to just above 0.03 in 2017 and 2021, and 0.048 in 2025. This increase mirrors an increase in AfD vote shares. The correlation between predicted and observed vote shares is above 0.9 in the later years, and about 0.6 in 2013. Overall, these results suggest a reasonable model fit, especially in the later years.

The correlations of predicted and observed municipal AfD shares with municipalities' demographic characteristics are almost identical (Figure [G2](#)).

To sum up, both the model fit and the observed heterogeneity in the data suggest that the model performs well in explaining observed AfD voting patterns.



Notes. Bars plot the values for correlation between predicted and observed vote shares, and R^2 from a regression of one on another. Predicted AfD vote shares are chosen at a random seed.

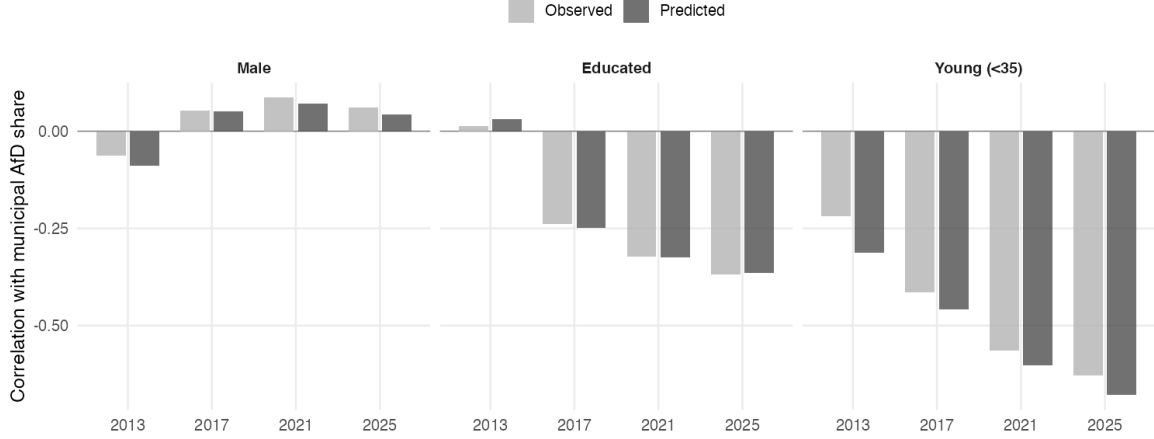
Figure G1. Correspondence between predicted and observed AfD vote shares across municipalities.

G.5 Statistical inference

The target is the *Right share* of AfD support in the benchmark specification (Section [6.2.2](#)). In this section I account for the sampling uncertainty in the sample of municipalities. The analysis was run only for 2021 to ease the computation.

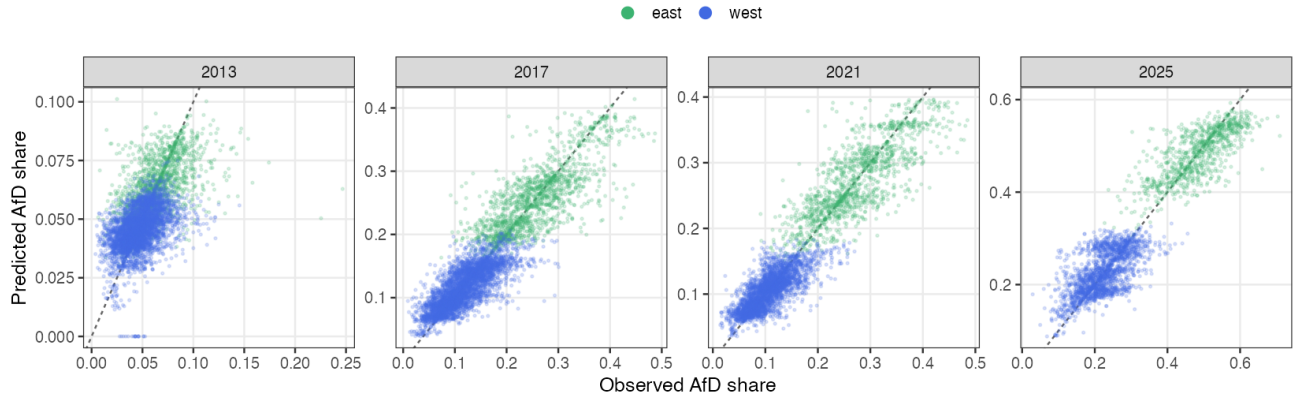
Sampling uncertainty is quantified with a nonparametric bootstrap. Each replication $b = 1, \dots, B$ ($B = 100$):

1. resamples municipality observations with replacement within each (state $\times$ election) group, preserving



Notes. Each bar is an estimate of a correlation between a predicted (observed) AfD vote share and a given municipality demographic characteristic.

Figure G2. Correlation between predicted and observed AfD vote shares with demographic characteristics of municipalities.



Notes. Each dot is a municipality, with the respective values for predicted and observed vote shares. Predicted AfD vote shares are chosen at a random seed.

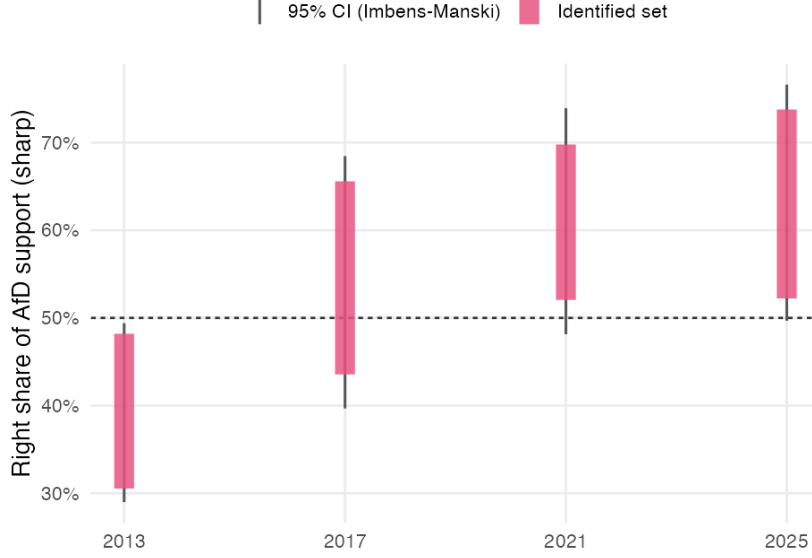
Figure G3. Correspondence between predicted and observed AfD vote shares in Eastern and Western Germany.

the rest of the design;

2. re-solves the full estimator on the resampled data, yielding endpoints $(\hat{\theta}_{(b)}^{lb}, \hat{\theta}_{(b)}^{ub})$.

From the bootstrap distribution of the endpoints I report an interval for the parameter following [Imbens and Manski \(2004\)](#), $[\hat{\theta}^{lb} - c \widehat{se}_{lb}, \hat{\theta}^{ub} + c \widehat{se}_{ub}]$ with c solving $\Phi_{\mathcal{N}}(c + \hat{\Delta} / \max(\widehat{se})) - \Phi_{\mathcal{N}}(-c) = 0.95$, where $\Phi_{\mathcal{N}}$ is the standard normal distribution function and $\hat{\Delta}$ the estimated width of the identified set.

Such 95% CI are presented on the Figure [G4](#); the results remain quantitatively similar to the main results in the paper.



Bars show the sharp identified set for the Right share of AfD support, θ_{Right} , in the main specification without second-choice moments, aggregated over East and West with AfD-vote weights within each bootstrap replication. Whiskers are 95% confidence intervals for the parameter following (Imbens and Manski, 2004). Endpoint standard errors are obtained from a nonparametric bootstrap ($B = 100$) that resamples municipalities with replacement within state-election strata and re-solves the full linear programme on each draw.

Figure G4. Bootstrapped 95% confidence intervals (Imbens–Manski), national

H The Role of the Supply Side: Shapley Values

Let $\theta_{Right}(Pref_t, Comp_t, Pos_t)$ denote the share of AfD voters whose ideal points fall in the *Right* region (Figure 8a), given voter preferences $Pref_t$ across demographic groups, the demographic composition of AfD voters $Comp_t$, and national median party positions Pos_t . I decompose the change in this share between t and t' as follows:

$$\begin{aligned} & \theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) \\ &= \phi(Pref) + \phi(Comp) + \phi(Pos) \end{aligned} \tag{H1}$$

where each Shapley value, e.g. $\phi(Pos)$, is the weighted average marginal contribution of this factor across all 2^2 coalitions of the other two factors:

$$\begin{aligned} \phi(Pos) = & \frac{1}{3} \left[\theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_t) \right] \\ & + \frac{1}{6} \left[\theta_{Right}(Pref_t, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_{t'}, Pos_t) \right] \\ & + \frac{1}{6} \left[\theta_{Right}(Pref_{t'}, Comp_t, Pos_{t'}) - \theta_{Right}(Pref_{t'}, Comp_t, Pos_t) \right] \\ & + \frac{1}{3} \left[\theta_{Right}(Pref_t, Comp_t, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) \right], \end{aligned}$$

and $\phi(Comp)$, $\phi(Pref)$ are defined analogously. The weights $\frac{1}{3}, \frac{1}{6}$ follow from $\frac{|S|!(n-|S|-1)!}{n!}$ with $n = 3$ ($|S| = 0, 2$ give $\frac{1}{3}$; $|S| = 1$ gives $\frac{1}{6}$). By construction $\phi(Pref) + \phi(Comp) + \phi(Pos)$ equals the left-hand side of (H1) exactly.

To evaluate each counterfactual share, I combine three ingredients. Party positions (Pos_t) are the

estimated national median positions in year t ; voter preferences ($Pref_t$) enter as group-specific ideal-point distributions, estimated as a primitive of the model, $f(x | s, d)$.

Composition ($Comp_t$) comes from the Representative Electoral Statistics. These are unique data on voting choices for 12 age–gender groups per federal state.⁶⁷ These data do not permit disaggregation by education, which I therefore omit.

Table H1 presents the results for $\phi(Pos)$ across every pair of consecutive years and regions of the country. The last row provides conservative bounds aggregated over the two parts of the country. Table J8 also presents the Shapley values for the other two channels (composition and preferences), but bounds for those channels are wide and uninformative.

	2013→2017	2017→2021	2021→2025
<i>West</i>			
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
<i>East</i>			
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
<i>National (AfD-electorate weighted, conservative)</i>			
Positions	[−0.188, −0.025]	[−0.038, +0.004]	[+0.008, +0.092]

Note: Bounds on the Shapley contribution of the positions channel, derived from a single linear program over the pinned feasible set with both position-year catchments held jointly. Positive values raise the Right share. The national row combines the East and West position bounds by each year’s AfD electorate-population weights, taking the worst case over the two year-weights of the pair; it is conservative relative to separate regional bounds.

Table H1. Shapley Decomposition of the Change in the Right Share of AfD Support

The bounds on the Shapley values for the Positions channel suggest that the supply-side channel contributes positively to the increase in the share of Right-motivated voters over 2021–2025, while its contribution for the other pairs of years leans negative. What these results make clear is that the evolution over time in the share of Right-motivated voters documented in this paper is not driven by parties changing their positions.

I Coalition-building and the firewall against cooperation with the far right

The sections in the main text asked how mainstream parties could foreclose the rise of populism by repositioning in a two-dimensional policy space. This section studies the welfare consequences of a different measure the establishment parties could use in their electoral competition with populists: the firewall against cooperation with them. Since the AfD entered the Bundestag in 2017 there has been an unwritten rule of no cooperation with it⁶⁸; comparable rules were enacted in France, Sweden and Belgium, and in most cases abandoned once the radical-right party became pivotal for some governing majority.

Until 2025 the German rule was largely non-binding at the national level: the AfD’s vote share was modest and its parliamentary group small. In the 2025 elections it outpolled the SPD (historically one of the two largest establishment parties) in the party-list vote, and, absent the firewall, would have been a plausible coalition partner for the CDU (i.e. CDU/CSU).

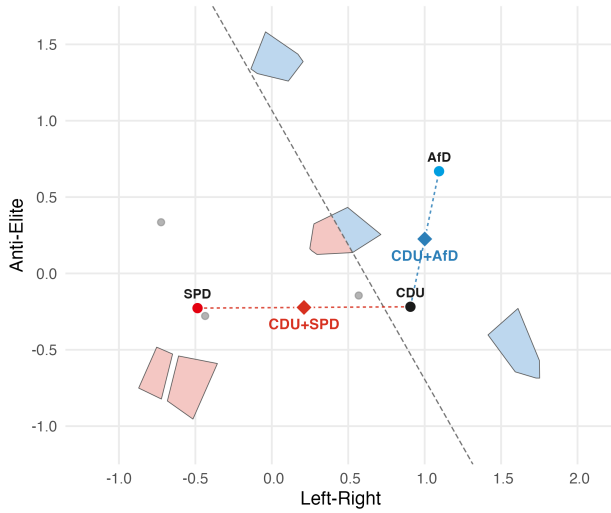
Among other things, critics of the firewall policy argue that it leaves a large share of the electorate unrepresented, and this section tests that argument via coalition formation. Given the vote shares, only two

⁶⁷They are based on a sample of polling districts in which marked ballots allow votes and turnout to be tabulated by gender and age group. Hence, these data are based on actual ballots, not surveys.

⁶⁸The rule is informal at the federal level, but the CDU codified it in its 2018 incompatibility resolution (*Unvereinbarkeitsbeschluss*). At the municipal level compliance is high but not universal: Schroeder, Ziblatt and Bochet (2025) find no cooperation in about 81% of district-council votes in which cooperation was feasible between 2019 and 2024.

coalitions were feasible: the centrist CDU–SPD and the right-wing CDU–AfD. Taking the estimated 2025 voter preferences as given, I ask (i) which coalition maximizes social welfare of the whole electorate and (ii) which coalition is preferred by a majority of voters.

Figure I1a illustrates the logic behind the coalition analysis. The policy implemented by the coalition is assumed to be the midpoint of its members’ positions – blue and red diamonds for the CDU+AfD and CDU+SPD coalitions, respectively. As before, the analysis uses cells as the unit of analysis; some cells are unambiguously closer to one coalition policy than to the other, while some cells are divided (as illustrated in the figure). I evaluate the coalitions using two different metrics – social welfare and majoritarian choice. I focus in this exercise on voters of all parties.



<i>Welfare difference, $CDU+AfD - CDU+SPD$</i>	
National	$[-0.18, 0.03]$
East	$[-0.29, -0.10]$
West	$[-0.13, 0.08]$
<i>Share of the electorate closer to $CDU+AfD$</i>	
National	$[61.9\%, 72.8\%]$
East	$[69.9\%, 83.8\%]$
West	$[58.5\%, 68.1\%]$

(b) Bounds on the target parameter values, θ_{Welfare} and θ_{Majority}

(a) Illustration: Coalition policy positions and random cells of voters’ locations

Notes: Illustration of the coalition-welfare exercise (Panel (a)). Dots mark the 2025 national median positions of the parties. Each coalition is assumed to implement the midpoint of its members’ positions (diamonds): CDU+AfD in blue, CDU+SPD in red. The dashed grey line is the set of voters indifferent between the two coalition policies. The polygons are illustrative cells of the preference partition: a cell is shaded blue (red) when every point in it is closer to the CDU+AfD (CDU+SPD) policy; the cell crossed by the indifference line is split, with each part shaded by the closer policy – within-cell locations are not identified, so such cells can only be bounded. Panel (b) reports bounds over the identified set of preference distributions. Distances are in the region-specific w_2 -weighted metric; no assumption is made on the location of mass within a cell. The welfare-difference rows bound $\mathbb{E}[d_W(x, Y_{CDU+AfD}) - d_W(x, Y_{CDU+SPD})]$ directly in one program per bound (sharp); negative values mean the CDU+AfD policy is closer on average. The majority rows bound the mass of cells lying entirely closer to the CDU+AfD policy (lower) and with any point closer (upper); they are invariant to any increasing transformation of the distance. National bounds combine East and West by their electorate weights (0.30/0.70); the regional programs are independent, so the combination is sharp.

Figure I1. Welfare difference and majority preference between the two 2025 coalitions

Social welfare from coalitions. Let $Y_{CDU+AfD}$ and $Y_{CDU+SPD}$ denote the two coalitions’ implemented policies; and as before let $f(x | s, d)$ be the density of ideal points among all eligible voters in state s and demographic group d , and $F_{S,D}$ their joint distribution. The target parameter is the difference of weighted

Euclidean distances $d_W(\cdot, \cdot)$ from voters' bliss points to the two coalitions' policies:⁶⁹

$$\theta_{\text{Welfare}} = \iint_{\mathcal{R}} \left[d_W(x, Y_{\text{CDU+AfD}}) - d_W(x, Y_{\text{CDU+SPD}}) \right] f(x | s, d) dx dF_{S,D}(s, d).$$

The bounds for θ_{Welfare} are in the upper part of Figure 11(b), where negative values mean that the CDU+AfD policy is closer to voters on average than the CDU+SPD policy. The estimated bounds at the national level straddle zero, so under this welfare rule neither coalition can be ranked unambiguously, although the bounds lean negative. This negative lean is driven by Eastern Germany, where the difference is bounded strictly below zero; in the West the bounds are roughly centered around zero, favoring neither coalition.

Note that this welfare estimation is not invariant to the distance metric; Table K1 provides the same welfare analysis for the squared Euclidean distance.

Majoritarian choice of the coalition. Another way to estimate which coalition is preferred by society is via the share of the electorate strictly closer to one coalition policy than to the other. To estimate this, I formulate the target parameter as the share of voters preferring the CDU+AfD policy,

$$\theta_{\text{Majority}} = \iint_{\mathcal{R}} \mathbb{1} \left\{ d_W(x, Y_{\text{CDU+AfD}}) < d_W(x, Y_{\text{CDU+SPD}}) \right\} f(x | s, d) dx dF_{S,D}(s, d).$$

For θ_{Majority} , the lower bound counts the mass of voters lying in the cells c with $D_{\max}(c) < 0$ (entirely closer to CDU+AfD, as in the colour coding in Figure 11a) and the upper bound the mass of cells with $D_{\min}(c) \leq 0$.

The results are in the lower part of Figure 11(b). For every preference distribution in the identified set, a majority of voters in both parts of the country is closer to the CDU+AfD policy: at least 69.9% in the East and at least 58.5% in the West. Hence the CDU+AfD coalition would win a pairwise vote against the CDU+SPD coalition. The underrepresentation argument is therefore nuanced and depends on the decision rule one has in mind when making such claims. This analysis nevertheless highlights once again the regional differences: in the worst scenario only 16.2% of voters in Eastern Germany actually prefer the coalition with the SPD over the one with the AfD; this highlights that coalition building under the firewall policy could create an additional feeling of exclusion and underrepresentation among Eastern Germans in particular.

The contrast between the two measures of coalition preferences highlights an essential fact about the present-day political landscape in Germany: many voters on the left of the political spectrum hold very strong preferences against a coalition of the center with the AfD — strong enough to keep the welfare comparison ambiguous even though a clear majority sides with CDU+AfD.⁷⁰

Figure K20 provides the same analyses as above, but under the assumption that the implemented coalition policy is the vote-share-weighted average of the coalition parties' own positions.

⁶⁹ As before, I report sharp bounds: for each cell c of the partition I compute the range $[D_{\min}(c), D_{\max}(c)]$ of the difference $D(x) = d_W(x, Y_{\text{CDU+AfD}}) - d_W(x, Y_{\text{CDU+SPD}})$ over $x \in c$, and the lower (upper) bound on θ_{Welfare} is the minimum (maximum) of the cell-mass-weighted sum of $D_{\min}(c)$ ($D_{\max}(c)$) over the identified set of the main linear program. These bounds are sharp: they impose a common preference distribution for both coalitions and, within each cell, a common bliss-point location for both distances.

⁷⁰ The results broadly align with survey data, both in terms of regional heterogeneity and in terms of the strong dislike of a CDU–AfD coalition among some groups of voters. Figure K21 plots the survivor functions of the coalition desirability ratings in the West and the East. In both regions many respondents place the CDU–AfD coalition at the very bottom of the scale (47% in the West, 36% in the East, against 21% and 26% for CDU–SPD). But in the West the CDU–SPD curve dominates over most of the scale, whereas in the East the CDU–AfD curve lies above it at every positive threshold.

J Tables & Figures

	<i>Perceived Party Position, L-R</i>			<i>Own L-R Position</i>
	All Parties	Chosen Parties	Chosen AfD	Chosen Parties
	(1)	(2)	(3)	(4)
<i>Panel A: no individual controls</i>				
Candidate, L-R	−0.007 (0.080)	0.276** (0.135)	1.207*** (0.213)	0.106 (0.166)
Demographic controls	No	No	No	No
<i>Panel B: with demographic controls</i>				
Candidate, L-R	−0.015 (0.078)	0.359** (0.140)	1.243** (0.408)	0.174 (0.169)
Demographic controls	Yes	Yes	Yes	Yes
State F.E.	Yes	Yes	Yes	Yes
Party F.E.	Yes	Yes	No	Yes
Observations	19,992	2,269	210	2,219

Note: Coefficient on the estimated L–R position of the respondent’s district candidate of the party; GLES 2021 pre- and post-election cross-sections, survey weights. Columns (1)–(3): dependent variable is the respondent’s placement of the party on the 1–11 left–right scale, for all parties, for the party chosen with the second vote (party-list), and for AfD voters only; column (4): the respondent’s own left–right self-placement. Panel B adds gender, age, age² and education level controls. Sample: districts of the estimation sample (districts with at most three imputed candidates). Standard errors clustered by party $\times$ state in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table J1. Correlation of LR Positions between Candidates (estimated) and Parties (reported), GLES 2021, estimation sample

	<i>Candidate LR Position</i>		
	All Parties	Chosen Parties	Chosen AfD
	(1)	(2)	(3)
<i>Panel A: no individual controls</i>			
Self, LR	0.001 (0.001)	0.002 (0.004)	0.000 (0.017)
Demographic controls	No	No	No
<i>Panel B: with demographic controls</i>			
Self, LR	0.002 (0.001)	0.004 (0.004)	0.000 (0.017)
Demographic controls	Yes	Yes	Yes
State F.E.	Yes	Yes	Yes
Party F.E.	Yes	Yes	No
Observations	19,450	2,219	205

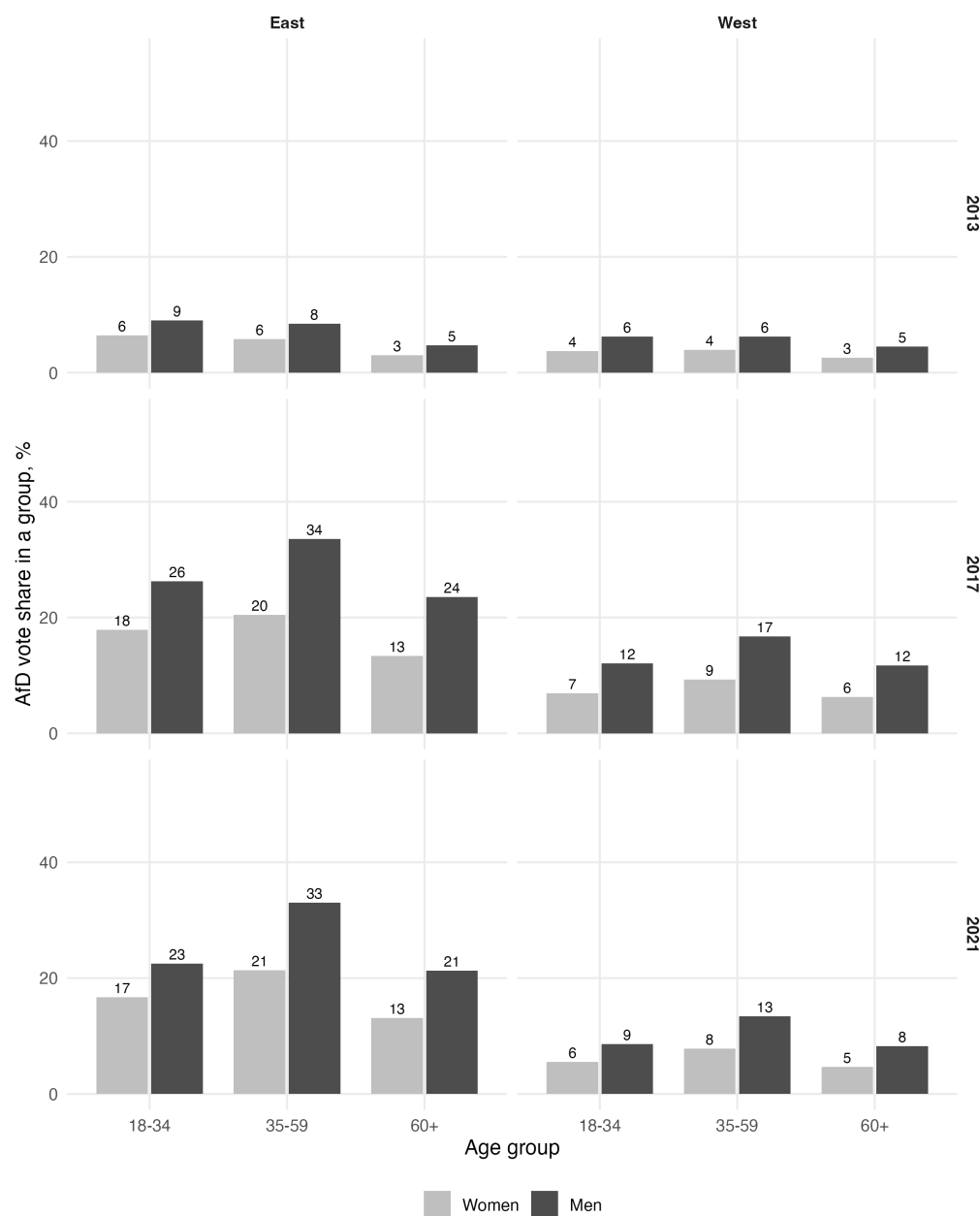
Note: Estimation is based on the GLES 2021 (pre- and post-election cross-sections), which records each individual's electoral district; survey weights. Dependent variable: estimated L–R position of the party's district candidate. Self-reported views take values from 1 (very left) to 11 (very right). Panel B adds gender, age, age² and school education (four groups). Sample: districts of the estimation sample (districts with at most three imputed candidates). Standard errors clustered by party $\times$ state in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table J2. Candidates' LR positions are uncorrelated with voters' self-reported views

	CDU/CSU	SPD	Grüne	FDP	Linke	AfD
	(1)	(2)	(3)	(4)	(5)	(6)
Raw difference	4.37 (0.24)	−3.16 (0.19)	−2.38 (0.16)	−0.60 (0.06)	−1.00 (0.17)	2.09 (0.18)
Within state	4.44 (0.18)	−1.77 (0.13)	−2.23 (0.12)	−0.34 (0.05)	−1.47 (0.06)	1.11 (0.11)
Within district	3.55 (0.15)	−1.59 (0.11)	−1.53 (0.09)	−0.07 (0.04)	−1.33 (0.05)	0.72 (0.09)

Each cell is the coefficient on $\mathbf{1}\{\text{eligible voters} < 10,000\}$, in percentage points, from a separate regression of the municipality's second-vote share on that indicator, pooling the 2013, 2017, 2021 and 2025 elections. *Raw difference* holds only the election year fixed; *within state* adds a state $\times$ year effect; *within district* adds a district $\times$ year effect. Standard errors clustered by municipality in parentheses.

Table J3. Small-minus-large difference in second-vote shares, pooled over 2013–2025, at three levels of comparison



Notes. Bars show the AfD share of valid second votes cast within each group. Source: Repräsentative Wahlstatistik of the Federal Returning Officer, 2013-2021, an official statistic in which ballots in a random sample of polling districts are marked by the voter's gender, birth-year group and state. The six published age groups (18-24, 25-34, 35-44, 45-59, 60-69, 70+) are pooled into three by summing votes for visualisation. East comprises Brandenburg, Mecklenburg-Vorpommern, Saxony, Saxony-Anhalt and Thuringia; West comprises the remaining states, including Berlin.

Figure J1. AfD party vote share by region, age group and gender, 2013, 2017 and 2021.

	2013	2017	2021	2025
Mean	0.001	0.001	0.001	-0.001
SD	0.016	0.033	0.032	0.048
P10	-0.017	-0.036	-0.034	-0.062
Median	0.000	0.000	0.000	0.000
P90	0.020	0.041	0.040	0.058

Note: The unit of observation is the municipality. Residual = observed AfD vote share – model-predicted share.

Table J4. Distribution of Residuals Across Municipalities

Year	Region	Total SD		Between-party		Within-party		AfD only	
		econ	anti-el.	econ	anti-el.	econ	anti-el.	econ	anti-el.
2013	East	0.63	0.60	0.66	0.58	0.21	0.29	0.22	0.11
2013	West	0.64	0.64	0.65	0.61	0.24	0.31	0.25	0.38
2017	East	0.66	0.58	0.67	0.56	0.24	0.28	0.29	0.26
2017	West	0.72	0.64	0.75	0.62	0.22	0.30	0.26	0.28
2021	East	0.72	0.63	0.71	0.56	0.32	0.36	0.48	0.43
2021	West	0.71	0.61	0.74	0.58	0.24	0.30	0.29	0.29
2025	East	0.69	0.42	0.72	0.33	0.22	0.29	0.23	0.32
2025	West	0.75	0.48	0.80	0.41	0.19	0.30	0.19	0.35

Notes: Standard deviations of district-level (Wahlkreis) candidate positions; between-party = SD of party means, within-party = SD of positions around the own-party mean. The between-party dispersion on the anti-elite dimension compresses sharply in 2025, reflecting party convergence and the thinner 2025 item battery.

Table J5. Dispersion of estimated candidate positions by year and region

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current instit.	c20	1–4	0.175
Parties are citizens’ central intermediaries	Current instit.	d6c	1–5	0.144
Legislation reflects citizens’ interests	Current instit.	d6b	1–5	0.137
Citizens have lost trust in politics	Current instit.	d6h	1–5	0.137
Referendums lead to badly designed laws	Direct democracy	d6g	1–5	0.138
There are enough opportunities to participate	Direct democracy	d6a	1–5	0.117
There should be federal-level referendums	Direct democracy	d6f	1–5	0.098
<i>Left-right dimension</i>				
Left-right self-placement	X	c3	1–11	0.038
Own position on immigration (self-placement)	Immigration	c6	1–11	0.032
Immigrants are good for the German economy	Immigration	c2i	1–5	0.093
Immigrants should adapt to German culture	Immigration	c2a	1–5	0.073
Same-sex marriage should be banned	Culture	c2d	1–5	0.090
Women should decide themselves about abortion	Culture	c2j	1–5	0.085
Providing social security is a core government task	Econ. policy	c2g	1–5	0.081
Government spending on development aid	Econ. policy	c18h	1–5	0.080
The government should stay out of the economy	Econ. policy	c2b	1–5	0.071
The government should reduce income differences	Econ. policy	c2h	1–5	0.070
Government spending on defense	Econ. policy	c18c	1–5	0.070

Notes. σ is the change in the standardized dimension score (Left-Right or Anti-Elite) induced by moving a single item one scale point, for a candidate answering all items; it is identical across parties because the score is linear in the answers. Moving a dimension score by a 0.2/0.5 step with a single item requires about $0.2/\sigma$ and $0.5/\sigma$ scale points. Units and item set are held fixed across years; the 2017 survey items are provided as an example.

Table J6. Item-level potency (2017): shift in the dimension score from a one-point move on each survey item

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current instit.	c25	1–4	0.245
Parties are citizens’ central intermediaries	Current instit.	d4c	1–5	0.201
Legislation reflects citizens’ interests	Current instit.	d4b	1–5	0.192
Citizens have lost trust in politics	Current instit.	d4g	1–5	0.192
There should be federal-level referendums	Direct democracy	d4f	1–5	0.138
<i>Left-right dimension</i>				
Left-right self-placement	X	c3	1–11	0.041
Own position on immigration (self-placement)	Immigration	c6	1–11	0.035
Immigrants are good for the German economy	Immigration	c2i	1–5	0.102
Immigrants should adapt to German culture	Immigration	c2a	1–5	0.080
Marriage should be only for heterosexual couples	Culture	c2d	1–5	0.099
Women should decide themselves about abortion	Culture	c2j	1–5	0.093
Providing social security is a core government task	Econ. policy	c2g	1–5	0.089
The government should stay out of the economy	Econ. policy	c2b	1–5	0.078
The government should reduce income differences	Econ. policy	c2h	1–5	0.076
Own position on taxes vs. social spending	Econ. policy	c9	1–11	0.030

Notes. σ is the change in the standardized dimension score (Left-Right or Anti-Elite) induced by moving a single item one scale point, for a candidate answering all items; it is identical across parties because the score is linear in the answers. Moving a dimension score by a 0.2/0.5 step with a single item requires about $0.2/\sigma$ and $0.5/\sigma$ scale points. Items are standardized with their pooled cross-year standard deviation; σ is larger in 2025 than in earlier years because fewer items are asked (five instead of seven on the Anti-Elite dimension). The 2025 items are shown as an example.

Table J7. Item-level potency (2025): shift in the dimension score from a one-point move on each survey item

	2013→2017	2017→2021	2021→2025
<i>West</i>			
Composition	[−0.102, +0.141]	[−0.138, +0.119]	[−0.122, +0.106]
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
Preferences	[−0.287, +0.506]	[−0.432, +0.466]	[−0.345, +0.295]
<i>East</i>			
Composition	[−0.090, +0.048]	[−0.066, +0.042]	[−0.031, +0.045]
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
Preferences	[−0.384, +0.234]	[−0.233, +0.312]	[−0.302, +0.157]

Notes: Bounds on the Shapley contribution of Composition, Positions and Preferences channels, derived from a single linear program over the pinned feasible set with both position-year catchments held jointly. Positive values raise the Right share (θ_{Right} from Section 6.2.2).

Table J8. Shapley Decomposition of the Change in the Right Share of AfD Support

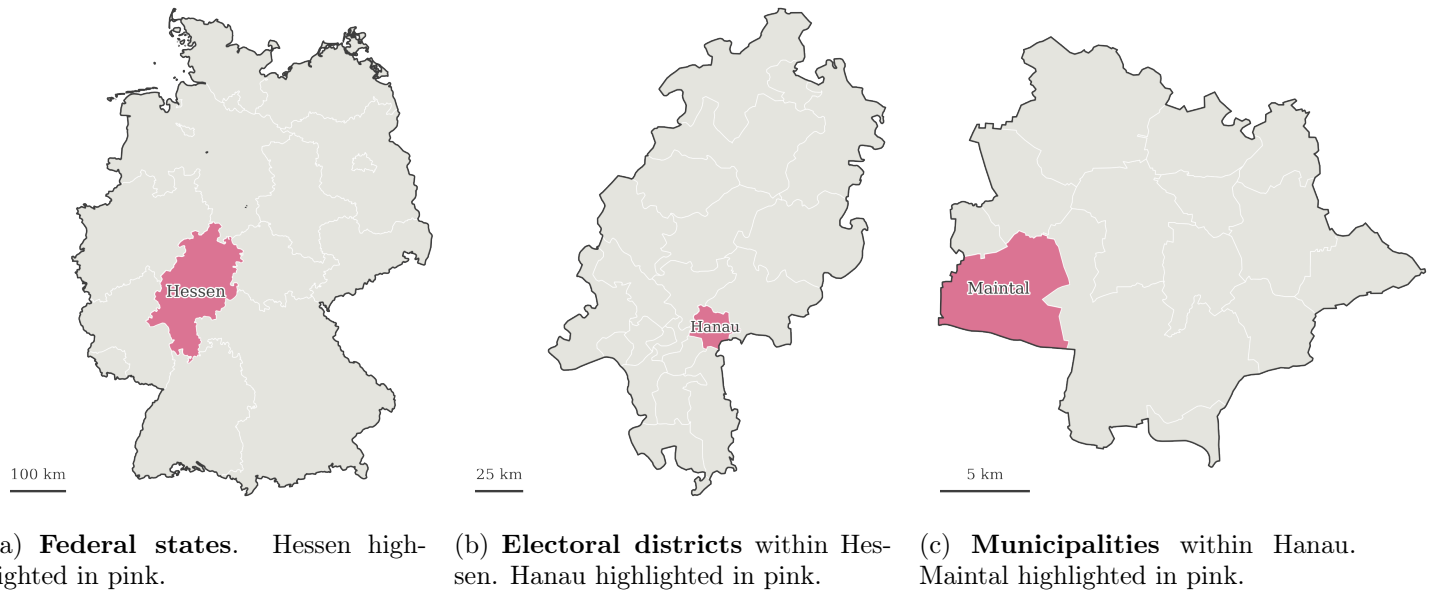
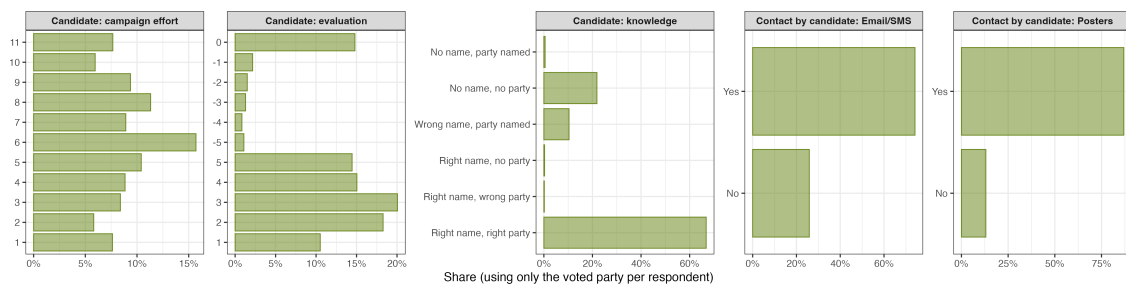


Figure J2. Administrative and electoral hierarchy in Germany.

Municipalities are nested within electoral districts, which are in turn nested within federal states. The figure illustrates these levels for one example state and one example electoral district.



Notes. **Campaign effort (1–11):** 1 = mostly about the candidate; 11 = mostly about the party; **Candidate knowledge:** respondents' ability to name the constituency candidate and their party; **Contact:** whether the respondent was contacted via the listed means. Data are for the 2021 Bundestag election.

Figure J3. Voters' knowledge of and interaction with SMD candidates

Partei	Insgesamt		davon Wählerinnen und Wähler in der Geburtsjahresgruppe											
			1997 – 2003		1987 – 1996		1977 – 1986		1962 – 1976		1952 – 1961		1951 und früher	
			entspricht Altersgruppe etwa von ... bis ... Jahren											
			18 – 24		25 – 34		35 – 44		45 – 59		60 – 69		70 und mehr	
	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%

Mecklenburg-Vorpommern

Insgesamt

Insgesamt	935,0	100	52,1	100	91,9	100	135,3	100	241,1	100	205,3	100	209,4	100
ungültig	17,0	1,8	0,9	1,7	0,9	1,0	2,2	1,6	3,2	1,3	4,1	2,0	5,8	2,8
gültig	918,1	98,2	51,2	98,3	91,0	99,0	133,1	98,4	237,9	98,7	201,2	98,0	203,7	97,2
CDU	180,7	19,7	5,3	10,4	11,2	12,3	22,2	16,7	48,9	20,6	42,9	21,3	50,0	24,6
SPD	261,4	28,5	9,7	18,9	18,8	20,7	29,0	21,8	59,6	25,1	64,4	32,0	80,0	39,3
AfD	170,0	18,5	7,4	14,4	19,2	21,1	33,5	25,2	52,7	22,1	35,7	17,7	21,5	10,6
FDP	66,3	7,2	8,4	16,4	9,1	10,0	11,5	8,6	18,8	7,9	11,5	5,7	7,0	3,4
DIE LINKE	116,0	12,6	6,7	13,1	9,7	10,7	12,5	9,4	25,3	10,7	28,3	14,1	33,5	16,4
GRÜNE	62,7	6,8	8,4	16,5	12,7	14,0	11,4	8,5	15,0	6,3	8,9	4,4	6,3	3,1
CSU	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Sonstige	61,0	6,6	5,3	10,3	10,3	11,3	13,1	9,9	17,5	7,3	9,5	4,7	5,4	2,6
darunter:														
FREIE WÄHLER ¹	18,5	2,0	0,9	1,7	2,9	3,2	3,2	2,4	5,5	2,3	4,1	2,0	1,9	0,9
dieBasis ¹	15,4	1,7	0,8	1,5	1,9	2,1	3,8	2,8	5,3	2,2	2,2	1,1	1,4	0,7

Figure J4. Representative Electoral Statistics: Example

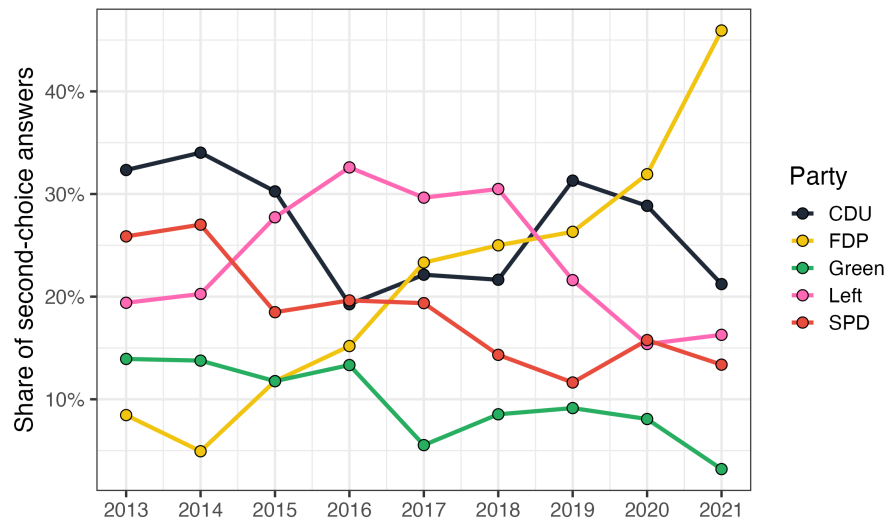
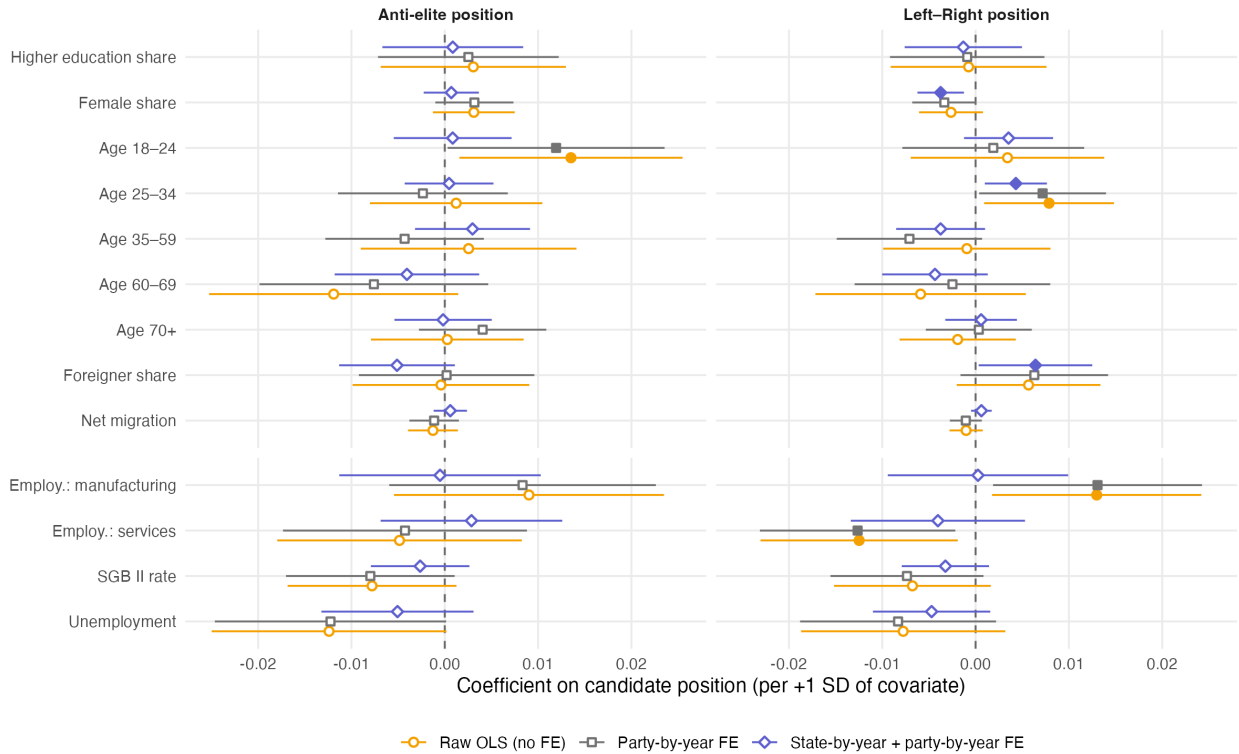
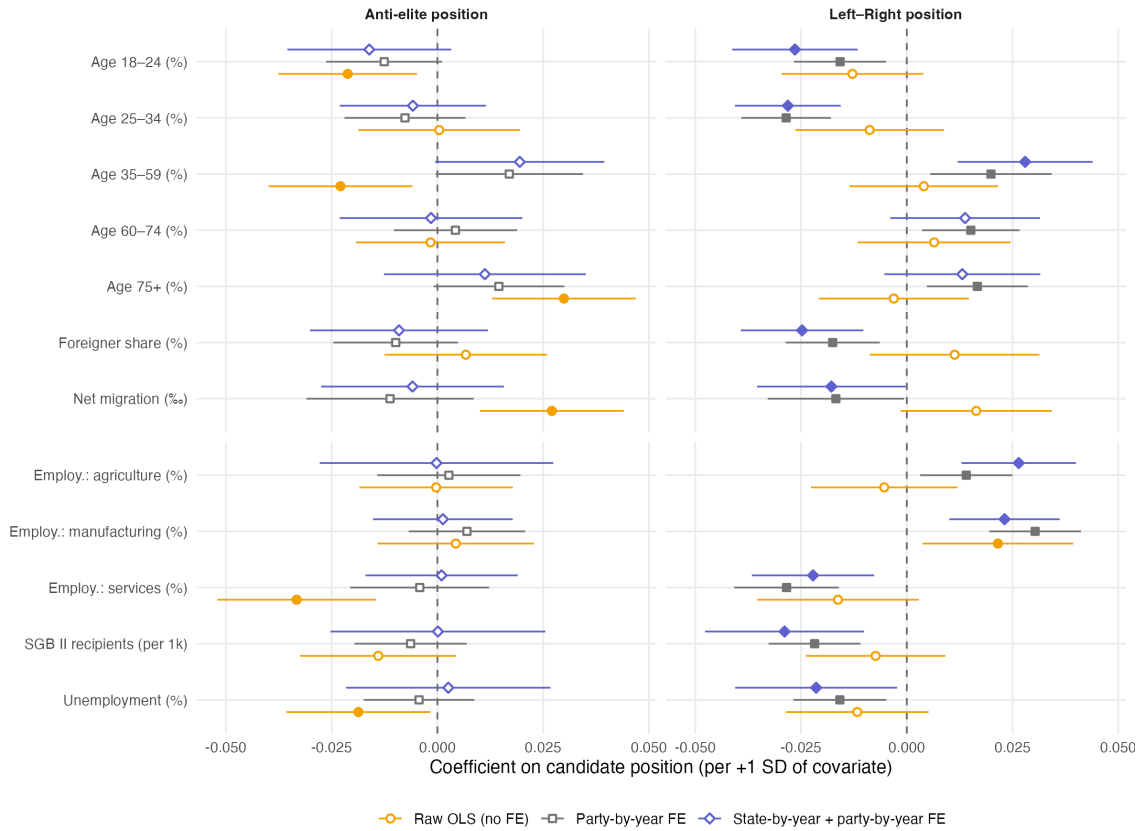


Figure J5. Second-best party choices of AfD voters, federal level



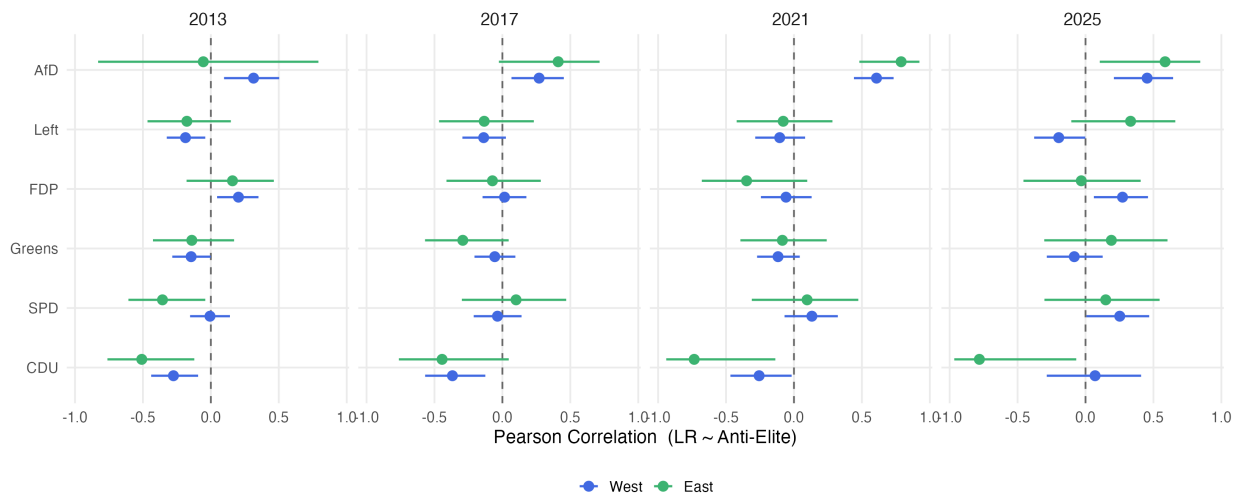
Notes. Each marker is the coefficient from a separate regression of a candidate position on one characteristic. The unit of observation is a municipality-by-party cell in the 2013–2025 elections. The sample consists of municipalities used in the estimation of the model (see Sections 4.2 and 5 for details). All covariates are standardized to mean zero and unit standard deviation. The specifications are (i) Raw OLS with no fixed effects, (ii) party-by-year fixed effects, and (iii) state-by-year plus party-by-year fixed effects. Standard errors are clustered by district-by-year. Horizontal bars are 95% confidence intervals. Filled markers denote coefficients significant at the 5% level.

Figure J6. Association between candidates' positions on each dimension and municipalities' demographic and socio-economic characteristics.

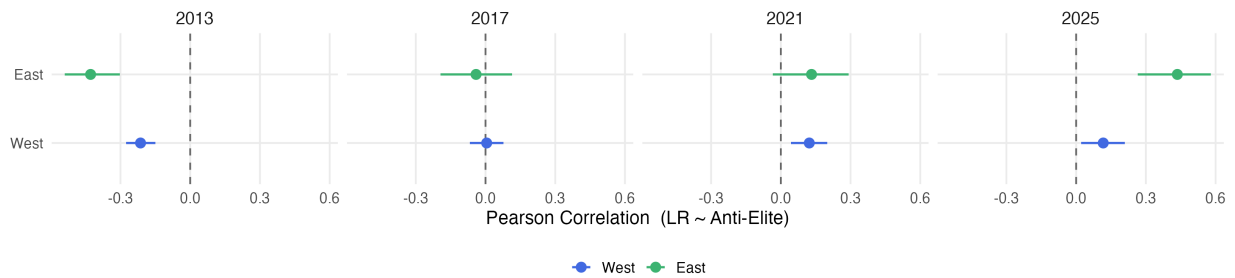


Notes. Each marker is the coefficient from a separate regression of a candidate position on one characteristic. The unit of observation is an electoral district-by-party cell in the 2013–2025 elections. The sample consists of electoral districts used in the estimation of the model (see Sections 4.2 and 5 for details). All covariates are standardized to mean zero and unit standard deviation. The specifications are (i) Raw OLS with no fixed effects, (ii) party-by-year fixed effects, and (iii) state-by-year plus party-by-year fixed effects. Standard errors are clustered by district-by-year. Horizontal bars are 95% confidence intervals. Filled markers denote coefficients significant at the 5% level.

Figure J7. Association between candidates' positions on each dimension and their electoral districts' demographic and socio-economic characteristics.



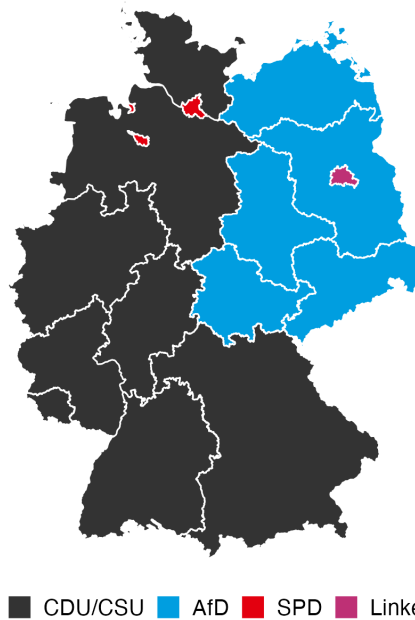
(a) Within parties



(b) Across parties

Figure J8. Correlation between candidates' Left–Right and Anti-Elite positions

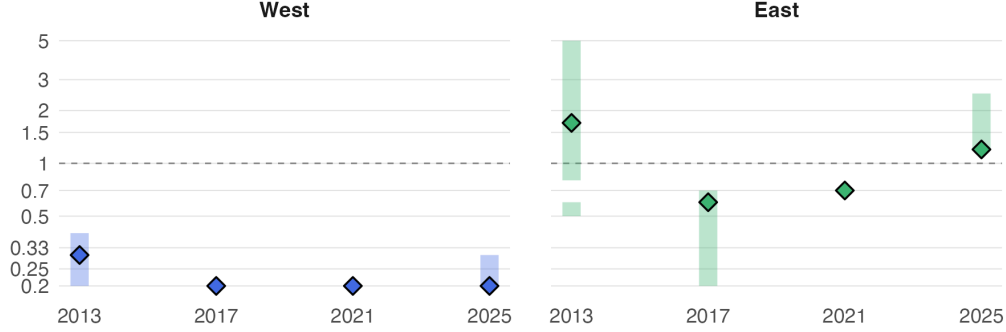
Leading party in the 2025 party-list vote



Notes: Each federal state is coloured by the party that received the largest share of party-list (second) votes in the 2025 Bundestag election. Vote counts come from the Official Electoral Returns. The AfD leads in all five East German states (Brandenburg, Mecklenburg-Vorpommern, Sachsen, Sachsen-Anhalt, Thüringen), with 32–39% of the vote and the CDU/CSU in second place at 18–20%. The CDU/CSU leads in every West German state except Berlin, Bremen and Hamburg.

Figure J9. Leading party in the 2025 federal election, by federal state

K Additional Results



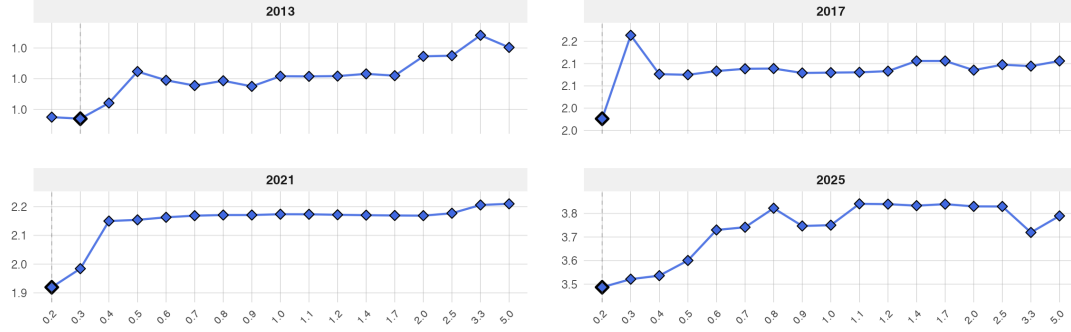
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Points denote the point estimates corresponding to the best-fitting weight, shaded intervals denote weights within 1% of the minimum objective-function value. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See Figure K2 for full results over the grid.

Figure K1. Sets of best-fitting weights for the Anti-Elite dimension w_{AE} relative to Left-Right $w_{LR} = 1$ across years and parts of the country

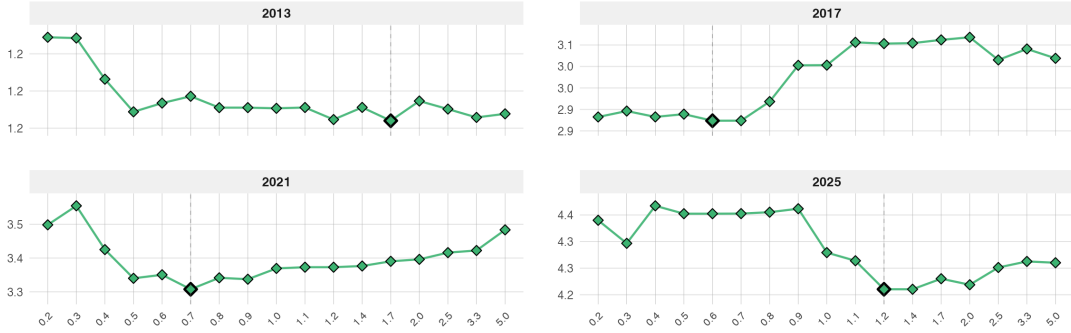
<i>Welfare difference (quadratic), CDU+AfD – CDU+SPD</i>	
National	[−1.43, 0.07]
East	[−3.33, −0.53]
West	[−0.62, 0.33]

Notes: Sharp bounds over the identified set of preference distributions on $\mathbb{E}[d_W^2(x, Y_{CDU+AfD}) - d_W^2(x, Y_{CDU+SPD})]$; negative values mean the CDU+AfD policy is closer to voters on average. Each coalition implements the simple average of its members' 2025 national median positions. National bounds combine East and West by their electorate weights; the regional programs are independent, so the combination is sharp.

Table K1. Welfare difference between the two 2025 coalitions under quadratic loss: equal-weighted policy



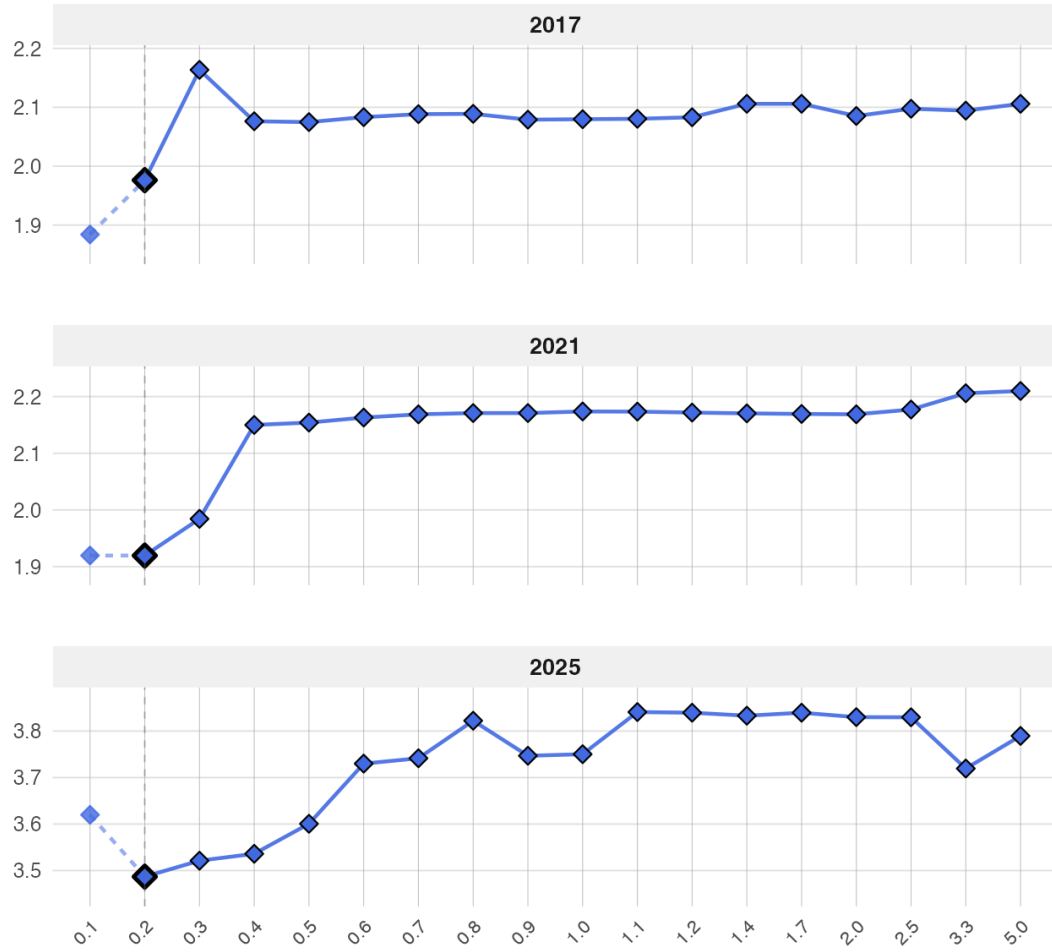
(a) West



(b) East

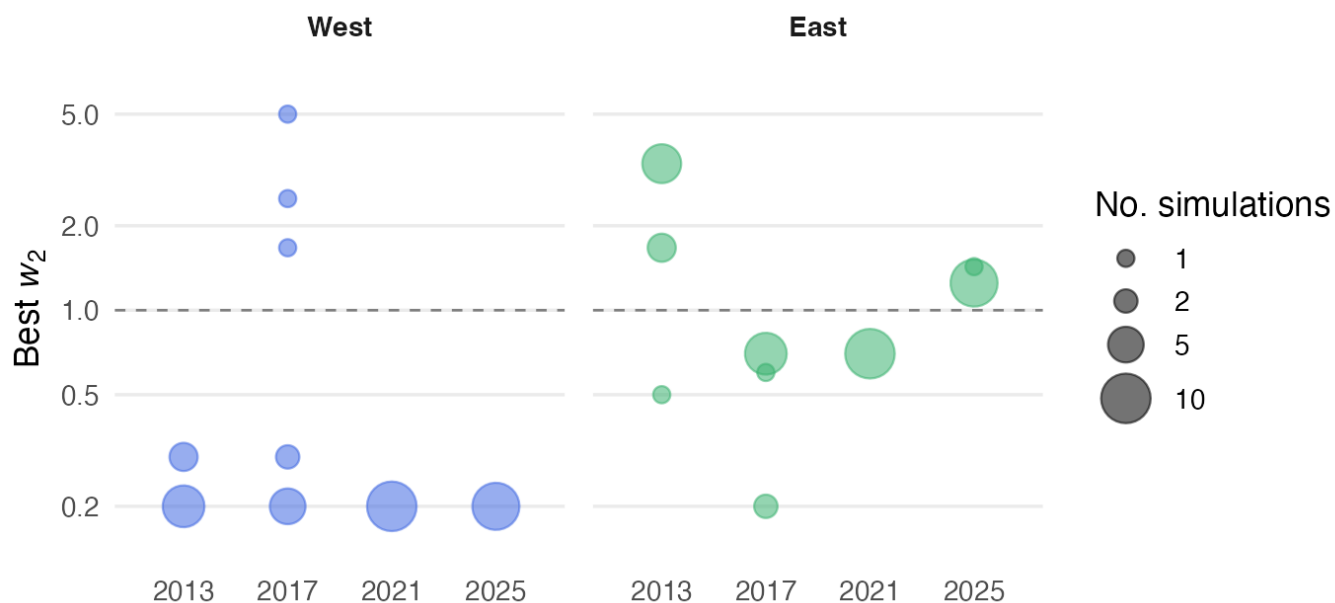
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.

Figure K2. Model Error Value by different relative weight values, years and parts of the country



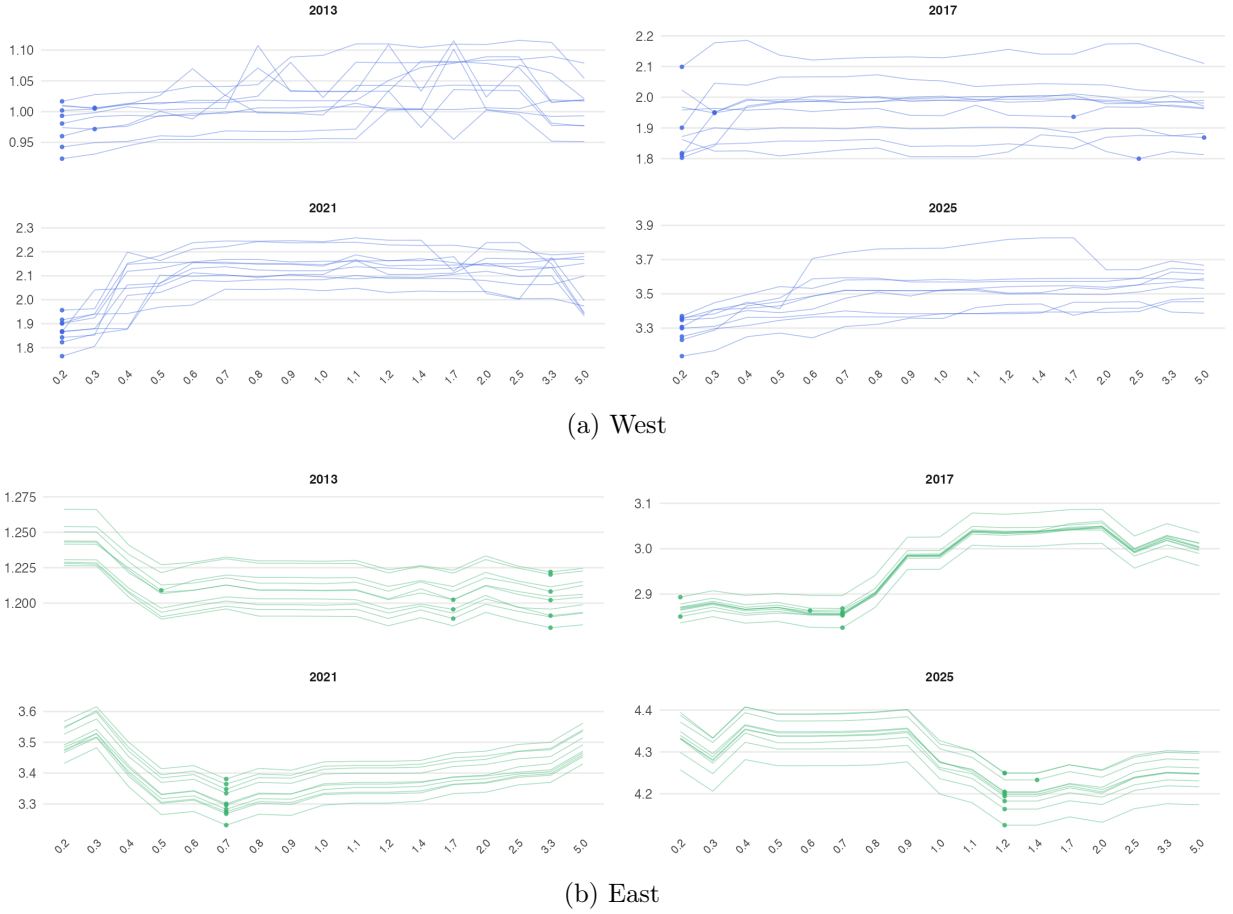
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{AE} for all groups within a given year and region.

Figure K3. Model Error Value by different relative weight values for the West 2017–2025 with a new point $w = 0.1$ (where the corner $w = 0.2$ is the best-fitting weight).



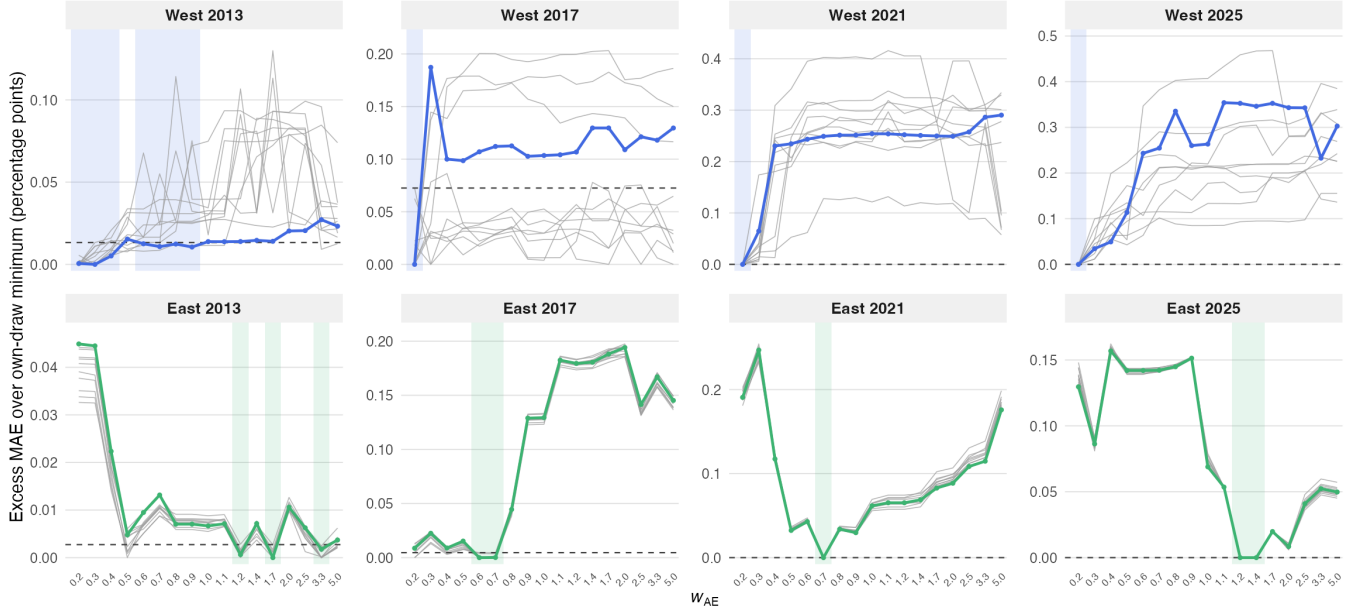
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure K4. Optimal relative weight values by years and parts of the country



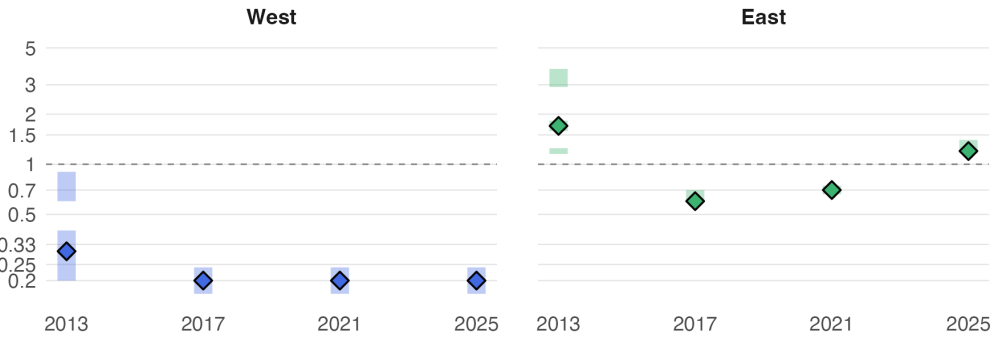
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure K5. Model Error Value by different relative weight values, years and parts of the country by subsamples



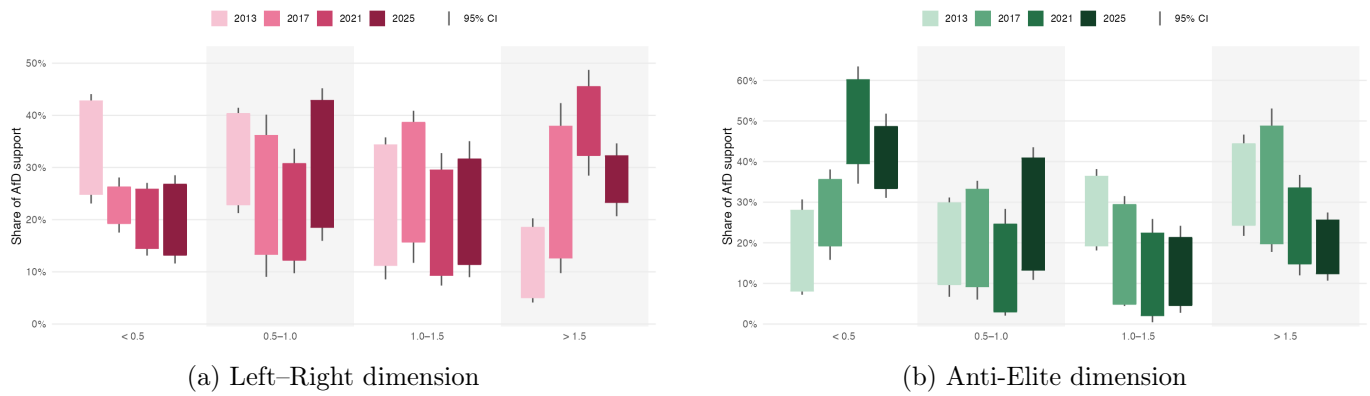
Notes. One panel per region-year. Gray lines are the $B = 10$ subsample MAE curves of Figure K5, each recentered on its own minimum ($D_{bj} = E_{bj} - \min_j E_{bj}$); recentering removes the vertical level shift between draws, which is what makes the raw curves look near-parallel, and leaves only the shape. The coloured line is the full-sample curve of Figure K2, recentered the same way ($\hat{d}_j$). The dashed horizontal line is the cutoff: the $(1 - \alpha)$ inverse-ECDF quantile of D_{bj^*} across draws at the full-sample minimizer j^* , with $\alpha = 0.05$.

Figure K6. Calibration of the weight confidence sets from the subsample re-runs



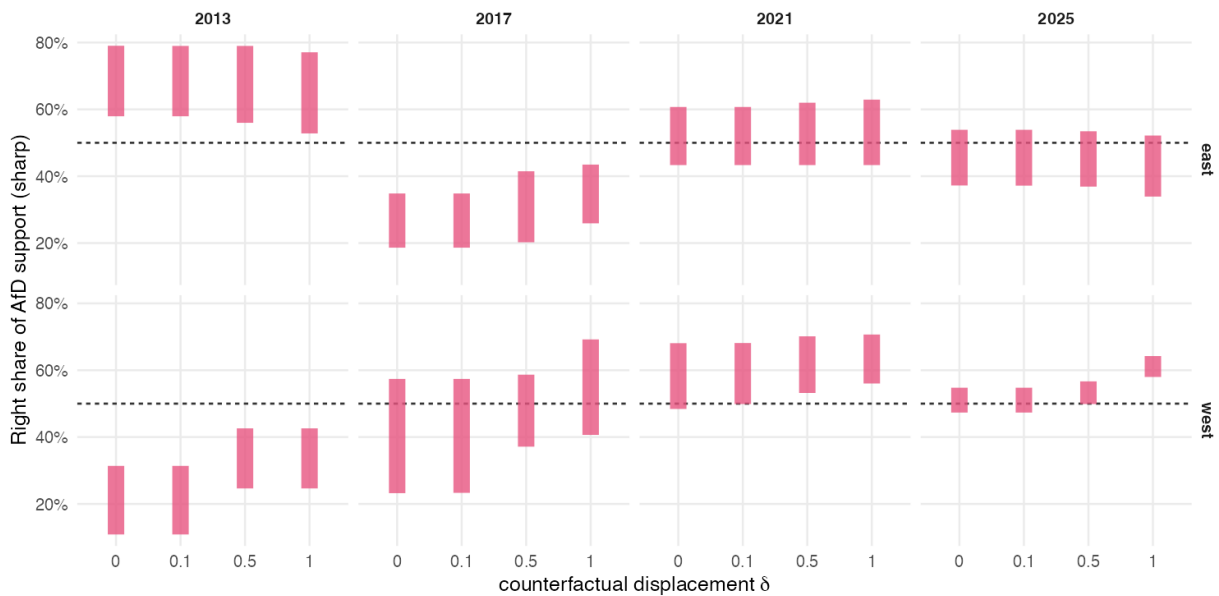
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Diamonds denote the point estimates corresponding to the best-fitting weight, identical to those in Figure K1. Shaded bars denote the confidence set CS , calibrated from the $B = 10$ subsample re-runs of Figures K4 and K5 rather than from the fixed 1% tolerance used in Figure K1. Each draw's MAE curve is recentered on its own minimum, $D_{bj} = E_{bj} - \min_j E_{bj}$, which removes the level shift between draws and retains only the curvature; the cutoff is the $(1 - \alpha)$ empirical quantile (inverse-ECDF, $\alpha = 0.05$) of D_{bj^*} . Separate exercises are run for East and West Germany. See Figure K6 for the calibration itself.

Figure K7. Confidence sets for the Anti-Elite weight w_{AE} , calibrated from the subsample re-runs



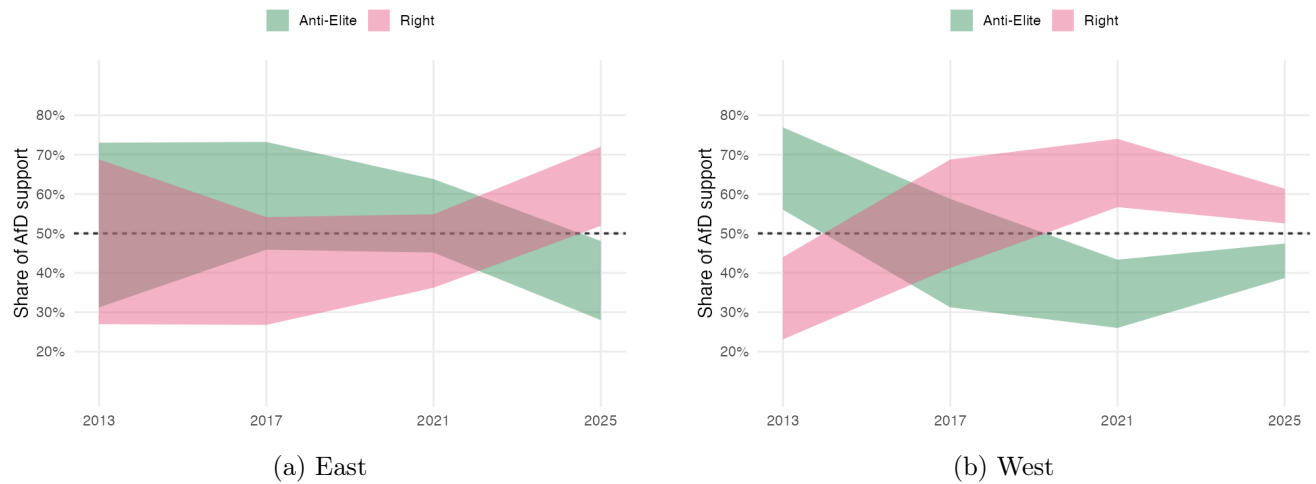
Notes: Thin gray lines: 95% confidence intervals from a municipality bootstrap: the municipalities are resampled with replacement, positions and partitions held fixed, and the bounds re-estimated; 50 replications.

Figure K8. Sampling uncertainty behind the dimension bins



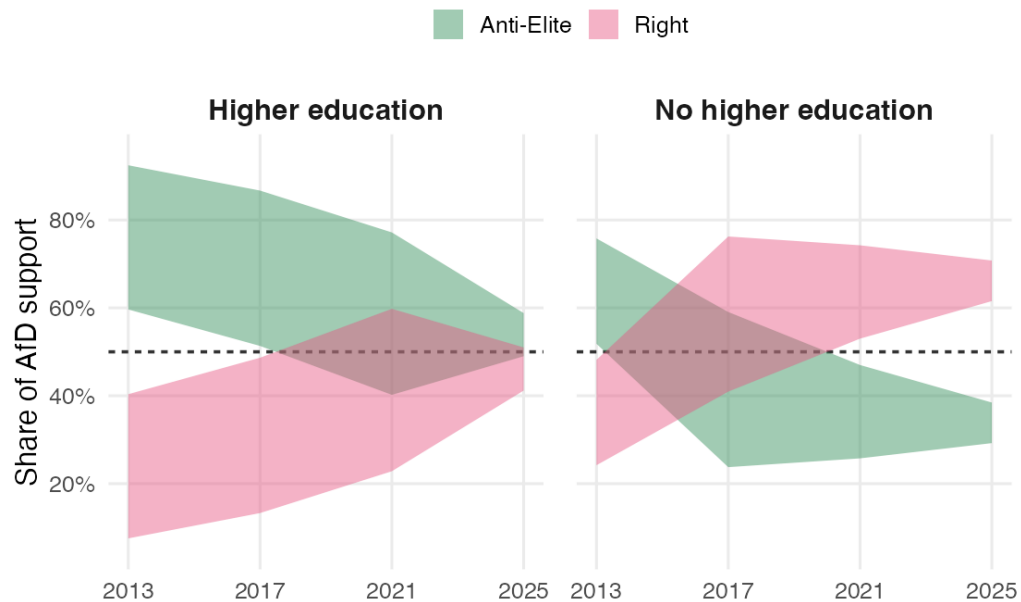
Notes: The smallest δ used is 10^{-6} (not exactly 0).

Figure K9. Bounds on the share of AfD voters preferring the hypothetical Right party, for different steps δ



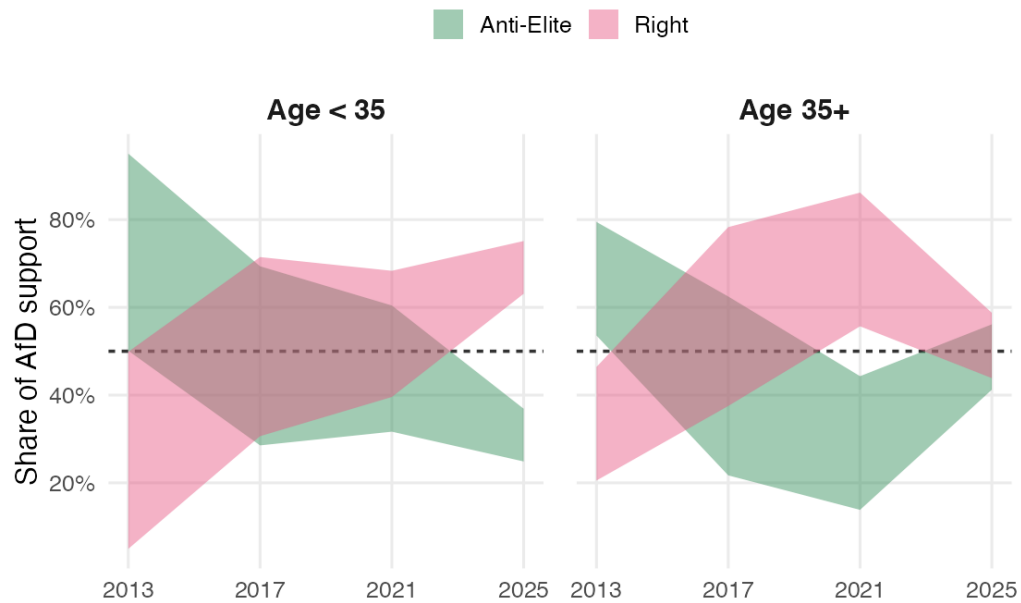
Notes: Panel plots bounds on the shares of voters whose bliss points lie in the attraction areas of two hypothetical parties that retain the AfD's median national position on one dimension while adopting a more moderate position on the other: the *Anti-Elite* party is more left-wing (green), whereas the *Right* party is more pro-elite (pink). Together, these areas recover the full AfD support. See Figure 8 for an illustration.

Figure K10. Regional Heterogeneity: Results separately for West and East Germany.



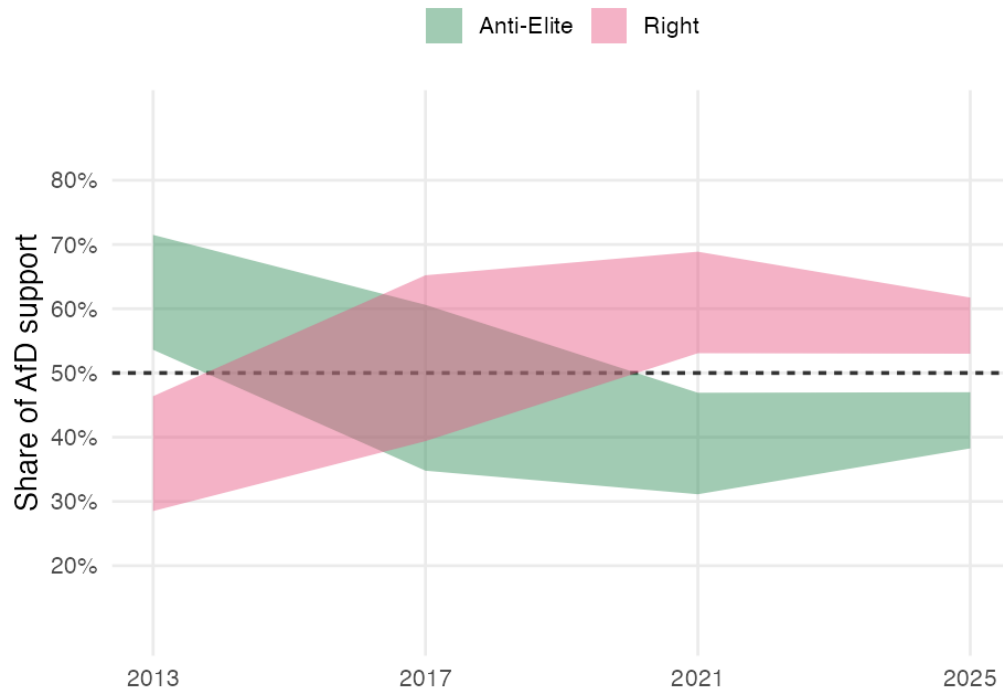
Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions across years are plotted. Results are for Western Germany only.

Figure K11. Estimated shares of AfD voters in both subspaces of policy space by educational attainment



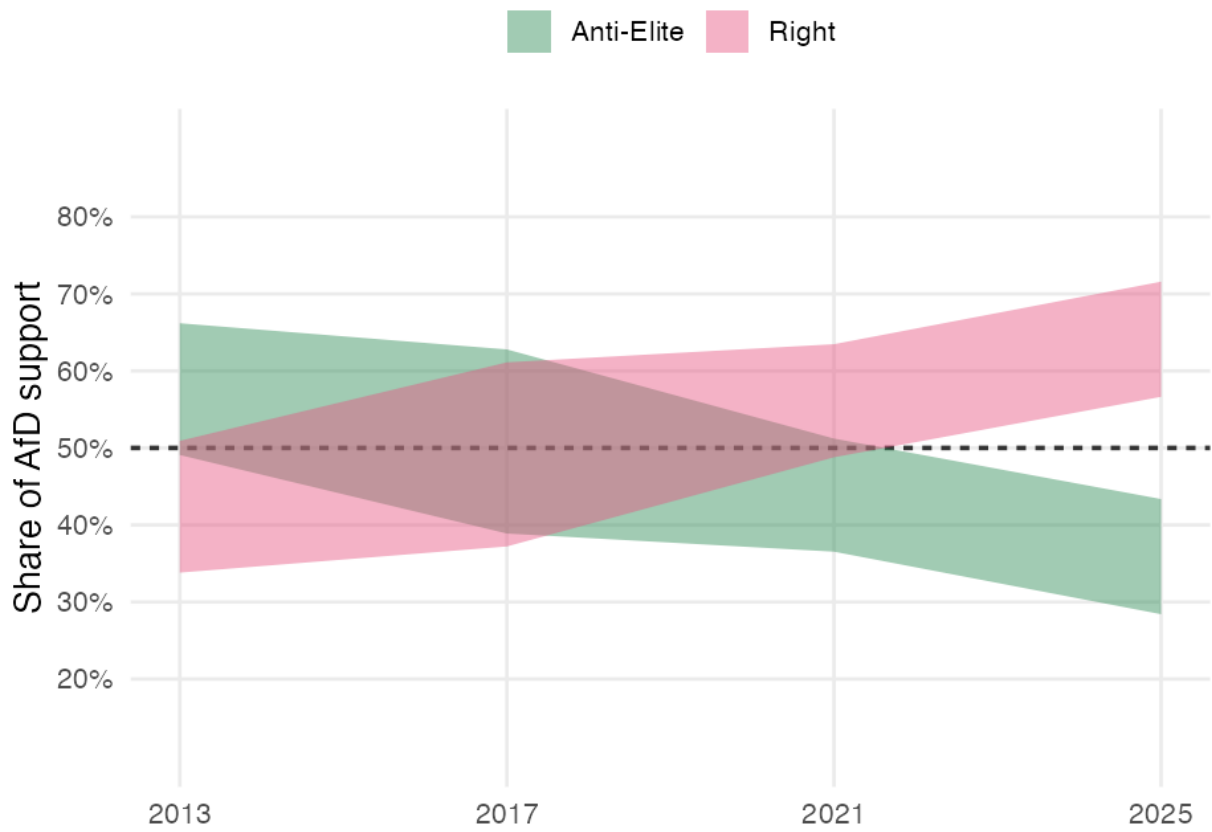
Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 8a across years are plotted. Results are for Western Germany only.

Figure K12. Estimated shares of AfD voters in both subspaces of policy space by age



Notes. The figure plots bounds on the shares of voters whose bliss points lie in the green and pink regions of Figure 8a, across years. The estimation uses survey-based second-choice moments, constructed for each state and election year: among respondents who name the AfD as their most preferred party, the shares naming each of the other parties as their second choice.

Figure K13. Estimated shares of AfD voters in both subspaces, estimation with second-choice moments.



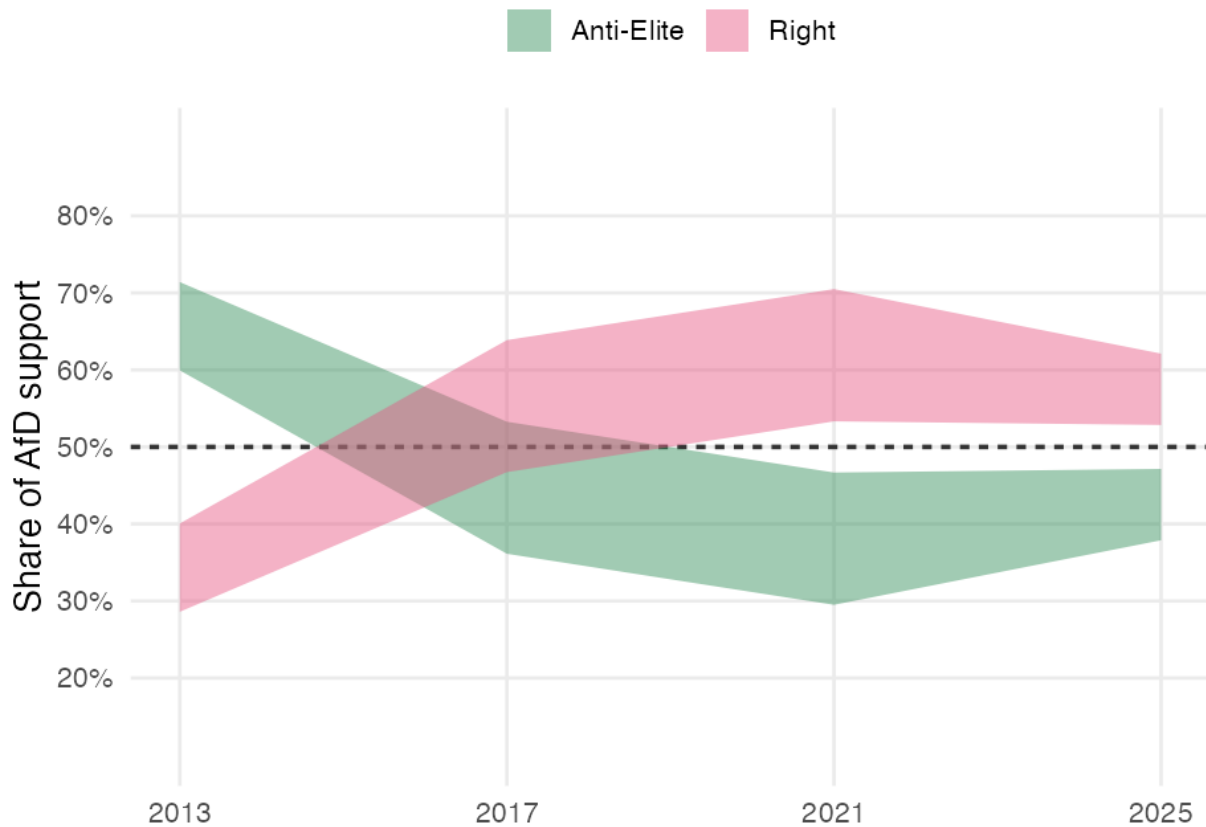
Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 8a across years are plotted.

Figure K14. Estimated shares of AfD voters in both subspaces on the sample of municipalities with fewer than 5,000 inhabitants.



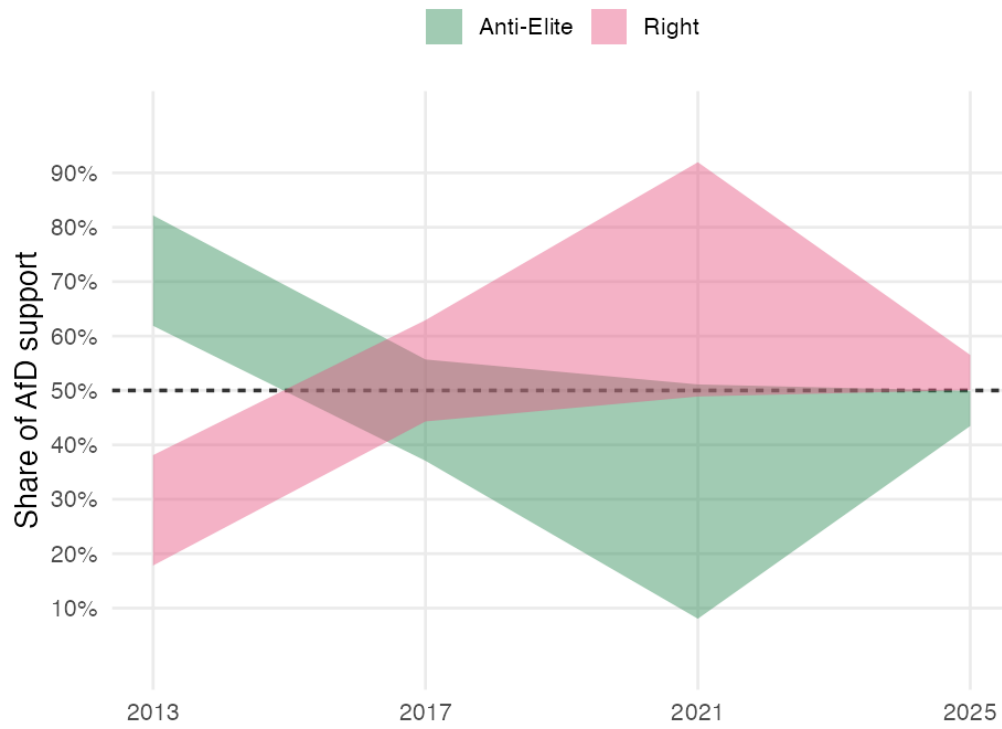
Notes: The figure plots bounds, across years, on the shares of voters whose bliss points lie in the green and pink regions of Figure 8a. Only districts in which the AfD candidate's vote share falls short of the winner's by more than 5% (10%) in the East (West) are retained, for the years 2017 and 2021 (2025). These restrictions drop 12 districts in the East in 2017 and 28 in 2021, and 22 districts in the West in 2025. The East in 2013 and the West in 2013–2021 are omitted because there were no serious contenders, so the results from the main text stand; the East in 2025 is omitted because the AfD won in almost every district, making the analysis infeasible.

Figure K15. Estimated shares of AfD voters in both subspaces on the sample of non-contender electoral districts.



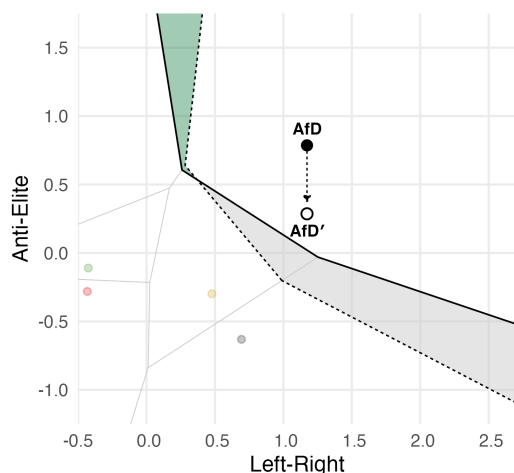
Notes: Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 8a across years are plotted. The analysis allows the estimated ideal-point mass of each demographic group, in each partition cell, to differ by up to a factor of three between conservative and moderate electoral districts of the same state—i.e. the conservative-district mass may range from one-third to three times the moderate-district mass in every cell. A district is classified as conservative if its AfD candidate is positioned above the within-state median AfD position on both the Left–Right and the anti-elite dimension, and moderate otherwise. This analysis is similar to an instrument in Tebaldi, Torgovitsky and Yang (2023).

Figure K16. Estimated shares of AfD voters in both subspaces allowing for the differences in ideal points distributions between *conservative* and *moderate* electoral districts.

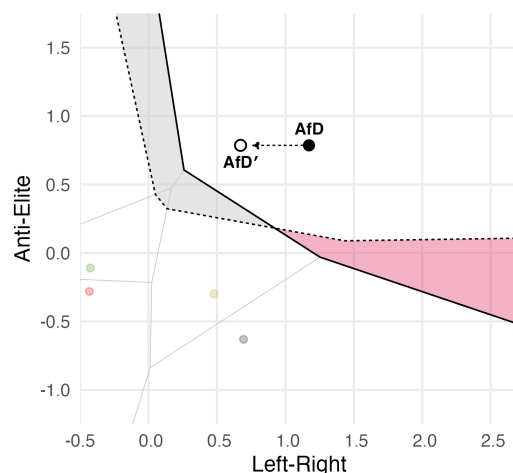


Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 8a across years are plotted. The sample is restricted to municipalities adjacent to a border between two electoral districts within the same federal state. The preference distribution of each demographic group is imposed to be identical across the municipalities of the same border pair, and is allowed to differ across border pairs within a state by up to 20%; across states the coupling is the same as in the baseline specification.

Figure K17. Estimated shares of AfD voters in both subspaces, estimated on the sample of border municipalities.



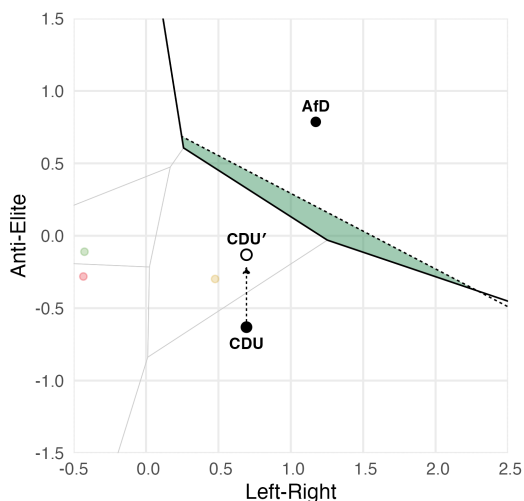
(a) AfD shift of 0.5 towards a more pro-elite position



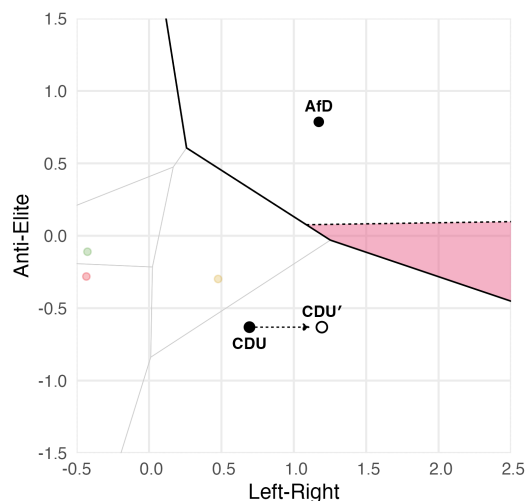
(b) AfD shift of 0.5 towards a more left-wing position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD. **AfD** denotes the AfD's median position across years, and **AfD'** denotes the shifted AfD position—either towards the pro-elite position (Panel (a)) or towards a more left-wing position (Panel (b)). Solid lines mark the existing partition, dashed lines the counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

Figure K18. Illustration of counterfactual moderating shifts by AfD by 0.5 units separately on two dimensions



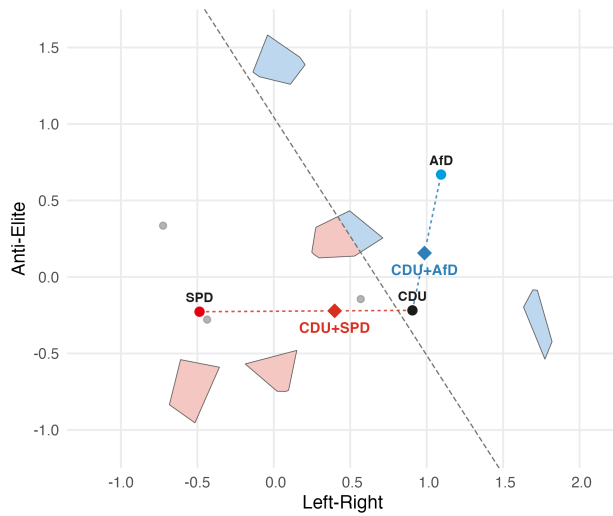
(a) CDU shift of 0.5 towards a more anti-elite position



(b) CDU shift of 0.5 towards a more right-wing position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD and CDU. **CDU** denotes the median CDU position across years, and **CDU'** denotes the shifted CDU position—either towards the anti-elite position (Panel (a)) or towards a more right-wing position (Panel (b)). Solid lines mark the existing partition, dashed lines the counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

Figure K19. Illustration of counterfactual shifts by CDU by 0.5 units separately on two dimensions



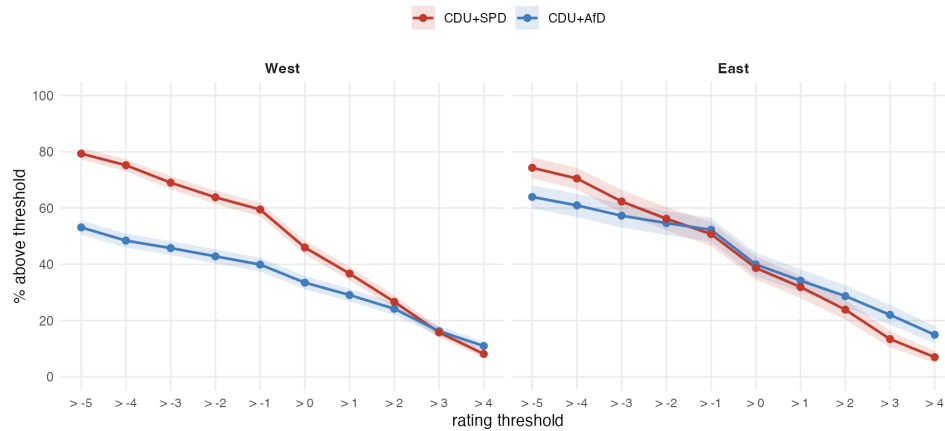
(a) Illustration: Coalition policy positions and voters' locations

Notes: Same objects as Figure I1, but each coalition implements the vote-share-weighted average of its members' positions (diamonds in Panel (a)), with weights from the 2025 second-vote shares. Both policy points shift toward the CDU. Panel (b) reports bounds over the identified set of preference distributions, constructed exactly as in Figure I1.

<i>Welfare difference, $CDU+AfD - CDU+SPD$</i>	
National	$[-0.13, 0.05]$
East	$[-0.24, -0.08]$
West	$[-0.08, 0.10]$
<i>Share of the electorate closer to $CDU+AfD$</i>	
National	$[54.8\%, 70.8\%]$
East	$[69.9\%, 78.6\%]$
West	$[48.3\%, 67.5\%]$

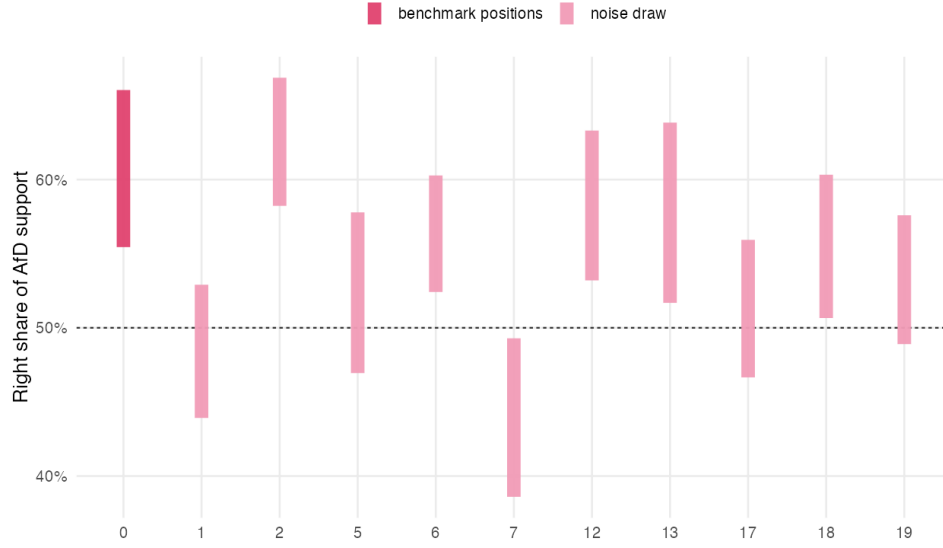
(b) Bounds under the vote-share-weighted policy

Figure K20. Welfare difference and majority preference between the two 2025 coalitions: vote-share-weighted policy



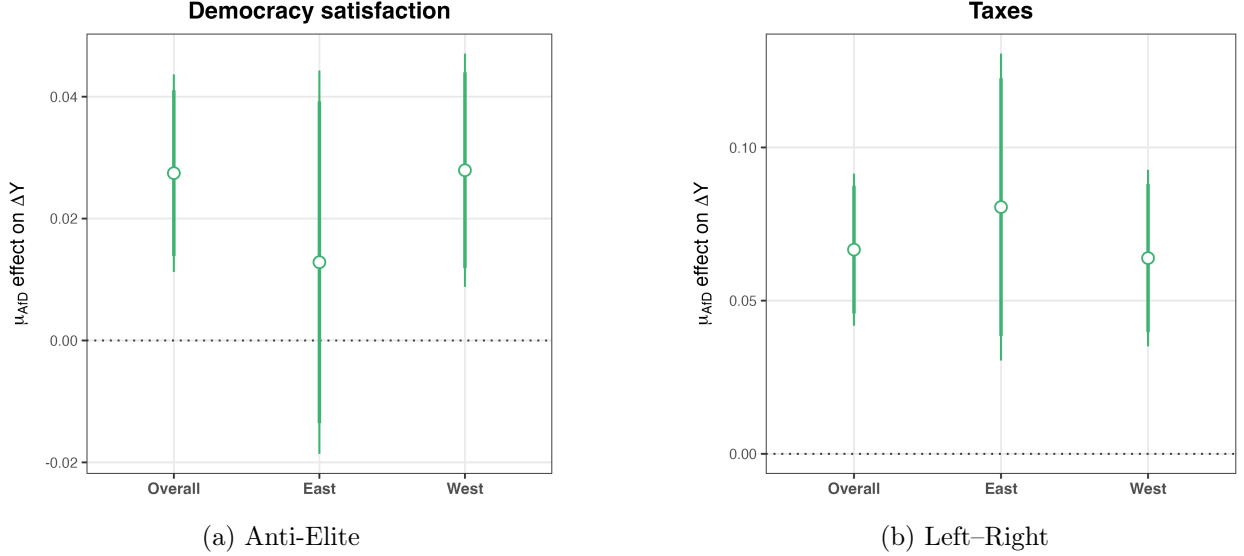
Notes. The figure plots the share of respondents rating each coalition strictly above a given threshold, for every threshold on the eleven-point desirability scale. Respondents are asked how desirable each coalition would be, on a scale from -5 ("not at all desirable") to $+5$ ("extremely desirable"). The same respondents rate both coalitions, so the two curves within a panel are directly comparable. Data are from the GLES Tracking, fielded 12–20 February 2025, one month before the federal election. Shaded bands are 95% confidence intervals.

Figure K21. Desirability of a CDU–SPD and a CDU–AfD coalition, West and East Germany



Notes. The figure shows the behaviour of the target parameter from Section 6.2.2, estimated on the West 2025 data, when candidate positions are perturbed by measurement error. In each of 10 replications, an i.i.d. $N(0, \delta^2)$ shock with $\delta = 0.01$ is added to every estimated candidate position on both dimensions and to the anchor of the hypothetical parties, and the parameter is re-estimated; the resulting values are plotted against the baseline obtained without perturbation. Relative to the spread of the West 2025 district positions, δ equals $0.013 \times \text{s.d.}(X_{LR})$ (s.d. 0.76) and $0.021 \times \text{s.d.}(X_{AE})$ (s.d. 0.48); relative to the AfD's own candidates, $0.05 \times \text{s.d.}$ and $0.03 \times \text{s.d.}$ respectively.

Figure K22. Robustness of the target parameter to measurement error in candidate positions



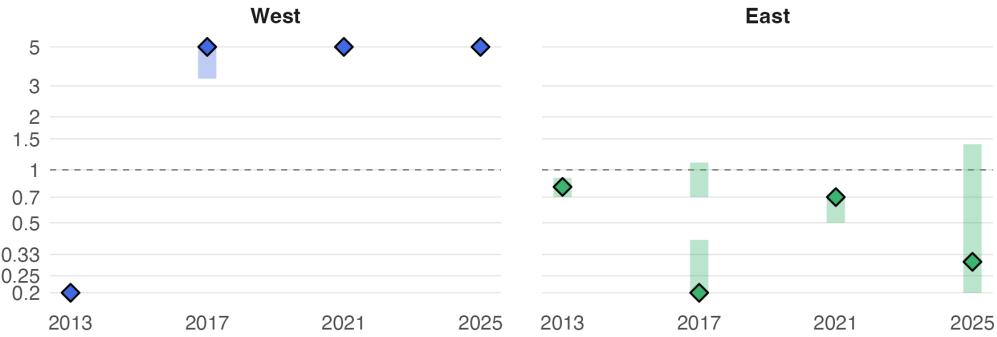
Notes: The panels plot the average yearly within-person evolution of preferences of repeat AfD voters relative to voters of other parties. Based on self-reported positions $Y_{i,t}$ I construct $\Delta Y_{i,t} = Y_{i,t} - Y_{i,t-1}$ for consecutive years $t-1, t$, and drop any respondent-year pair for which AfD status changes across the two waves. That is, I keep only pairs in which either $\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 1$ (consistent AfD voters) or $\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 0$ (consistent non-AfD voters). Plotted values are μ_{AfD} from: $\Delta Y_{i,t} = \mu + \mu_{\text{AfD}} \cdot \mathbf{1}\{\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 1\} + \delta_t + \varepsilon_{i,t}$. Higher values correspond to positions closer to the AfD. The point estimate of μ_{AfD} , together with 90% (thick) and 95% (thin) confidence intervals, is reported for the Overall, East, and West samples. Standard errors clustered at the respondent level.

Figure K23. Results for within-person preference evolution for repeat AfD voters relative to voters of other parties

	Overall		East		West	
	No FE	Year FE	No FE	Year FE	No FE	Year FE
μ	0.0229*** (0.0047)	0.0260*** (0.0048)	0.0165* (0.0099)	0.0183* (0.0102)	0.0249*** (0.0053)	0.0282*** (0.0054)
Year FE	No	Yes	No	Yes	No	Yes
Obs. (ΔY)	4806	4806	1155	1155	3651	3651
Individuals	2182	2182	506	506	1688	1688

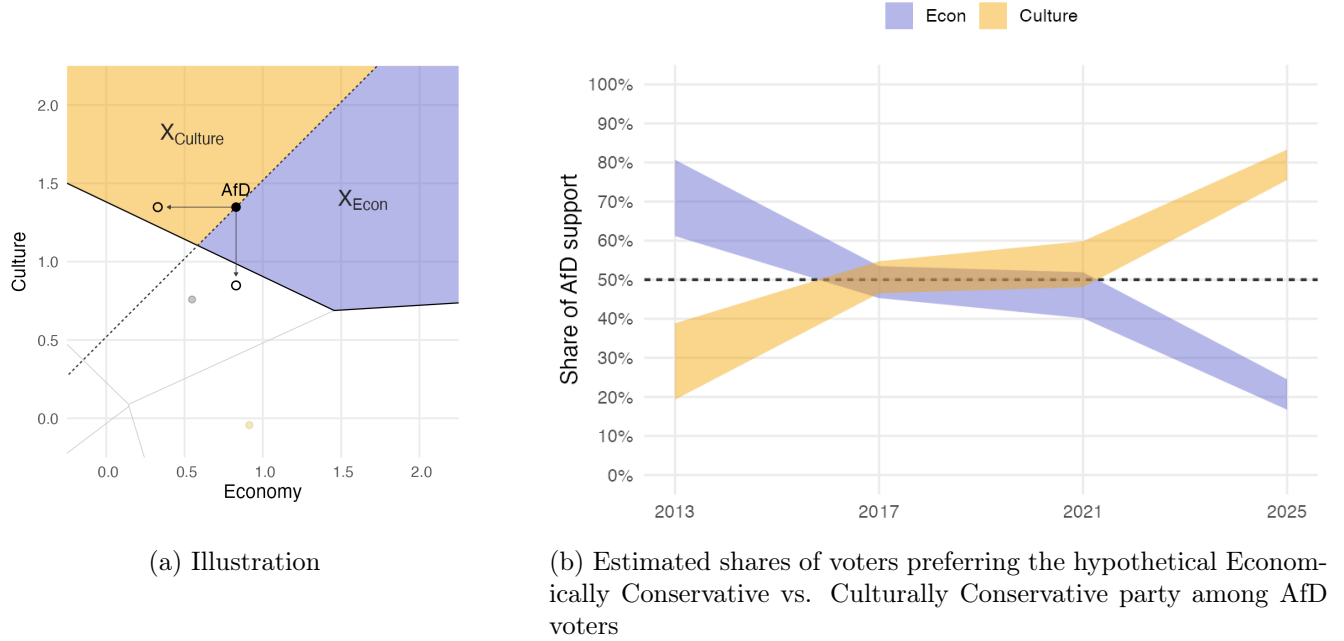
Notes: Sample: AfD voters only (from their first AfD vote onward). For each year t I compute the *median* AfD voter's position in the normalized (taxes, democracy-satisfaction) space, where both outcomes are oriented so that higher values are closer to the AfD, and draw a 45 degree line through the point, and define a dummy $\text{Below45}_{i,t}$ for being "below" the 45 degree line. I estimate, on consecutive-year pairs in which i is an AfD voter in t , $\text{Below45}_{i,t+1} - \text{Below45}_{i,t} = \mu + \delta_t + \varepsilon_{i,t}$. Standard errors are clustered at the respondent level. Columns are estimated separately on the Overall, East, and West subsamples.

Table K2. Annual drift in AfD voters' "below 45°" dummy



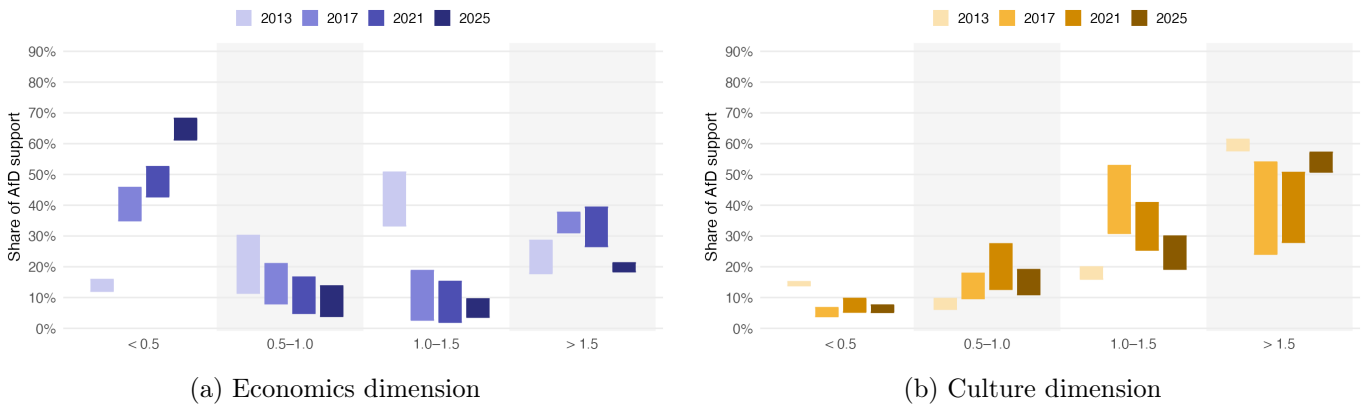
Notes. $w_{econ} = 1$ is fixed throughout; w_{cult} should be interpreted as the relative weight placed on the cultural dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each state s has its own bliss-point distribution $f(x | s, d)$, while imposing the same weight w_{cult} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.

Figure K24. Weights for the Cultural dimension w_{cult} relative to Economic $w_{econ} = 1$ across years and parts of the country



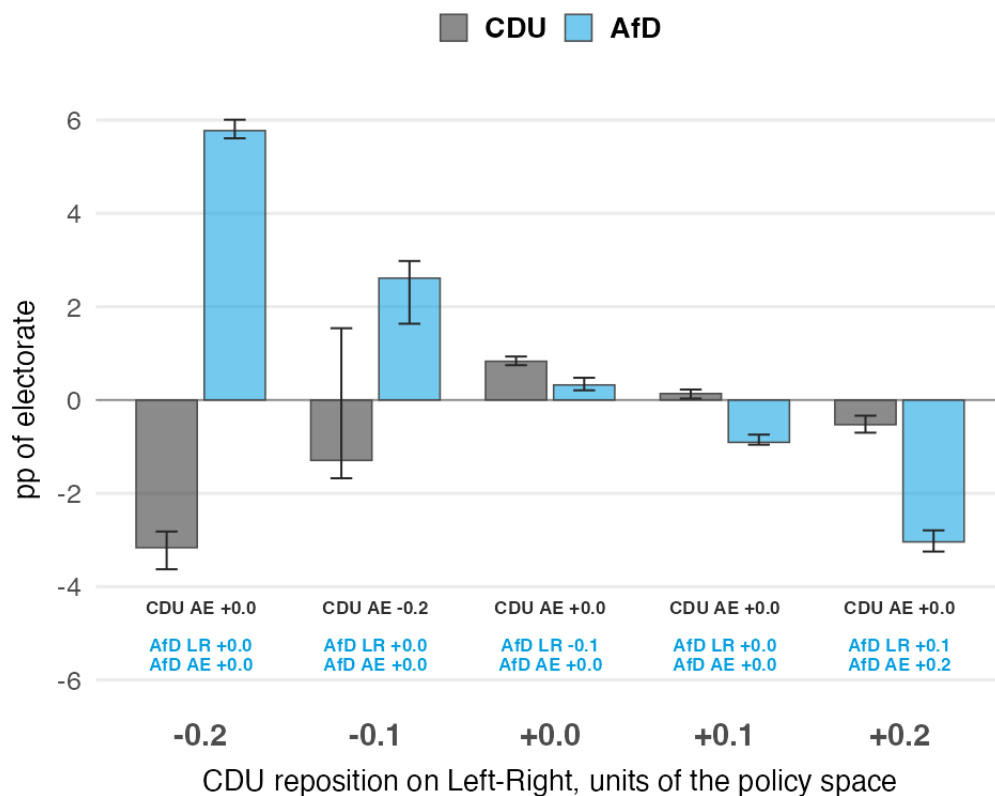
Notes: Panel (a) illustrates party positions at their national median, and hollow circles illustrate the counterfactual AfD *shifts*. The areas of AfD voters motivated more by economic (purple) or cultural (yellow) ideological stance of AfD are plotted. The illustrative weight used is $w_{cult} = 1$. Panel (b) plots bounds on the shares of voters whose bliss points lie in the yellow and purple regions across years. The results are aggregated across the two parts of the country.

Figure K25. Illustration of a target parameter (a) and results (b)



Notes: The analogue of Figure 7 for the Economics $\times$ Culture split of the Left–Right dimension. Each panel plots sharp bounds on the share of AfD voters whose bliss points lie in each 0.5-wide interval of one dimension, the other dimension left unrestricted; within each interval, bars run from 2013 (lightest) to 2025 (darkest). The strip below each panel places the intervals on the dimension, with the AfD's national median position by year (open circles) and the pooled medians of the other parties (faded dots). Bounds are computed separately for East and West Germany and aggregated; adjacent intervals share mesh cells, so the shares can sum to more than one within a year.

Figure K26. Distribution of AfD supporters' bliss points along the Economics and Culture dimensions, 2013–2025



Notes. Bars show the median net change in each party's vote share across 100 draws from the identified set, in percentage points of the vote share, after the AfD's best reply to each Left-Right move of the CDU (paired with the best Anti-Elite move in a draw); whiskers show interquartile ranges. National payoffs are population-weighted sums of the East and West changes. Both parties move both on Left-Right and Anti-Elite, in steps of 0.1 up to ± 0.2 . The labels under the bars give the modal reply across draws (the AfD responds only to the CDU's +0.2, with +0.1).

Figure K27. Estimated net CDU and AfD gain relative to the status quo by CDU Left-Right moves

L Candidate data

L.1 Candidate nominations across state lists and single-member districts

This subsection provides descriptive information on nomination patterns across state lists and single-member districts (SMDs). The data on all nominated politicians, with their names and party affiliations, were obtained by scraping the non-partisan platform *Wen Wählen?* (*Whom to Vote For?*). I focus on candidates from the main parties considered in the paper (AfD, CDU/CSU, FDP, Left, Greens, SPD).

Under the mixed system, party nominations proceed through two channels: single-member district nomination and state-list nomination. The nomination of *single-member district* candidates is formally decentralized to constituency-level members' or delegates' assemblies, and state party actors do not nominate district candidates themselves, though they retain gatekeeping power. This is because the composition of state lists is fully controlled by state party actors, and the two channels are closely linked: about 64% of candidates who appear on state lists also run in an SMD, and about 74% of those who run in an SMD also appear on a state list (Table L1 reports the exact shares across parties and years). The state lists, chosen by the state leadership, therefore constrain the district-level choices made by local party units.

Candidates nominated in single-member districts also occupy higher list positions: across all years and lists their median position is 10, against 22 for list-only candidates. District strength predicts both access and rank. Candidates placed first or second in their district hold a list place in 84% of cases, against 68% of those placed third or lower; conditional on being listed, they stand at the 28th percentile of their list at the median, against the 44th for the remainder. The pattern also holds within lists: among candidates on the same state-party list, the top tercile of district vote share sits at a median position of 7, against 12 for the bottom tercile, and an additional percentage point of district vote share is worth 0.44 list places (s.e. 0.09).

I use these data to explore the patterns of candidate nomination and renomination to single-member districts. Across 2013, 2017, 2021, and 2025 there were 7,001 unique nominations at the candidate-election level.⁷¹ Of these, 1,546 were repeat nominations from one electoral cycle to the next: the same candidate could be nominated in 2013 and 2017, in 2017 and 2021, or in 2021 and 2025. Repeat nominations account for 29.6% of all nominations in 2017, 2021, and 2025. Among them, only 91 (5.9%) are in a different electoral district, and only 9 (0.6%) in a different federal state.

Renomination patterns differ by electoral success. Among candidates who win their district, 56.3% are renominated, against 24.1% of those who lose; within each of the largest parties the rate rises with the first-vote share—from 17.1% in the bottom quartile of a party's candidates to 35.4% in the top, or 1.0 percentage points per percentage point of first-vote share (s.e. 0.08). Winning candidates who are renominated return to the same district in 99.6% of cases.⁷² Among losing candidates who are renominated, 91.5% return to the same district.

The strong attachment of candidates to their electoral districts (and federal states) is due in part to where they reside at the time of the elections. To examine this, I use the question in the GLES Candidate Survey that asks Bundestag candidates how long they have lived in their electoral district. Participation in the survey is voluntary, so not all candidates respond, and the information is self-reported. The share of candidates born in the district in which they run was 28%, 16%, and 31% in 2013, 2017, and 2021, respectively,

⁷¹The maximum possible number is 7,176. It is slightly lower because some parties—chiefly the AfD and the Greens—did not field candidates in all districts in some years.

⁷²There are two cases in which a candidate was nominated to a *different* district following a change in district boundaries. Albert Helmut Weiler won the district of Gera-Jena-Saale-Holzland-Kreis in 2013, which was dissolved before the 2017 elections; he was nominated in, and won, the district of Saalfeld-Rudolstadt-Saale-Holzland-Kreis-Saale-Orla-Kreis, which was partly composed of the district in which he had run in 2013. Kay-Uwe Ziegler won the district of Anhalt in 2021, which was merged into Anhalt-Dessau-Wittenberg before the 2025 elections; he was nominated in, and won, the neighbouring district of Mansfeld, whose boundaries were redrawn at the same time.

and the mean number of years of residence in the district was 25, 25.3, and 25.9 across the same years. About 20% of candidates had lived in their district for less than ten years in every year. While most candidates thus have long-standing ties to their district, a non-negligible share moved not long before the elections, possibly for reasons related to their political careers.

	2013	2017	2021	2025
SPD	69%	62%	67%	65%
Left	76%	71%	71%	77%
Greens	76%	77%	64%	78%
FDP	80%	80%	77%	82%
CSU	41%	41%	25%	28%
CDU	52%	47%	46%	46%
AfD	28%	57%	76%	71%
All	62%	63%	63%	66%

	2013	2017	2021	2025
SPD	96%	99%	96%	97%
Left	48%	46%	46%	45%
Greens	72%	71%	85%	85%
FDP	86%	90%	88%	95%
CSU	64%	64%	50%	55%
CDU	81%	79%	81%	82%
AfD	49%	47%	45%	56%
All	73%	72%	73%	76%

(a) Of state-list candidates, share also nominated in a district

(b) Of district candidates, share also nominated on a state list

Notes. The data are drawn from the non-partisan candidate platform *Wen Wählen?*, which provides information on the candidate lists and state lists for federal elections. These data provide the universe of all party nominations. Panel (a) is computed over state-list candidacies, panel (b) over district candidacies, matched by candidate name and year. The Piraten are excluded from both panels, including the *All* rows.

Table L1. Overlap between state-list and single-member-district nominations

L.2 Estimation of candidates' positions

1. Estimation of candidates' positions using GLES Candidate Survey Data Candidate positions on two dimensions—left–right and anti-elite—are estimated from the GLES Candidate Studies for 2013, 2017, 2021 and 2025. For each dimension, a set of survey items is first harmonized (see Table L2 for the full list of questions). Each item is then z-scored using a pooled mean and standard deviation computed across all waves in which it appears. This strategy places all items on a common metric and preserves comparability across years.

Each candidate's score on a dimension is the unweighted mean of all available z-scored items within that dimension. The anti-elite dimension comprises seven questions on satisfaction with democracy and current institutions; the left-right dimension uses questions covering redistribution, state intervention, immigration, and social and cultural issues, such as abortion and same-sex marriage. It is also supplemented by a left–right self-placement scale.

Because of the low response rate to this survey (around 35% across all years and parties), the survey was complemented with a non-partisan candidate survey, which also asks candidates about their positions on different issues in each electoral year.

2. Complementing GLES Candidate Survey Data with non-partisan candidate survey *Wen Wählen*. For each election year, candidates from the GLES Candidate Survey are merged with candidate data from the non-partisan platform *Wen Wählen*. Because the two surveys cannot be linked at the level of the individual candidate, the merge is at the district–party level: each *Wen Wählen* candidate is matched to the mean GLES-derived score of their party's candidates in their district. The GLES scores themselves come from the pooled cross-year normalization, so the ideological scale is comparable across waves. The elastic net procedure is then used to impute ideological scores for those candidates missing in the GLES Survey but present in *Wen Wählen*. Since the *Wen Wählen* questionnaire changes from election to election, the models

are estimated separately by wave (2013, 2017, 2021, 2025).

Two separate elastic net models are estimated per wave, one for each ideological dimension. The outcome y_i is the GLES-derived district–party score attached to candidate i . Only candidates whose district–party cell contains at least one GLES-surveyed candidate enter the training set.

The feature matrix $\mathbf{X}$ contains, for candidate i , the full battery of that wave’s policy questions (65 in 2013, 76 in 2017, 69 in 2021, 50 in 2025; the same set enters both dimension models). Each question is encoded as one-hot indicators for the five response categories: *Yes*, *Rather Yes*, *Undecided*, *Rather No*, *No*, plus an explicit indicator for a missing answer. No year indicators are needed since each wave is fit separately.

The elastic net minimizes a penalised least-squares criterion that interpolates between Ridge (ℓ_2) and Lasso (ℓ_1) regularisation:

$$\hat{\beta}(\lambda, \alpha) = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \lambda \left[\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2} \sum_j \beta_j^2 \right]$$

where n is the number of candidates in the training set, $\lambda \geq 0$ controls the overall amount of regularisation, and $\alpha \in [0, 1]$ determines the mix between the two penalty types. Setting $\alpha = 1$ recovers the Lasso, which drives many coefficients to exactly zero (implicit variable selection); $\alpha = 0$ recovers Ridge, which shrinks all coefficients towards zero while retaining all predictors. Intermediate values of α combine both effects.

Both λ and α are selected jointly by 10-fold cross-validation on the labelled candidates (those with a GLES-based score). The procedure is:

1. The labelled training set is randomly partitioned into 10 roughly equal folds.
2. A full Cartesian grid of 550 hyperparameter combinations is constructed:
 - $\lambda \in \{10^{-4}, \dots, 10^2\}$ (50 values, log-uniform spacing);
 - $\alpha \in \{0.0, 0.1, 0.2, \dots, 1.0\}$ (11 values).
3. For each of the $550 \times 10 = 5\,500$ fold–hyperparameter combinations the fit is run exclusively on the nine training folds and then applied to the held-out validation fold.
4. The root mean squared error (RMSE) between predicted and observed scores is recorded for each held-out fold.
5. The pair $(\hat{\lambda}, \hat{\alpha})$ that minimizes the mean RMSE across the 10 folds is selected.

The winning model is then refit on the wave’s full labelled training set using $(\hat{\lambda}, \hat{\alpha})$, and this final fit is used to predict scores for the wave’s unlabelled candidates. The selected mixtures are Ridge-leaning ($\hat{\alpha}$ between 0.1 and 0.5 across waves and dimensions), retaining 20–119 non-zero coefficients per model. The most potent questions for each year and dimension are presented in Table L3. The most potent questions and directions are as expected given the underlying dimensions.

The model performs well, explaining a large share of variation in observed GLES-based Left–Right position scores (cross-validated $R^2 = 0.76$ pooled across waves, with a root mean squared error of 0.336 against a standard deviation of the target score of 0.689 (RMSE/s.d. = 0.49). This indicates that the elastic net predictions are able to capture meaningful within-party variation in candidate positions across districts.

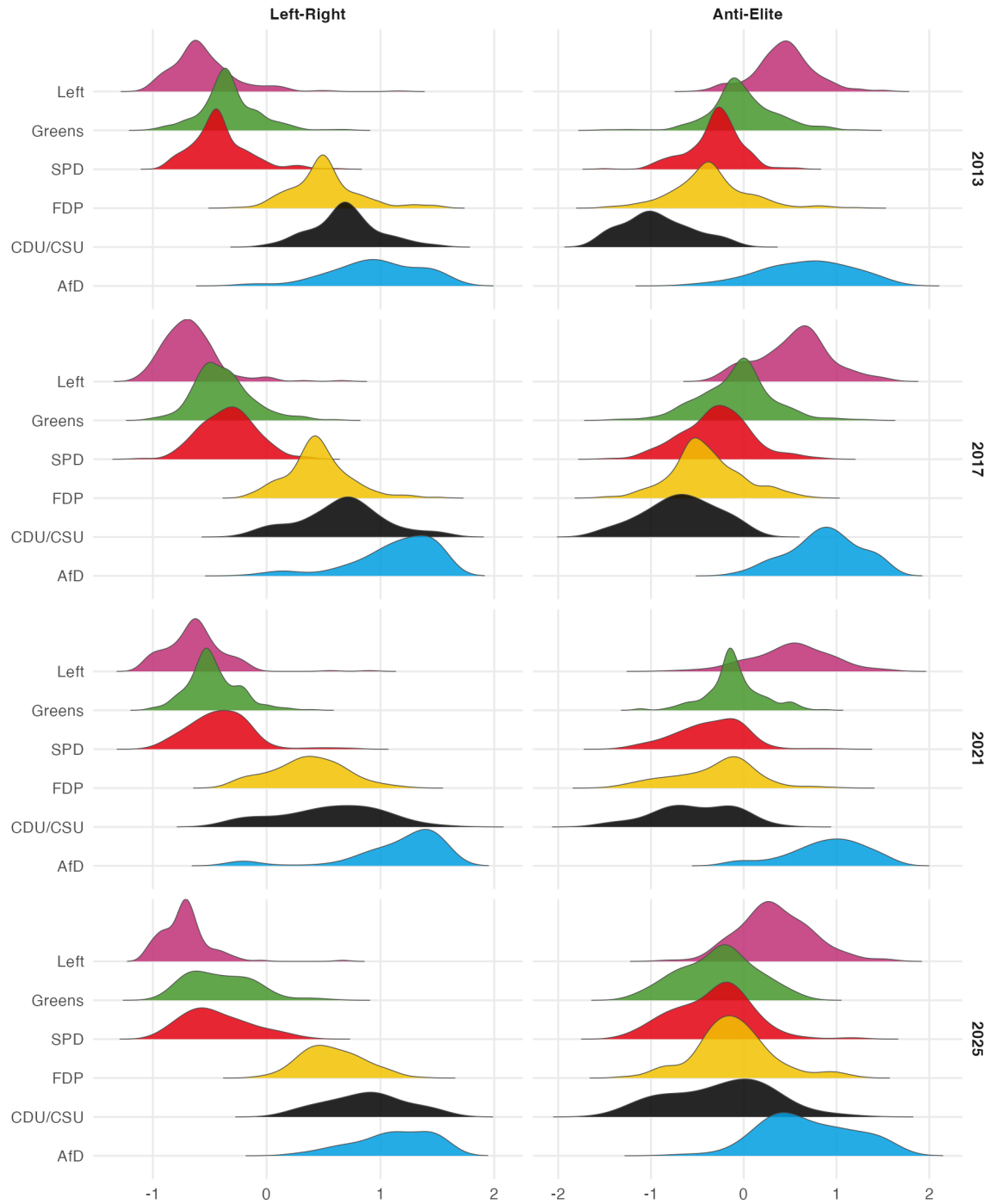
Performance is somewhat lower for the anti-elite score: an RMSE of 0.450 against a standard deviation of 0.648 (RMSE/s.d. = 0.69) and a cross-validated $R^2 = 0.53$ pooled. The anti-elite score is harder to map from the policy-stance questions available in *Wen Wählen*. Nonetheless, the model explains about half of the variance in observed anti-elite scores, a clear improvement over mean imputation.

To mitigate overfitting, all performance metrics are computed from out-of-fold cross-validation predictions. Across the four waves, 1,147 *Wen Wählen* candidates carry a GLES-based training score and 1,348 receive an elastic-net imputed score; imputed scores are clamped to $[-1.5, 1.5]$ before entering the estimation data. Combining the two candidate surveys yields a locally estimated position for about 60% of all district×party cells. Missing positions are imputed with the within-state-wave median; if that median is also unavailable, the national within-wave median serves as a fallback.⁷³

Dimension	Question	Values
Anti-Elite	Satisfaction with democracy in Germany	1–4
Anti-Elite	Democracy offers enough opportunities for citizens to participate	1–5
Anti-Elite	Laws reflect citizens’ interests	1–5
Anti-Elite	Political parties are the central intermediary between citizens and government	1–5
Anti-Elite	There should be nationwide referendums (direct votes) at the federal level	1–5
Anti-Elite	Referendums tend to lead to bad laws	1–5
Anti-Elite	Citizens have lost trust in politicians	1–5
<i>Economy:</i>		
Left–Right	In a crisis, the government should spend to stimulate economic growth	1–5
Left–Right	In a crisis, Germany should provide financial support to other countries/EU partners	1–5
Left–Right	Government should stay out of the economy	1–5
Left–Right	Providing social security should be a key goal of government	1–5
Left–Right	Government should reduce income differences	1–5
<i>Culture:</i>		
Left–Right	Immigrants should adapt to German culture	1–5
Left–Right	Same-sex marriage should be banned	1–5
Left–Right	Immigration is good for the German economy	1–5
Left–Right	Women should be free to decide about abortion	1–5
Left–Right	Immigration policy: make immigration easier vs. restrict it	1–11
<i>Overall:</i>		
Left–Right	Self-placement on the left–right scale	1–11

Table L2. Survey questions used to estimate candidate positions (GESIS Candidate Survey Data)

⁷³The global median fallback is applied in 5% of the used electoral districts, and only in Eastern Germany in 2013 and 2017.



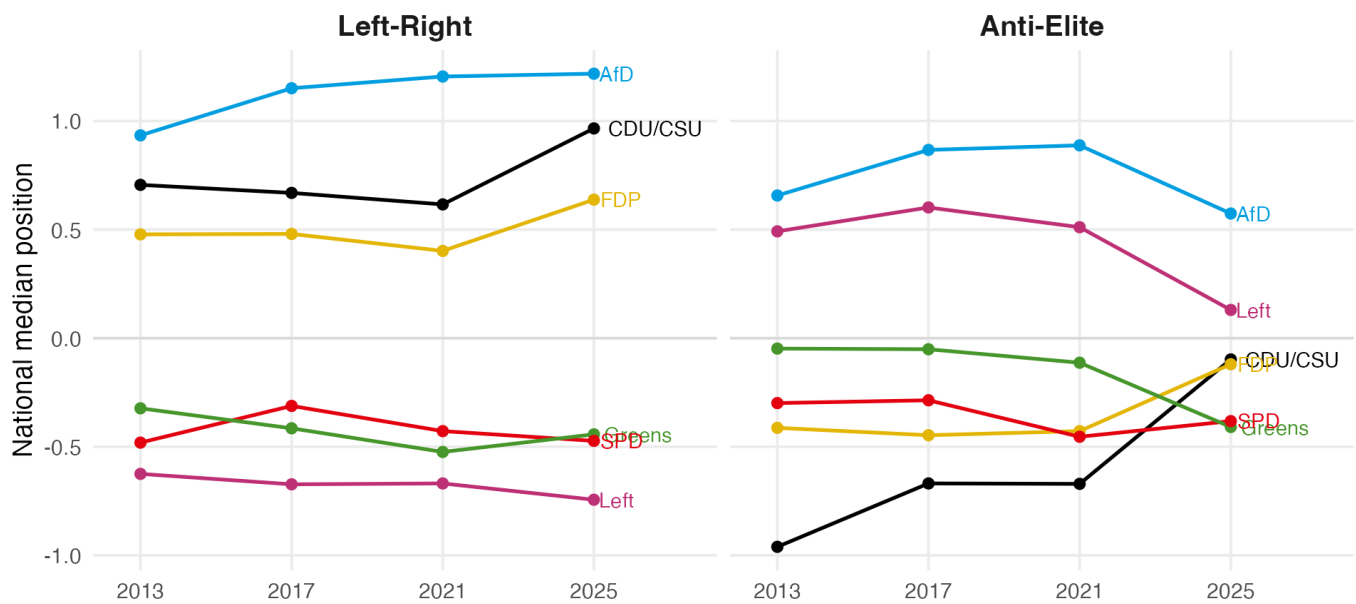
Notes. Kernel density estimates of the positions of single-member district candidates by party, separately for each election (rows) and dimension. Positions are estimated from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Each density uses a Gaussian kernel with the bandwidth selected by Scott's rule computed separately for each party, year and dimension.

Figure L1. Distribution of candidate positions by party and election year

Year	Item (translated)	Answer	Coef.
<i>Left-Right (positive = further right)</i>			
2013	The welfare state is more important than reducing debt!	Yes!	-0.08
	More rights for homosexual couples!	Yes!	-0.08
2017	Repeal “marriage for all”!	Yes!	+0.12
	Reintroduce a wealth tax!	No!	+0.07
2021	Ban gender-inclusive language!	Yes!	+0.20
	Legalise soft drugs!	No!	+0.19
2025	Raise the minimum wage substantially!	Yes!	-0.39
	Statutory health insurance for everyone!	Yes!	-0.19
<i>Anti-Elite (positive = more anti-elite)</i>			
2013	Referendums at the federal level!	Yes!	+0.13
	Stop foreign deployments of the Bundeswehr!	Yes!	+0.16
2017	Referendums at the federal level!	Yes!	+0.25
	Stop the sanctions against Russia!	Yes!	+0.18
2021	Build a common European army!	No!	+0.14
	Reintroduce the Deutsche Mark!	Yes!	+0.13
2025	Invest more in defence!	Rather yes	-0.17
	Leave NATO!	No!	-0.12

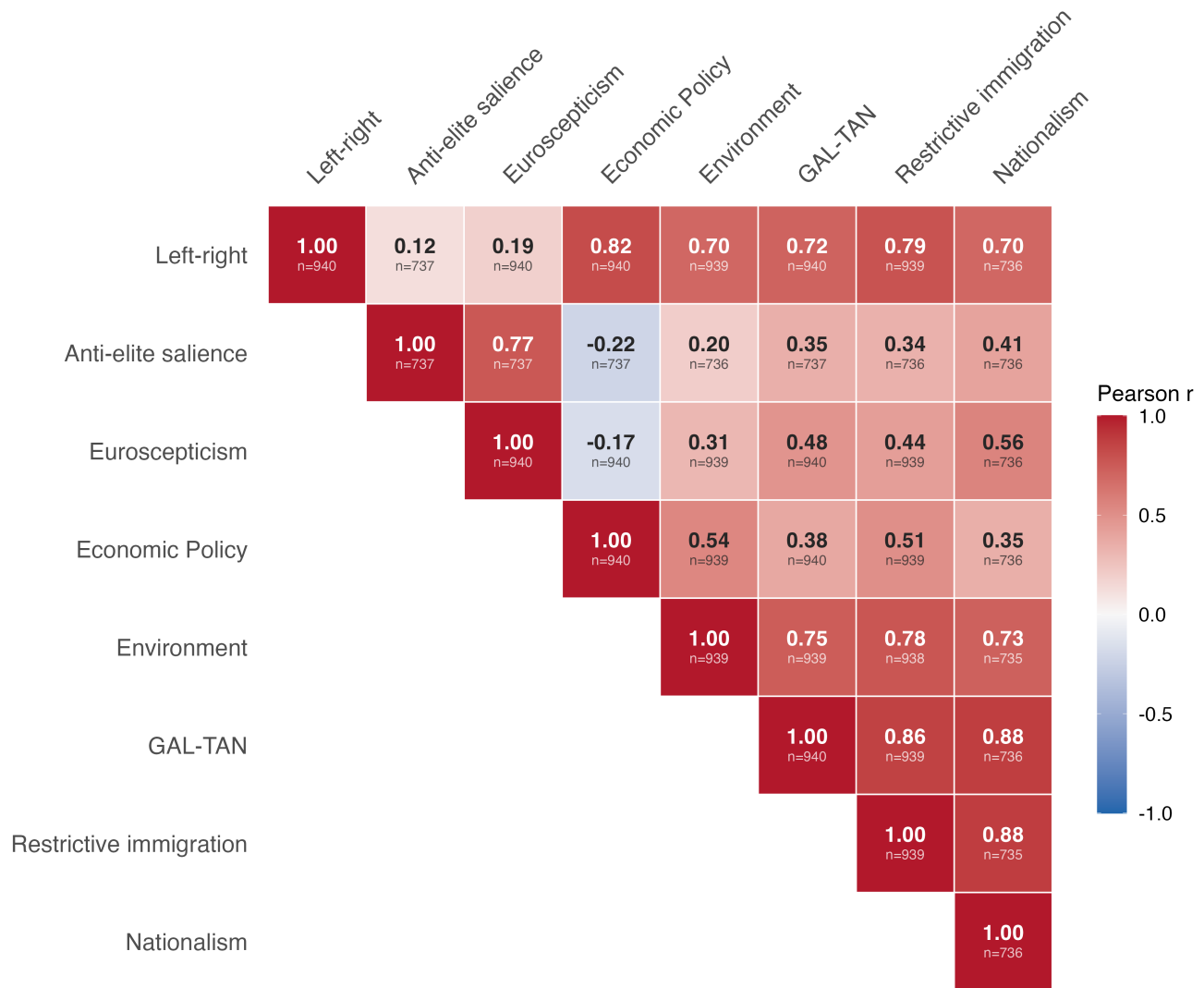
Notes. For each election year and dimension, the two *Wen Wählen* items with the largest contribution to the variance of the imputed score, measured as $\sum_a |\beta_a| \text{sd}(D_a)$ over the item’s answer dummies D_a ; the answer with the largest single contribution is shown. Coef. is the elastic-net coefficient β_a on that answer dummy: the change in the predicted GLES score (whose items are standardized, so roughly one standard deviation of the candidate score per unit) for a candidate giving this answer rather than any other answer to the item, holding all other answers fixed. The prediction targets are the district×party means of the GLES Left-Right and Anti-Elite scores; one elastic net per wave and dimension, λ and α chosen by 10-fold cross-validation. Answer categories “No answer”, “Undecided”, missing and pooled-rare levels are excluded from the ranking.

Table L3. Most Predictive *Wen Wählen* Items in the Elastic-Net Imputation



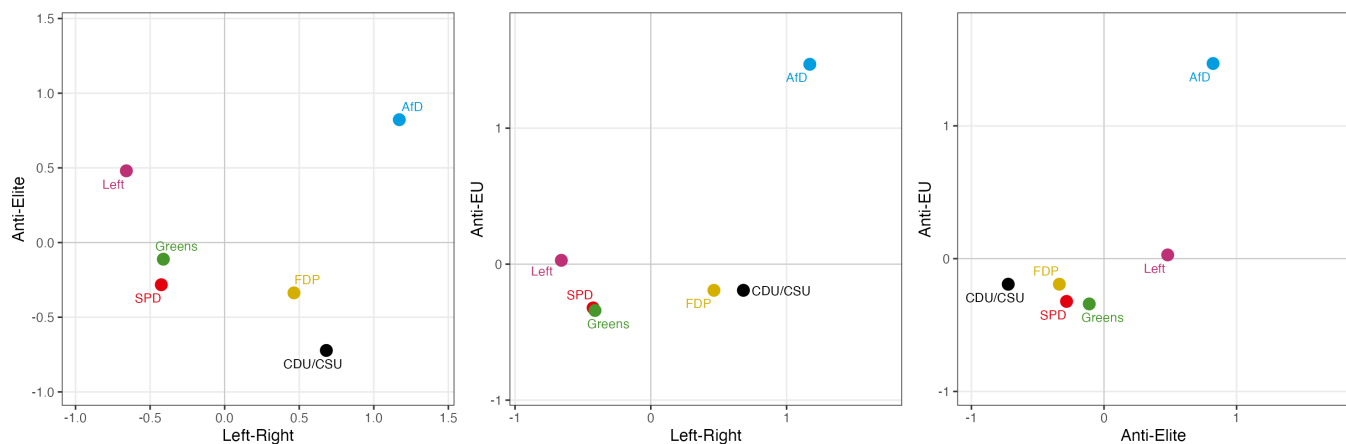
Notes. The data are based on the estimated single-member district candidate positions from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Within-party dynamics broadly follow the Chapel Hill Expert Survey (Bakker et al., 2015): for instance, the rightward and anti-elite shift of the CDU and the moderation of the AfD's anti-elite positioning between 2021 and 2025 are mirrored in the change between the 2019 and 2024 CHES waves.

Figure L2. Medians of left-right and anti-elite positions across parties over years



Notes. Each cell reports the pairwise-complete Pearson correlation between two expert-estimated party-position dimensions, computed across party-year observations. Data are the Chapel Hill Expert Survey 1999–2024 (party-level expert means), restricted to European countries and to the waves from 2010 onwards ($N = 940$ party-year observations, 408 parties, 28 countries). All CHES policy scales run 0–10 and are coded so that higher values are more right, conservative or restrictive; *Economic Policy* is the solely economic left–right scale and *Euroscepticism* reverses the EU-position item. *Anti-elite salience* and *Nationalism* were not fielded in the 2010 wave. Selected correlations (95% Fisher- z CIs in brackets): Left–Right $\sim$ Anti-elite salience 0.12 [0.05, 0.19]; Left–Right $\sim$ Euroscepticism 0.19 [0.13, 0.25]; Anti-elite salience $\sim$ Euroscepticism 0.77 [0.74, 0.80].

Figure L3. Correlations between expert-estimated party positions across policy dimensions



Notes. Each dot is the median position of a party's single-member district candidates, pooled across all four elections (2013–2025) and all federal states. Positions are estimated from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Raw Pearson correlations across all pooled candidates (95% Fisher- z CIs in brackets): Left–Right $\sim$ Anti-Elite -0.00 $[-0.04, 0.03]$; Left–Right $\sim$ Anti-EU 0.46 $[0.44, 0.49]$; Anti-Elite $\sim$ Anti-EU 0.60 $[0.58, 0.62]$. After removing party-by-year means, the latter two fall to 0.26 and 0.35 , while Left–Right $\sim$ Anti-Elite stays at 0.01 .

Figure L4. Party median positions on Left–Right, Anti-Elite and Anti-EU dimensions, pairwise